Radio Navigation Grimoire

BY CAPT. PANKAJ PAHIL



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ABOUT THE AUTHOR

Capt. Pankaj Pahil

Capt. Pankaj Pahil is the founder of Ghost Aviator, a premier platform dedicated to providing comprehensive DGCA exam preparation for pilots in India. With a mission to make high-quality aviation education accessible to everyone, he has developed a structured, chapter-by-chapter learning system designed to perfectly mirror the actual DGCA exam format.

Through Ghost Aviator, Capt. Pahil provides invaluable study materials, including these concise chapter notes, question banks with thousands of MCQs, and detailed lectures—all to ensure that students from every background can achieve their dreams of flying.



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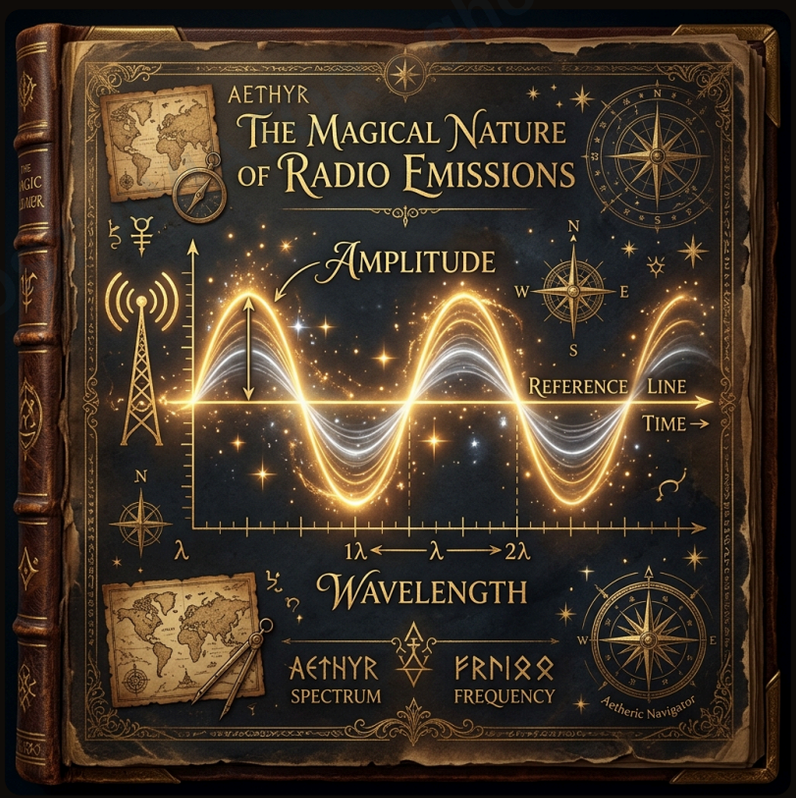
Properties of Radio Waves

CHAPTER 1

1. INTRODUCTION

NOTE

What this section covers: The historical context of radio and radar in aviation, the transition from visual to instrument-based navigation, and why understanding radio fundamentals is essential for every pilot.



PROPERTIES OF A RADIO WAVE

Radio and radar systems are now an integral and essential part of aviation, without which the current intensity of air transport operations would be unsustainable. In the early days of aviation, aircraft were flown with visual reference to the ground; flight at night, in cloud, or over the sea was not possible.

As aircraft complexity increased, it became necessary to design navigational systems that permit aircraft to operate without reference to terrain features. Early systems were very basic and inaccurate by modern standards — providing reasonable en route accuracy over land, but only limited service over the oceans. Until approximately 40 years ago, oceanic flight relied on traditional seafarer techniques of astro-navigation — taking sights on the sun, moon, stars, and planets to determine position.

Developments that commenced in the 1910s accelerated during the 1930s and 1940s , continuing to the present day, producing long-range systems that by the 1970s provided a global navigation service.

> ⚠ ✈️ **EXAM PERSPECTIVE**
>
> It is perhaps ironic that, having forsaken navigation by the stars, the most widely used modern navigation systems — Global Navigation Satellite Systems (GNSS) — are once again space-based. The ICAO position (important for the exam) is that certain terrestrial systems must be retained to back up GNSS, both for en route navigation and runway approaches.

The development of radar in the 1930s allowed air traffic control systems to be developed, providing a service capable of identifying and monitoring aircraft at a much higher intensity than would otherwise be possible. Modern satellite technology extends this capability over oceans and areas where conventional radar is not practical.

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2. THE RADIO NAVIGATION SYLLABUS

> ⓘ **NOTE**
>
> What this section covers: An overview of the syllabus structure, explaining why the foundational chapters (radio wave properties, propagation, modulation, antennae) must be mastered before the applied navigation chapters.

The syllabus begins with the nature of radio waves and how they travel through the atmosphere. This is essential to understand:

- Why different radio frequencies are selected for particular applications
- The limitations those frequencies impose
- How radio waves are produced, transmitted, received, and how information is encoded and decoded

✅ SYLLABUS LOGIC

Chapters 1–4 (waves → propagation → modulation → antennae) build the physics foundation. Chapters 5–18 apply that foundation to specific navigation and radar systems. The Q&A in Chapter 19 tests all chapters. Mastering Chapters 1–4 makes everything else easier to understand and retain.

3. ELECTROMAGNETIC (EM) RADIATION

NOTE

What this section covers: How radio waves are generated from an alternating current in a wire, the two-component nature of EM energy (electric and magnetic fields), and the concept of polarization.

If a direct electric current (DC) is passed through a wire, a magnetic field is generated around the wire perpendicular to the current flow.

If an alternating electric current (AC) is passed through the wire, because the direction of current flow is changing, the polarity of the magnetic field also changes — reversing polarity as the current direction reverses. At low frequencies the magnetic field collapses back to zero with the current; but as frequency increases , the magnetic field will not have collapsed completely before the reversed field starts to establish itself, and energy begins to travel outward from the wire in the form of electromagnetic radiation — i.e., radio waves.

The resulting EM energy is made up of two components:

- ✦ An Electrical (E) field — parallel to the wire
- ✦ A Magnetic (H) field — perpendicular to the wire

🔵 KEY CONCEPT — THE TWO-COMPONENT WAVE

A radio wave is NOT just an electrical or just a magnetic field — it is both simultaneously, at right angles to each other and at right angles to the direction of propagation . In Figure 1.1, blue lobes (E field) are in the vertical (X) plane; red lobes (H field) are in the horizontal (Y) plane; the wave travels along the Z axis. This perpendicular relationship is fundamental — it cannot be otherwise for EM radiation.

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4. POLARIZATION

NOTE

What this section covers: Linear polarization (vertical and horizontal), circular polarization, the importance of aerial orientation for maximum signal strength, and the exam-relevant advantages of circular polarization in aviation.

The polarization of radio waves is defined as the plane of the electric field and is dependent on the plane of the aerial:

- A vertical aerial → electrical field in the vertical plane → vertically polarized wave
- A horizontal aerial → electrical field in the horizontal plane → horizontally polarized wave

⚠ CRITICAL RULE — AERIAL ALIGNMENT

To receive maximum signal strength from an incoming radio wave, the receiving aerial must be in the same plane as the polarization of the wave. A vertically polarized wave requires a vertical aerial. Misalignment between transmit and receive polarization causes significant signal loss.

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Circular Polarization

Circular polarization can be produced in a variety of ways — one method uses a helical antenna . In circular polarization, the electrical (and hence magnetic) field rotates at the frequency of the radio wave . The rotation may be right-handed or left-handed, dependent on the orientation of the aerial array.

✅ TWO KEY ADVANTAGES OF CIRCULAR POLARIZATION (EXAM FAVOURITE)

Rain clutter elimination in radar: When circularly polarized energy is reflected from water droplets, the circularity of the returned signal is reversed . A receiver configured for the original rotation therefore rejects this return — eliminating precipitation clutter.
Orientation-independent reception: A simple dipole aerial can receive a circularly polarized wave regardless of its physical orientation. This is a major advantage in mobile systems — cellular phones, satellite communication, and satellite navigation (GNSS).

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💡 EXAM TIP

Questions on circular polarization almost always focus on one of these two advantages. Memorise: "Rain clutter out, dipole orientation freed."

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5. RADIO WAVES & THE PERIOD

NOTE

What this section covers: The sinusoidal nature of radio waves, the period (τ), how frequency is derived from period, the units Hz/kHz/MHz/GHz, and how to perform frequency–wavelength conversions using the core formula.

The length of time it takes to generate one complete cycle of a radio wave is known as the period , signified by the Greek letter tau (τ) , and measured in microseconds (μs) .

1 μs = 10 −6 second

Frequency

Frequency is the reciprocal of period:

FORMULA 1 — FREQUENCY FROM PERIOD

$$f = 1 / \tau$$

📐 WORKED EXAMPLE — PERIOD TO FREQUENCY

Given: $\tau = 0.125\ \mu s = 0.125 \times 10^{-6}$ s

$f = 1 / (0.125 \times 10^{-6})$

$f = 8{,}000{,}000$ Hz $= 8$ MHz

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Frequency Units

UNIT	SYMBOL	VALUE
Kilohertz	kHz	103Hz = 1,000 Hz
Megahertz	MHz	106Hz = 1,000,000 Hz
Gigahertz	GHz	109Hz = 1,000,000,000 Hz

6. WAVELENGTH

NOTE

What this section covers: The speed of radio waves, the wavelength formula, the simplified MHz–metre relationship, and worked examples for converting between frequency and wavelength.

The speed of radio waves (c) is the same as the speed of light (both are electromagnetic radiation):

c = 300,000,000 m/s (= 300 × 10 6 m/s) or 162,000 nautical miles per second

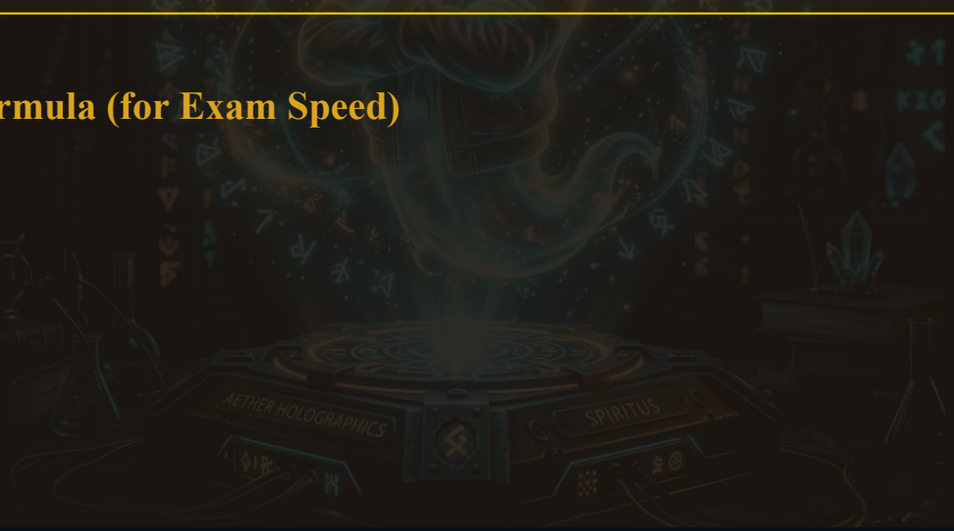
FORMULA 2 — WAVELENGTH FROM PERIOD & SPEED

$$\lambda = c \times \tau \quad (2)$$

$$\lambda = c / f \quad (3)$$

$$f = c / \lambda \quad (4)$$

Simplified Formula (for Exam Speed)



SIMPLIFIED — MHZ & METRES

$$\lambda\,(m) = 300 / f\,(MHz)$$

$$f\,(MHz) = 300 / \lambda\,(m)$$

WORKED EXAMPLES

> $\lambda = 300 / 121.5 = 2.47$ m

> $f = 300 / 1515 = 0.198$ MHz $= 198$ kHz

> $f = 300 / 0.032 = 9375$ MHz $= 9.375$ GHz

> $\lambda = 300 / 0.357 = 840$ m

⚠ COMMON ERROR — UNIT CONVERSION

Always convert to MHz and metres before using the simplified formula. kHz must be divided by 1000. GHz must be multiplied by 1000. cm must be divided by 100. Skipping this step is the single most common calculation error in this chapter.

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7. FREQUENCY BANDS

NOTE

What this section covers: The 8 frequency bands of the radio spectrum (VLF to EHF), their frequency and wavelength ranges, and their civil aeronautical applications. This table is directly examinable.

The radio part of the electromagnetic spectrum extends from 3 kHz to 300 GHz . It is divided into 8 frequency bands, each related to its neighbours by a factor of 10 .

FREQUENCY BAND	ABBREVIATION	FREQUENCIES	WAVELENGTHS	CIVIL AERONAUT USAGE
Very Low Frequency	VLF	3–30 kHz	100–10 km	Nil
Low Frequency	LF	30–300 kHz	10–1 km	NDB / ADF
Medium Frequency	MF	300–3000 kHz	1000–100 m	NDB / ADF, lo range communi
High Frequency	HF	3–30 MHz	100–10 m	Long-range communication
Very High Frequency	VHF	30–300 MHz	10–1 m	Short-range co VDF, VOR, ILS Localizer, Mark Beacons
Ultra High Frequency	UHF	300–3000 MHz	100–10 cm	ILS Glidepath, SSR, Satellite c GNSS, long-ra radars
Super High Frequency	SHF	3–30 GHz	10–1 cm	RADALT, AWI (Weather Radar MLS, short-ran radars
Extremely High Frequency	EHF	30–300 GHz	10–1 mm	Nil

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⚠ 💡 **MEMORY AID — "VERY LOVELY MEN HAVE VERY UGLY SCRUFFY ENEMIES"**

V LF · L F · M F · H F · V HF · U HF · S HF · E HF Key associations to memorise: NDB/ADF → LF and MF VOR, ILS Localizer, VHF Comms, Markers → VHF ILS Glidepath, DME, SSR, GNSS → UHF Weather Radar (AWR), RADALT, MLS → SHF

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8. PHASE COMPARISON

NOTE

What this section covers: The concept of phase, the requirements for a valid phase comparison (same frequency), how to determine phase difference graphically and mathematically, and why phase comparison is used in navigation systems such as VOR and ILS.

Some radio navigation systems use the comparison of phase between two signals to define navigational information. Two critical requirements:

⚠ TWO ABSOLUTE REQUIREMENTS FOR PHASE COMPARISON

Both signals must have the same frequency — otherwise any phase comparison is meaningless. The comparison must yield a positive result (0° to 360°). If a calculation gives a negative value, add 360°.

One signal is designated the reference signal and the other the variable signal .

Graphical Method

- Identify zero-phase on the reference wave (point A in Figure 1.4)
- Move in the positive direction from point A
- Measure the phase angle the reference wave travels through before zero-phase is reached on the variable wave (point B)
- That angle is the phase difference

In Figure 1.4: starting at zero phase on the reference wave (point A), the reference wave travels through 270° before zero phase is reached on the variable wave (point B). Phase difference = 270° .

Mathematical Method

PHASE DIFFERENCE FORMULA

$$\text{Phase Difference} = \text{Phase}_{\text{Reference}} - \text{Phase}_{\text{Variable}}$$

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✨📐 WORKED EXAMPLE FROM FIGURE 1.4

> At origin: Reference phase = 0° (= 360°)

> Variable phase = 090°

> Phase difference = 360° − 090° =270°✓ (positive — valid)

✨📐 WORKED EXAMPLE — NEGATIVE RESULT CORRECTION

> Reference = 110° Variable = 315°

> 110° − 315° = −205° (negative — invalid)

> Add 360°: −205° + 360° =155°✓

⚠

💡 EXAM TIP — WHERE IS PHASE COMPARISON USED?

Phase comparison is the fundamental operating principle of VOR (30 Hz reference vs 30 Hz variable), ILS Localizer (90 Hz vs 150 Hz amplitude), and LORAN-C (pulse phase comparison). Understanding this concept here makes those later chapters far easier.

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9. PRACTICE FREQUENCY–WAVELENGTH CONVERSIONS

ⓘ

NOTE

What this section covers: Worked solutions to the 10 practice conversion problems from the source, with frequency band identification for each.

#	GIVEN	FIND	SOLUTION	BAND
1	f =198 kHz	λ	300/0.198 =1515 m	LF
2	λ =2.7 m	f	300/2.7 =111.1 MHz	VHF
3	f =5.025 GHz	λ	300/5025 =0.0597 m = 5.97 cm	SHF
4	λ =137.5 m	f	300/137.5 =2.182 MHz = 2181.8 kHz	MF
5	f =137.5 MHz	λ	300/137.5 =2.18 m	VHF
6	λ =3 km = 3000 m	f	300/3000 =0.1 MHz = 100 kHz	LF
7	f =329 MHz	λ	300/329 =0.912 m = 91.2 cm	UHF
8	λ =29 cm = 0.29 m	f	300/0.29 =1034 MHz	UHF
9	f =500 kHz	λ	300/0.5 =600 m	MF
10	λ =5 cm = 0.05 m	f	300/0.05 =6000 MHz = 6 GHz	SHF

⚡ QUICK REVISION SUMMARY — CHAPTER 1

- ✧ EM wave =E field (parallel to wire) + H field (perpendicular), both at 90° to propagation direction
- ✧ Polarization = plane of theelectric field
- ✧ Maximum signal: receiving aerial must be in thesame plane as wave polarization
- ✧ Circular polarization advantages:rain clutter rejection+dipole orientation-free reception
- ✧ Speed of radio waves:300 × 10^6 m/sor162,000 NM/s
- ✧ Key formulas:f = 1/τ|λ = c/f| simplified:λ(m) = 300 / f(MHz)
- ✧ Radio spectrum:3 kHz to 300 GHz, 8 bands
- ✧ Phase comparison requirement:same frequency, result must bepositive
- ✧ ICAO position:terrestrial systems must be retainedas GNSS backup

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📝 Practice Questions & Detailed Answers How to Use This Section Cover the answer panel and attempt each question independently. Reveal the answer, read the explanation, and follow the anchor link back to the relevant study section if you need to review the concept. The distractor analysis tells you exactly why each wrong option is wrong — that depth is where exam marks are won.

VALUE	DESCRIPTION	SECTION
3 kHz – 300 GHz	Full radio spectrum range	§7
300×10^6 m/s	Speed of radio waves (metres)	§6
162,000 NM/s	Speed of radio waves (nautical miles)	§6
3–30 kHz / 100–10 km	VLF band	§7
30–300 kHz / 10–1 km	LF band (NDB/ADF)	§7
300–3000 kHz / 1000–100 m	MF band (NDB/ADF, comms)	§7
3–30 MHz / 100–10 m	HF band (long-range comms)	§7
30–300 MHz / 10–1 m	VHF band (comms, VOR, ILS LOC)	§7
300–3000 MHz / 100–10 cm	UHF band (ILS GP, DME, SSR, GNSS)	§7
3–30 GHz / 10–1 cm	SHF band (AWR, MLS, RADALT)	§7
30–300 GHz / 10–1 mm	EHF band (nil aviation use)	§7

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Chapter 1 — Properties of Radio Waves | For private study use only

Chapter Expert Quiz

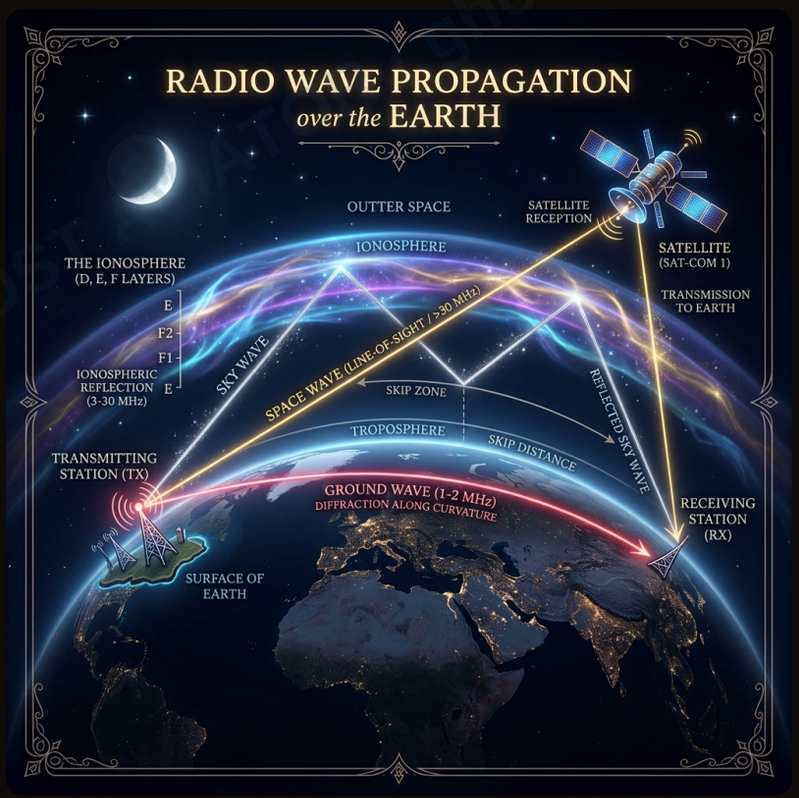
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Radio Propagation Theory

CHAPTER 2

1. INTRODUCTION

> **NOTE**
>
> What this section covers: The definition of propagation, why different frequency bands use different paths, and how propagation paths determine both the uses and limitations of specific frequency bands.



RADIO WAVE PROPAGATION PATHS

In the context of radio waves, propagation simply means how radio waves travel through the atmosphere. Different frequency bands use different propagation paths; the propagation path often determines the uses to which a particular frequency band can be put in either communication or navigation systems, and also imposes limitations on the use of those frequencies.

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2. FACTORS AFFECTING PROPAGATION

NOTE

What this section covers: Six factors that affect how far and how reliably a radio wave reaches a receiver: attenuation (absorption + inverse square law), static interference, fading, transmitter power, receiver sensitivity, and directivity.

2.1 ATTENUATION

Attenuation is the term for the loss of signal strength in a radio wave as it travels outward from the transmitter. There are two distinct aspects:

Absorption

As a radio wave travels outward, its energy is absorbed and scattered by: molecules of air and water vapour, dust particles, water droplets, vegetation, the surface of the earth, and the ionosphere.

⚠ ABSORPTION RULE

The effect of absorption (except ionospheric absorption) increases as frequency increases and becomes a very significant factor above approximately 1000 MHz .

Inverse Square Law

EM radiation from an aerial spreads out as the surface of a sphere, so power decreases with increasing distance.

Power available is proportional to the inverse of the square of the range: $P \propto 1/R^2$

💡 KEY EXAM IMPLICATION

To double the range of a transmitter, the power output must be increased by a factor of 4 ($2^2 = 4$). To triple the range → power × 9. To increase range by factor n → power × n^2.

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✨ 📐 WORKED EXAMPLE — POWER & RANGE

> Q: Increase VHF range from 50 NM to 100 NM. Factor = 100/50 = 2. Power factor = 2^2 =4

> Q: Increase range from 50 NM to 150 NM. Factor = 3. Power factor = 3^2 =9

2.2 STATIC INTERFERENCE

Large amounts of static electricity are generated in the atmosphere by weather, human activity, and geological activity. Key points:

- Effect of static interference is greater at lower frequencies
- At VHF and above , interference is generally negligible
- Radio waves travelling through the ionosphere collect interference at all frequencies
- Receiver and transmitter circuitry also produce static interference

The strength of the required signal relative to interference is expressed as the Signal-to-Noise Ratio (S/N) . Best clarity/accuracy requires minimising the noise floor.

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2.3 FADING

Transmissions following different paths (e.g. reflections) can arrive at a receiver simultaneously but not necessarily in phase. In extreme cases two signals arrive in anti-phase and cancel each other out. Signals going in and out of phase produce alternate fading and strengthening of the received signal.

2.4 POWER

An increase in transmitter power output will increase range, within the limits of the inverse square law. To double the range → power × 4.

2.5 RECEIVER SENSITIVITY

If internal noise in a receiver can be reduced, the receiver will process weaker signals — increasing effective range. However, reducing receiver noise is an expensive process.

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2.6 DIRECTIVITY

Concentrating power into a narrow beam increases range or reduces power required for a given range. However, the signal will only be usable in the direction of the beam .

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3. PROPAGATION PATHS OVERVIEW

NOTE

What this section covers: The four propagation paths used in aviation: surface wave, space wave, sky wave, and satellite comms. Which frequencies use which path.

Ionospheric propagation is affected by the properties of the ionosphere. For this chapter the focus is on sky wave . Satellite propagation is covered in Chapter 18. Knowledge of propagation below 30 kHz is not required.

4. NON-IONOSPHERIC PROPAGATION

4.1 SURFACE WAVE

NOTE

What this section covers: How surface waves travel, diffraction, factors affecting surface wave range (frequency, terrain, polarization), and the range formula for MF transmissions.

Surface wave propagation exists at frequencies from approximately 20 kHz to 50 MHz (upper VLF to lower VHF). The portion of the wave in contact with the Earth's surface is retarded, causing the wave to bend around the surface — a process known as diffraction .

Factors Affecting Surface Wave Range

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⚠ THREE KEY FACTORS — ALL EXAMINABLE

Frequency: As frequency increases → surface attenuation increases → surface wave range decreases. Surface wave is effectively non-existent above HF (>30 MHz). Surface type: Losses are greater over land than over sea (sea has good electrical conductivity). Greater ranges are achievable over sea . Polarization: A horizontally polarized wave is attenuated very quickly. Therefore vertical polarization is generally used at lower frequencies.

This is the primary propagation path for the LF frequency band and the lower part of the MF frequency band (i.e. 30 kHz to 2 MHz).

SURFACE WAVE RANGE FORMULA — MF AT 300 KHZ

$$\text{Sea:}\quad \text{Range} \approx 3 \times \sqrt{\text{Power (kW)}} \quad [\text{NM}]$$

$$\text{Land:}\quad \text{Range} \approx 2 \times \sqrt{\text{Power (kW)}} \quad [\text{NM}]$$

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WORKED EXAMPLE

300 kHz transmitter, Power = 10 kW

Sea range = $3 \times \sqrt{10} = 3 \times 3.162 = \sim 9.5$ NM $\times 10 = \sim 300$ NM

Land range = $2 \times \sqrt{10} = 2 \times 3.162 = \sim 6.3 \times 10 = \sim 200$ NM

($\sqrt{\text{Power}}$ where Power in kW, result in hundreds of NM for 300 kHz)

4.2 SPACE WAVE

NOTE

What this section covers: The two components of the space wave (direct + reflected), line-of-sight limitation at VHF and above, atmospheric refraction, and the LOS range formula.

The space wave comprises two paths: a direct wave and a reflected wave .

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At frequencies of VHF and above , radio waves start to behave more like visible light — there is a radio horizon just as there is a visual horizon. The only atmospheric propagation at these frequencies is line of sight .

⚠ CRITICAL VHF RULE

At VHF and above, regardless of transmitter power, if the receiver is below the line-of-sight range , it will receive nothing . Power is irrelevant below the radio horizon.

There is some atmospheric refraction which bends radio waves towards the Earth's surface, extending range slightly beyond the geometric horizon. For the EASA syllabus, the line-of-sight range formula is:

LOS RANGE FORMULA (VHF AND ABOVE)

$$\text{Range (NM)} = 1.23 \times (\sqrt{hTX} + \sqrt{hRX})$$

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📐 WORKED EXAMPLES

Range = $1.23 \times (\sqrt{1024} + \sqrt{1600}) = 1.23 \times (32 + 40) = 1.23 \times 72 = 88.6$ NM

$200 = 1.23 \times (\sqrt{1700} + \sqrt{hAC})$

$200/1.23 = 162.6 = \sqrt{1700} + \sqrt{hAC} = 41.23 + \sqrt{hAC}$

$\sqrt{hAC} = 162.6 - 41.23 = 121.37$

$hAC = 121.37^2 \approx 14{,}731$ ft $\approx 15{,}000$ ft

💡 EXAM TIP — NOTE 1

"Regardless of the possible propagation paths, if a receiver is in line of sight with a transmitter, then the space wave will be received." This means at lower frequencies where surface wave or sky wave exist, the space wave is ALSO received if LOS exists — the space wave is always present when LOS is available.

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5. THE IONOSPHERE

NOTE

What this section covers: Formation of the ionosphere, why ionization occurs, what affects ionization intensity (time of day, season, latitude, solar flares), and the dangers of solar flare activity to aviation systems.

The ionosphere extends upward from approximately 60 km to the limits of the atmosphere (notionally 1500 km). At these altitudes, atmospheric pressure is very low (0.22 hPa at 60 km) and gaseous atoms are widely dispersed.

Incoming solar radiation at ultra-violet and shorter wavelengths interacts with atoms, raising their energy levels and causing electrons to be ejected from atomic shells. Since an atom is electrically neutral, the result is:

- ✦ Negatively charged free electrons
- ✦ Positively charged ions (the remaining part of the atom)

Ionization Intensity — Key Relationships

🔵 FACTORS CONTROLLING IONIZATION LEVELS

Time of day: Highest levels ~ 1400 local time (balance of ionization vs decay); lowest just before sunrise Season: Higher in summer than winter (greater solar radiation intensity) Latitude: Increases as latitude decreases (closer to equator = more solar intensity) Solar flares: Unpredictable but can cause exceptionally high ionization levels

⚠ SOLAR FLARE HAZARD TO AVIATION

Intense solar flare activity → severe disruption of communication and navigation systems, particularly space-based systems . It is not unusual for communication satellites to be shut down during periods of intense solar flare activity to avoid damage.

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Ionization levels do not increase linearly with altitude. Gravitation and terrestrial magnetism affect the distribution of gases at high altitudes (where normal atmospheric mixing is absent), causing ionized particles to form into discrete layers .

6. IONOSPHERIC LAYERS (D, E, F)

NOTE

What this section covers: The three ionospheric layers, their average altitudes, day/night behaviour, seasonal variations, and how ionization is not uniform within each layer.

LAYER	AVG ALTITUDE	DAY BEHAVIOUR	NIGHT BEHAVIOUR	AVIATION SIGNIFICANCE
D Layer	~75 km	Forms at sunrise; absorbs LF/MF frequencies below ~2 MHz	Disappears at sunset	Blocks sky wave during day; night absence allows LF/MF sky wave
E Layer	~125 km	Reduces in altitude at sunrise	Increases in altitude after sunset; present 24 hours	Refracts sky wave up to ~2 MHz; max sky wave range ~1350 NM
F Layer	~225 km	Splits into F1 and F2 at sunrise	Rejoins at sunset	Refracts 2–50 MHz sky wave; max range ~2200 NM
F2 (summer)	Up to>400 km	Increases in altitude in summer	Reduces in winter	Greater skip distances in summer

⚠ DAWN & DUSK INSTABILITY

Around dawn and dusk the ionosphere is in a transitional state — "electrical turbulence." Radio navigation and communication systems using the ionosphere are subject to excessive interference and disruption during these periods.

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7. SKY WAVE PROPAGATION

7.1 CRITICAL ANGLE & SKIP DISTANCE

NOTE

What this section covers: How ionization refracts radio waves back to Earth, the concept of the critical angle, skip distance, and what "first returning sky wave" means.

As a radio wave transits an ionized layer, it encounters an increasing density of ions toward the layer centre, then decreasing density toward the far edge:

- Waves at right angles to the layer → retarded but maintain a straight path (pass through)
- Waves at an angle → refracted away from the normal entering, refracted back toward normal exiting

If the radio wave refracts to the Earth horizontal before reaching the layer centre → it undergoes total internal refraction and returns to the Earth's surface as sky wave .

🔵 KEY DEFINITIONS — MUST KNOW

Critical angle: The angle from the vertical at which the first sky wave returns to Earth (total internal refraction occurs) Skip distance: The distance from the transmitter to the point where the first returning sky wave appears at the surface Dead space: The area between where the surface wave is totally attenuated and where the first returning sky wave appears — no detectable signal

7.2 DEAD SPACE

Sky wave occurs in the LF, MF, and HF frequency bands. Frequencies above 50 MHz (VHF) do not produce sky waves under normal conditions — VHF navigation systems are therefore free of sky wave interference.

Sky wave only likely above 50 MHz during abnormal ionospheric conditions associated with intense sunspot or solar flare activity.

7.3 EFFECT OF CHANGES IN IONIZATION & FREQUENCY

CHANGE	CRITICAL ANGLE	SKIP DISTANCE	DEAD SPACE	SURFACE WAVE RANGE
↑ Ionization	Decreases (smaller)	Decreases	Decreases	No direct effect
↓ Ionization	Increases	Increases	Increases	No direct effect
↑ Frequency	Increases	Increases	Increases	Decreases
↓ Frequency	Decreases	Decreases	Decreases	Increases
↑ Layer altitude	—	Increases	Increases	No direct effect

⚠

💡 MEMORY AID — FREQUENCY & DEAD SPACE

As frequency increases : skip distance ↑ AND surface wave range ↓. Both effects make the dead space larger from both ends simultaneously. This is the most tested relationship in this chapter.

7.4 LF & MF SKY WAVE

⚠

⚠ EXAM-CRITICAL: LF/MF NIGHT-TIME SKY WAVE PROBLEM

During the day: D-layer absorbs sky wave at frequencies below ~ 2 MHz . Only surface wave is present for LF/MF — reliable for NDB/ADF navigation. At night: D-layer disappears. Sky waves refracted from the E-layer appear at relatively short ranges and cause interference with short-range navigation systems (NDB/ADF) relying on surface wave reception. This makes ADF unreliable at night — covered in detail in Chapter 7.

7.5 ACHIEVABLE RANGES & MULTI-HOP

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✅ SKY WAVE MAXIMUM RANGES

E-layer (avg 125 km): average max sky wave range = 1350 NM F-layer (avg 225 km): average max sky wave range = 2200 NM Multi-hop sky wave: wave refracted at ionosphere → reflected from Earth → refracted again. Can achieve ranges of half the diameter of the Earth

8. HF COMMUNICATIONS (MUF & OWF)

NOTE

What this section covers: Why HF sky wave is currently the only practical long-range communication method over oceans, the Maximum Usable Frequency (MUF), the Optimum Working Frequency (OWF), day/night frequency selection, range vs frequency selection, and the classic mid-Atlantic HF example.

Over oceans and uninhabited land areas, VHF is impractical (line-of-sight limitation). To achieve ranges of 2000–3000 NM :

- Surface wave at these ranges requires VLF/LF frequencies → complex, heavy equipment, high static, low data rates
- Currently the only practical solution is HF Communications using sky wave propagation

Maximum Usable Frequency (MUF)

The MUF for a given range is the frequency of the first returning sky wave . This is the ideal frequency because it has had the shortest path through the ionosphere — less attenuation, less static interference.

⚠ PROBLEM WITH USING MUF DIRECTLY

Since ionization levels fluctuate, a decrease in ionization increases the skip distance. If operating at the MUF, even a small reduction in ionization can cause loss of signal. Therefore MUF is not used as the operating frequency.

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Optimum Working Frequency (OWF)

OWF = 0.85 × MUF — determined by decades of experimentation and experience.

The OWF provides a safety margin below the MUF so that normal ionization fluctuations do not cause loss of signal.

Day vs Night Frequency Selection

🔵 DAY/NIGHT RULE OF THUMB

Ionization is lower at night → frequency required at night is roughly half the frequency required by day for the same range.

Range vs Frequency Selection

✅ RANGE–FREQUENCY RELATIONSHIP

Short ranges → require lower frequencies (smaller skip distance needed) Long ranges → require higher frequencies (larger skip distance needed)

💡 CLASSIC EXAM SCENARIO — MID-ATLANTIC AT SUNRISE

An aircraft in mid-Atlantic at sunrise communicates with UK on 12 MHz . What frequency to communicate with USA? Answer: 6 MHz Reasoning: Midway between aircraft and UK → it is day (high ionization → high frequency needed). Midway between aircraft and USA → it is night (low ionization → roughly half the frequency). Half of 12 = 6 MHz .

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9. PROPAGATION SUMMARY

NOTE

What this section covers: A consolidated table of which propagation path is used (and which is present but not normally utilised) for each frequency band.

FREQUENCY BAND	PRIMARY PATH	SECONDARY PATH (PRESENT BUT NOT NORMALLY USED)
LF	Surface Wave	(Sky Wave)
MF	Surface Wave	(Sky Wave)
HF	Sky Wave	(Surface Wave)
VHF	Space Wave	—
UHF	Space Wave	—
SHF	Space Wave	—
EHF	Space Wave	—

10. SUPER-REFRACTION & SUB-REFRACTION

NOTE

What this section covers: Abnormal atmospheric refraction at VHF and above — super-refraction (extended range, duct propagation) and sub-refraction (reduced range), their causes, and where they occur.

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10.1 SUPER-REFRACTION

Significant at frequencies above 30 MHz (VHF and above). Radio waves experience greater than normal downward bending towards the Earth's surface, giving range increases of up to 40% above the usual LOS range .

✅ **CONDITIONS FOR SUPER-REFRACTION**

Decrease in relative humidity with height Temperature falling more slowly with height than standard (reduced lapse rate) Fine weather and high pressure systems Warm air flowing over a cooler surface

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🔵 DUCT PROPAGATION

In extreme cases — a low-level temperature inversion with marked decrease in humidity with height (warm dry air above cool moist air) — a low-level duct forms that traps radio waves above 30 MHz, giving extremely long ranges . Classic real-world example: UK television interference from continental stations during summer high pressure. Most common locations: warm desert areas bordering oceanic areas — Mediterranean and Caribbean seas. Also in temperate latitudes during winter high pressure (descending adiabatically warmed air over cool moist surface air).

10.2 SUB-REFRACTION

Much rarer than super-refraction. Causes a reduction in normal refraction, giving a decrease in LOS range of up to 20% .

⚠ CONDITIONS FOR SUB-REFRACTION

Increase in relative humidity with increasing height Temperature decreasing with height at a greater rate than standard (super-adiabatic lapse rate) Poor weather with low pressure systems Cold air flowing over a warm surface

FEATURE	SUPER-REFRACTION	SUB-REFRACTION
Range effect	Up to +40%	Up to −20%
Frequency	>30 MHz (VHF+)	>30 MHz (VHF+)
Pressure	High pressure	Low pressure
Humidity profile	Decreases with height	Increases with height
Lapse rate	Less than standard	Greater than standard
Air movement	Warm over cool	Cold over warm
Extreme form	Duct propagation	N/A

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⚡ QUICK REVISION SUMMARY — CHAPTER 2

- ✧ Attenuation:absorption + inverse square law (P ∝ 1/R²; double range → power ×4)
- ✧ Absorptionincreases with frequency; significant above1000 MHz
- ✧ Staticworse at lower frequencies; negligible at VHF+ (unless through ionosphere)
- ✧ Surface wave:20 kHz–50 MHz; vertical polarisation; sea > land range; used 30 kHz–2 MHz
- ✧ Space wave (VHF+):line-of-sight only; Range = 1.23 × (√hTX+ √hRX); power irrelevant below horizon
- ✧ Ionosphere:60–1500 km; D (~75 km), E (~125 km), F (~225 km)
- ✧ D layer:day only; absorbs <2 MHz; absence at night → LF/MF sky wave interference
- ✧ Critical angle→ skip distance → dead space (zone of silence)
- ✧ ↑ Ionization → skip distance ↓; ↑ Frequency → skip distance ↑, surface wave range ↓, dead space ↑↑
- ✧ E-layer max range: 1350 NM; F-layer max range: 2200 NM
- ✧ OWF = 0.85 × MUF; night frequency ≈ ½ day frequency; short range → lower frequency
- ✧ Super-refraction:+40% range; high pressure; duct possible; Mediterranean/Caribbean
- ✧ Sub-refraction:−20% range; low pressure; cold over warm

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📝 Practice Questions & Detailed Answers How to Use This Section Cover the answer panel and attempt each question independently. Read the explanation and distractor analysis thoroughly — this is where understanding is built, not just answers confirmed.

VALUE	DESCRIPTION	SECTION
$P \propto 1/R^2$	Inverse square law	§2.1
×4 power	Power needed to double range	§2.1
1000 MHz	Absorption becomes very significant above this	§2.1
20 kHz – 50 MHz	Surface wave frequency range	§4.1
30 kHz – 2 MHz	Surface wave normally used (LF–lower MF)	§4.1
Range = 1.23×(√hTX + √hRX)	LOS range formula (heights in feet, NM)	§4.2
60 km	Base of ionosphere	§5
1500 km	Notional upper limit of ionosphere	§5
0.22 hPa	Atmospheric pressure at 60 km	§5
~1400 local	Peak ionization time (daily)	§5
~75 km	D-layer average altitude	§6
~125 km	E-layer average altitude	§6
~225 km	F-layer average altitude	§6
>400 km	F2-layer max altitude in summer	§6
2 MHz	Boundary: D absorbs below this / E refracts above	§7.4
50 MHz	Upper limit of sky wave under normal conditions	§7.2
1350 NM	E-layer maximum sky wave range	§7.5
2200 NM	F-layer maximum sky wave range	§7.5
OWF = 0.85 × MUF	Optimum working frequency	§8
Night f ≈ ½ Day f	HF day/night frequency rule of thumb	§8
+40%	Super-refraction: max range increase	§10
−20%	Sub-refraction: max range decrease	§10

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Chapter Expert Quiz

Modulation

CHAPTER 3

1. INTRODUCTION

NOTE

What this section covers: The definition of modulation, its purpose, and a summary of the five forms of modulation used in aviation: Keyed, AM, FM, Phase, and Pulse.

Modulation is the process of adding information to a radio wave, or the formatting of radio waves for other purposes. It is the critical step that makes radio useful for communications and navigation.

Before audio information can be transmitted, sound must first be converted to an electrical signal by a microphone (a device that converts sound waves to electrical current). This electrical signal is the audio frequency (AF) ; the radio wave it is combined with is the carrier wave at radio frequency (RF) .

🔵 FIVE FORMS OF MODULATION USED IN AVIATION

Keyed Modulation — interrupting the carrier (Morse code; NDB ident) Amplitude Modulation (AM) — AF amplitude varies RF amplitude Frequency Modulation (FM) — AF amplitude varies RF frequency Phase Modulation — AF varies phase of carrier (used in GPS, MLS) Pulse Modulation — intermittent carrier bursts (radar, DME)

Modulation is not only used for voice communications. Navigation systems such as VOR, ILS, DME, and GPS use modulation for bearing determination, distance measurement, and data transmission.

2. KEYED MODULATION

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NOTE

What this section covers: The simplest form of modulation — interrupting the carrier wave to create Morse code symbols.

The simplest way to convey information on a carrier wave is to interrupt it — creating short and long bursts of RF energy. This is keyed modulation . By varying the burst lengths to match Morse code dot/dash patterns, information can be transmitted.

🔵 MORSE CODE BASICS

Dot (·) — short burst Dash (–) — long burst (3× dot duration) Morse K = – · – (dash, dot, dash)

This form of transmission is called telegraphy . Until the development of AM, it was the only means of passing information by radio.

💡 AVIATION RELEVANCE — NDB IDENTIFICATION

Keyed modulation is still used today by Non-Directional Beacons (NDBs) for station identification. The NDB transmits its two or three-letter ICAO identifier in Morse code superimposed on the carrier. Pilots tune the ADF receiver, and the ident is heard in the headset, confirming the correct NDB is being received. This is covered in Chapter 7.

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3. AMPLITUDE MODULATION (AM)

NOTE

What this section covers: How the AF amplitude modifies RF amplitude, the heterodyning process, how sidebands are produced, bandwidth calculation, and power distribution in an AM signal.

In Amplitude Modulation , the amplitude of the audio frequency (AF) modifies (varies) the amplitude of the radio frequency (RF) carrier wave.

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AM — What Varies 〰 AF amplitude → RF amplitude FM — What Varies ∽∽ AF amplitude → RF frequency PM — What Varies φ AF amplitude → RF phase

3.1 HETERODYNING & SIDEBANDS

The process of combining RF and AF signals is called heterodyning . The result:

- The RF carrier frequency remains unchanged
- New frequencies are created at the sum (RF + AF) and difference (RF − AF) of the two frequencies
- These new frequencies form an Upper Sideband (USB) and a Lower Sideband (LSB)

✨ 📐 WORKED EXAMPLE — AM SIDEBAND CALCULATION

> RF = 2182 kHz | AF = 3 kHz
>
> USB extends from: 2182.001 kHz → 2185 kHz (RF + AF)
>
> LSB extends from: 2181.999 kHz → 2179 kHz (RF − AF)
>
> Total spread: 2179 kHz → 2185 kHz
>
> Bandwidth = 2185 − 2179 =6 kHz = 2 × AF

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AM BANDWIDTH FORMULA

$$\text{Bandwidth} = 2 \times \text{AF (audio frequency)}$$

$$\text{USB frequency} = \text{RF} + \text{AF}$$

$$\text{LSB frequency} = \text{RF} - \text{AF}$$

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📐 EXAM-STYLE AM BANDWIDTH CALCULATION

> Q1: RF = 4716 kHz, AF = 6 kHz → Bandwidth = 2 × 6 =12 kHz

> USB = 4716 + 6 = 4722 kHz; LSB = 4716 − 6 = 4710 kHz

> Spread: 4710 kHz → 4722 kHz = 12 kHz ✓

3.2 AM POWER DISTRIBUTION

In a standard double-sideband AM transmission:

COMPONENT	POWER	% OF TOTAL	CONTAINS INFO?
RF Carrier	100 W	67%	No
Upper Sideband (USB)	25 W	17%	Yes (full copy)
Lower Sideband (LSB)	25 W	17%	Yes (full copy)
Total	150 W	100%	—

⚠ EXAM-CRITICAL EFFICIENCY FACTS

Only 1/3 of total power carries information (the two sidebands combined) The two sidebands contain identical copies of the audio information The carrier has no information content — it has done its job of shifting AF into RF frequencies Each sideband = 1/4 of carrier power (25W vs 100W carrier)

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4. SINGLE SIDEBAND (SSB)

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NOTE

What this section covers: Why SSB is more efficient, how it works, and its three key advantages in HF/MF aviation communications.

Since both sidebands contain identical information, and the carrier contains no information, there is significant redundancy in standard double-sideband (DSB) AM. SSB solves this by:

- ✦ Removing one sideband (either USB or LSB)
- ✦ Suppressing the carrier
- ✦ Transmitting only the remaining sideband — which contains all the information

✅ THREE ADVANTAGES OF SSB OVER DSB (MUST KNOW ALL THREE)

Double the channels: Each SSB transmission uses only one sideband bandwidth → twice as many channels available in the same frequency band Better signal-to-noise ratio: The ionosphere introduces static interference at MF/HF; narrower bandwidth of SSB significantly reduces interference from this static Less power required: No power wasted on carrier or redundant sideband → lighter, more efficient equipment

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Additionally, when using sky wave propagation (HF communications), different frequencies within the AM bandwidth refract by different amounts, causing distortion. The narrower bandwidth of SSB significantly reduces this distortion.

💡 WHY SSB FOR HF?

HF frequencies (2–30 MHz) are in very high demand globally. SSB doubles the channel count, reduces power consumption, reduces static interference from the ionosphere, and reduces multipath distortion — all in one step. This is why all HF radio telephony in aviation uses SSB (emission designator J3E).

5. FREQUENCY MODULATION (FM)

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NOTE

What this section covers: How FM differs from AM, the relationship between AF amplitude/frequency and RF frequency deviation/rate of change, FM bandwidth figures, and why NBFM is not used in aviation.

In Frequency Modulation , the amplitude of the audio frequency modifies the frequency of the carrier wave (the amplitude of the carrier remains constant).

🔵 TWO FM RELATIONSHIPS — BOTH EXAMINABLE

Frequency deviation of carrier ∝ amplitude of AF (greater audio amplitude → greater frequency swing) Rate of frequency change = frequency of the AF modulating wave (higher audio pitch → faster frequency switching)

FM vs AM Bandwidth Comparison

SYSTEM	BANDWIDTH	NOTES
AM (DSB)	9 kHz	International standard allocation for AM broadcasting
FM Broadcast	150 kHz	By international agreement for music/broadcast
NBFM (voice)	8 kHz	Narrow Band FM — reduced bandwidth for voice only
Aeronautical Comms limit	6 kHz	ICAO standard for aeronautical voice channels
HF Comms limit	3 kHz	SSB occupies only one sideband

⚠ WHY NBFM IS NOT USED IN AVIATION

Even NBFM (Narrow Band FM) with a reduced bandwidth of 8 kHz is still greater than the 6 kHz permitted for aeronautical voice communications. Therefore NBFM communication systems are not used in aviation . Standard aviation VHF/UHF uses AM (A3E for VHF, J3E for HF).

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💡 FM VS AM SUMMARY

FM has better noise immunity than AM (noise appears as amplitude variations, which FM ignores) FM requires much wider bandwidth → impractical below VHF for aviation use FM is used in aviation navigation systems (e.g. VOR uses FM on the variable phase signal) but not for voice comms

6. PHASE MODULATION

NOTE

What this section covers: Analogue phase modulation, digital phase shift keying (PSK), and the specific systems that use phase modulation in aviation.

In Phase Modulation , the phase of the carrier wave is modified by the input signal. There are two cases:

Analogue Phase Modulation

The phase of the carrier is modified in proportion to the amplitude of the analogue input signal — similar to AM but it is the phase angle that changes rather than the amplitude or frequency.

Digital Phase Shift Keying (PSK)

With a digital (binary) input signal, the phase change reflects a binary 0 or 1 :

NOTE

🔵 Binary PSK (simplest case) 0° phase shift → represents binary 0 180° phase shift → represents binary 1 More complex systems use multiple phase shift values to represent more data per symbol (e.g. QPSK = 4 phase states, 8PSK = 8 phase states)

✅ AVIATION SYSTEMS USING PHASE MODULATION

GPS — uses Binary Phase Shift Keying (BPSK) to transmit navigation data and ranging codes MLS (Microwave Landing System) — uses Differential Phase Shift Keying (DPSK)

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7. PULSE MODULATION

NOTE

What this section covers: How pulse modulation works and where it is used in aviation.

Pulse modulation is formed by generating and transmitting a sequence of short-period pulses — the carrier wave is transmitted in intermittent bursts rather than continuously.

🔵 HOW PULSE MODULATION WORKS

Carrier wave is transmitted as a series of brief, high-power pulses Between pulses, the transmitter is silent — it can listen for replies Information can be encoded in the timing, amplitude, width, or position of pulses

✅ AVIATION SYSTEMS USING PULSE MODULATION

Radar systems (primary radar, SSR, AWR) — pulse transmitted, reflected pulse timed to determine range DME (Distance Measuring Equipment) — pulse pairs transmitted/received to measure slant range (emission designator P0N) SSR (Secondary Surveillance Radar) — pulse trains for interrogation and reply

8. EMISSION DESIGNATORS

NOTE

What this section covers: How to decode three-character emission designators, and the specific designators for key aviation equipment.

Emission designators are a standardised three-character code (letter-digit-letter) that describe the characteristics of an electronic transmission. You are not required to memorise the full table, but you must know the key aviation examples.

EMISSION DESIGNATOR STRUCTURE

Symbol 1 (Letter): Type of modulation of main carrier

Symbol 2 (Digit): Nature of signals used for modulation

Symbol 3 (Letter): Type of information transmitted

Symbol 1 — Modulation Type

LETTER	MODULATION
N	Unmodulated carrier (no modulation)
A	Amplitude Modulation — Double Sideband (DSB)
H	AM — Single Sideband, Full Carrier
J	AM — Single Sideband, Suppressed Carrier (SSB)
F	Frequency Modulation
G	Phase Modulation
P	Sequence of unmodulated pulses
K	Sequence of pulses modulated in amplitude

Symbol 2 — Nature of Modulating Signal

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DIGIT	SIGNAL TYPE
0	No modulating signal
1	Single channel — quantized/digital information, no sub-carrier
2	Single channel — quantized/digital information, with sub-carrier
3	Single channel — analogue information
7	Two or more channels — quantized/digital information
8	Two or more channels — analogue information
9	Composite: digital + analogue channels

Symbol 3 — Information Type

LETTER	INFORMATION TRANSMITTED
N	No information
A	Telegraphy for aural reception (Morse)
B	Telegraphy for automatic reception
D	Data transmission, telemetry, telecommand
E	Telephony (including sound broadcasting)
W	Combination of above
X	Cases not otherwise covered

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Key Aviation Emission Designators — Worked Examples

✨ 📐 A3E — VHF RADIO TELEPHONY

> A = Amplitude Modulation, Double Sideband

> 3 = Single channel, analogue information

> E = Telephony

> → AM carrier wave modulated with speech (VHF voice comms)

J3E — HF RADIO TELEPHONY

> J = AM, Single Sideband, Suppressed Carrier (SSB)

> 3 = Single channel, analogue information

> E = Telephony

> → SSB voice comms over HF (carrier + one sideband removed)

EQUIPMENT / SYSTEM	DESIGNATOR	MEANING
ADF / NDB	N0NA1A / N0NA2A	Unmod carrier + keyed Morse ident
VHF RTF	A3E	AM-DSB, analogue, telephony
HF RTF	J3E	SSB suppressed carrier, telephony
VOR	A9W	AM, composite analogue+digital, combination
ILS	A8W	AM, 2+ analogue channels, combination
Marker Beacons	A2A	AM, digital sub-carrier, Morse ident
DME	P0N	Unmod pulse sequence, no info (ranging only)
MLS	N0XG1D	Phase mod, digital data, guidance

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⚠ EXAM NOTE

"It is not necessary to know the details of the [full emission] table." However you must know A3E (VHF RTF) and J3E (HF RTF) and be able to decode them. Other designators (ADF, VOR, ILS, DME) are "unlikely to be examined" but may appear as distractors.

⚡ QUICK REVISION SUMMARY — CHAPTER 3

- ✧ Keyed:interrupts carrier; Morse code; used by NDBs for ident
- ✧ AM:AF amplitude → RF amplitude; bandwidth = 2 × AF; USB = RF+AF, LSB = RF−AF
- ✧ AM power:carrier = 2/3 total; each sideband = 1/6 total; only 1/3 carries info
- ✧ SSB advantages:2× channels, better S/N, less power
- ✧ FM:AF amplitude → RF frequency deviation; AF frequency → rate of change
- ✧ FM bandwidth:Broadcast = 150 kHz; NBFM = 8 kHz > 6 kHz limit → NOT used in aviation voice
- ✧ Phase modulation:analogue (amplitude → phase angle); digital PSK (0° = 0, 180° = 1)
- ✧ GPS = BPSK; MLS = DPSK
- ✧ Pulse modulation:radar systems (DME, SSR, primary radar)
- ✧ A3E= VHF RTF;J3E= HF RTF (SSB suppressed carrier)
- ✧ DME = P0N(pulse, unmod, no info — pure ranging)

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📝 Practice Questions & Detailed Answers How to Use This Section Cover the answer panel and attempt each question independently. Read the full explanation and distractor analysis — knowing why the wrong answers are wrong is as important as knowing the right answer.

✓ Correct Answer: (d) — All of the above Explanation: SSB has three advantages: (1) more frequencies/channels (each SSB uses half the bandwidth of DSB); (2) reduced power requirement (no power wasted on carrier or second sideband); (3) better signal/noise ratio (narrower bandwidth, less ionospheric static). All three options (a), (b), and (c) are correct individually, so (d) is the best answer. See Section 4 . Why the other options are not the best: (a) — True but incomplete. More channels is one advantage, but not the only one. (b) — True but incomplete. Reduced power is one advantage. (c) — True but incomplete. Better S/N ratio is one advantage. Instructor's Note: Any time a question asks "which is an advantage" and lists multiple valid advantages individually plus "all of the above" — the answer is almost certainly "all of the above." Know all three SSB advantages cold: channels × 2, power ↓, S/N ↑.

Chapter Expert Quiz

Antennae

CHAPTER 4

1. INTRODUCTION

NOTE

What this section covers: What antennae do, why the choice of antenna type matters, and what determines which antenna is used in each radio system.

Antennae (or aerials) are the means by which radio energy is radiated and received. Every radio system — from a cockpit VHF radio to a precision ILS array — uses an antenna specifically designed for its function.

🔵 KEY PRINCIPLE

The type of antenna used is determined by: (1) the frequency in use, (2) whether directional or omnidirectional radiation is required, and (3) the platform (aircraft, ground, ship). This chapter covers the principles common to all antennae, plus specialist types for navigation systems.

2. BASIC PRINCIPLES — DIPOLE & MARCONI AERIALS

NOTE

What this section covers: The two fundamental antenna types for basic communications — the half-wave dipole and the quarter-wave Marconi (which is used on aircraft).

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There are two basic aerial types for communications:

Figure 4.1 — Half-wave dipole: power fed to the centre; two λ/4 elements extend in opposite directions. Current (I) is maximum at the centre, voltage (V) maximum at the tips. Radiates perpendicular to the aerial axis (source p.55) Figure 4.2 — Marconi (quarter-wave) aerial: one λ/4 element above a conductive ground plane, which acts as the second half of a dipole via reflection. Used on aircraft for better aerodynamics (source p.55)

FEATURE	HALF-WAVE DIPOLE	QUARTER-WAVE MARCONI
Total aerial length	λ/2	λ/4
Ground plane	Not required	Required (metal surface)
Radiation direction	Perpendicular to aerial	Perpendicular to aerial
Aircraft use	Not ideal (aerodynamics)	Standard on aircraft (blade aerial)
How it works	Centre-fed, both halves radiate	Ground plane acts as second half of dipole

💡 WHY MARCONI AERIALS ON AIRCRAFT?

The Marconi aerial's single-element design has better aerodynamic qualities (blade aerial on aircraft fuselage). The metal aircraft skin acts as the ground plane. Dipoles need two exposed elements, which would create aerodynamic drag and structural problems at aircraft speeds.

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3. AERIAL LENGTH & VELOCITY FACTOR

NOTE

What this section covers: The wavelength formula, theoretical aerial lengths, the velocity factor correction, and the key calculation type for exam questions.

For maximum efficiency, an aerial must be the correct length for the wavelength of the frequency in use — ideally λ/2 (dipole) or λ/4 (Marconi).

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WAVELENGTH FORMULA (FROM CHAPTER 1)

$$\lambda \text{ (metres)} = 300 / f \text{ (MHz)}$$

$$\text{Dipole length} = \lambda / 2$$

$$\text{Marconi length} = \lambda / 4$$

Velocity Factor — Why the Actual Aerial Is Shorter

EM energy travels at the speed of light in free space . In a physical conductor (the aerial element), the speed is slightly slower — the denser medium reduces propagation velocity by approximately 5% . This means the electrical wavelength inside the conductor is shorter than the free-space wavelength.

Therefore, the actual optimum aerial length is approximately 95% of the theoretical value:

ACTUAL AERIAL LENGTH WITH VELOCITY FACTOR

$$\text{Actual length} = 0.95 \times \text{Theoretical length}$$

$$\text{Actual Marconi} = 0.95 \times (\lambda/4)$$

$$\text{Actual Dipole} = 0.95 \times (\lambda/2)$$

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✨ 📐 WORKED EXAMPLES — AERIAL LENGTH CALCULATIONS

> λ = 300/125 = 2.4 m

> Theoretical λ/4 = 2.4/4 = 0.6 m = 60 cm

> Actual = 0.95 × 60 cm = 57 cm

> λ = 300/406 = 0.739 m

> Theoretical λ/4 = 0.739/4 = 0.1847 m = 18.5 cm

> Actual = 0.95 × 18.5 =17.6 cm ≈ 17.5 cm✓ (answer c)

> λ = 300/75 = 4.0 m

> Theoretical λ/2 = 4.0/2 = 2.0 m

> Actual = 0.95 × 2.0 =1.9 m✓ (answer a)

4. AERIAL FEEDERS

NOTE

What this section covers: How energy travels between the aerial and the transmitter/receiver, and why different feeder types are needed at different frequency bands.

The means of carrying RF energy between the aerial and equipment depends on frequency and power level . As frequency increases, losses in simple feeders become unacceptable.

FREQUENCY BAND	FEEDER TYPE	WHY
LF & MF	Simple wire	Low losses at low frequency over reasonable distances
HF & VHF	Twin wire feeder	Single wire losses too high; twin wire more efficient
UHF	Coaxial cable	Twin wire losses unacceptably high; coax required
Upper UHF, SHF, EHF	Waveguide	Coax losses impractical; dipole/Marconi aerials not usable at these frequencies

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🔵 WAVEGUIDE — KEY FACTS

A hollow, rectangular metal tube — no inner conductor Internal dimensions determined by frequency: internal dimension = λ/2 (half the wavelength) Used in radar systems (UHF/SHF) to carry high-power pulses from transmitter to the parabolic dish or phased array antenna At SHF and EHF, dipole and Marconi aerials are impractical — the waveguide itself acts as the radiating element (horn feed)

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5. POLAR DIAGRAMS

NOTE

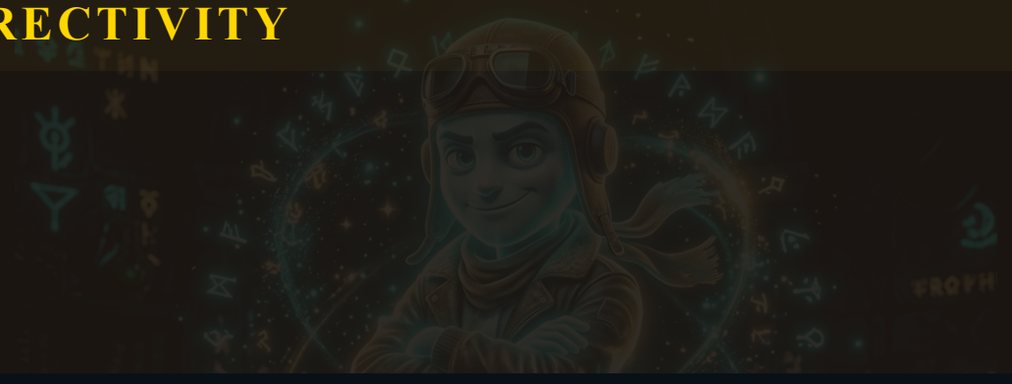
What this section covers: How to read and interpret polar diagrams, the radiation pattern of a dipole aerial, and the torus 3-D representation.

A polar diagram shows the radiation (or reception) pattern of an aerial — it is a line joining all points of equal signal strength, generally shown as a plan view perpendicular to the plane of radiation. (The same polar diagram applies to both transmission and reception — aerials are reciprocal.)

🔵 DIPOLE RADIATION PATTERN

Maximum radiation: perpendicular (90°) to the aerial Minimum radiation (nulls): at the tips of the aerial (along the aerial axis) 3-D pattern: a torus (doughnut shape) — symmetrical around the aerial axis Horizontal plan view of a vertical dipole: a perfect circle (omnidirectional)

6. DIRECTIVITY



NOTE

What this section covers: How directivity is achieved using parasitic elements (reflectors and directors), the side lobe problem, ILS localizer array design, and the ADF loop aerial.

6.1 REFLECTORS & DIRECTORS (YAGI ARRAY)

Many navigation systems require directional emission/reception. The simplest method is adding parasitic elements to the basic aerial:

🔵 REFLECTOR — HOW IT WORKS

A metal rod 5% longer than the aerial Placed at a distance of λ/4 behind the aerial, in the same plane Re-radiates energy 180° out of phase → cancels signal behind, reinforces signal in front Result: no radiation behind the reflector; increased forward gain

🔵 DIRECTORS — HOW THEY WORK

Metal rods shorter than the aerial Placed in front of the aerial (toward the desired direction) Act as focusing elements — concentrate the signal into (or out of) the aerial Multiple directors produce progressively narrower, stronger beams

⚠ THE SIDE LOBE PROBLEM

While a narrow main beam is produced, unwanted side lobes also appear. These side lobes: Receive and transmit signals in unintended directions Cause TV ghosting (reflections from buildings) Create significant problems in SSR and ILS Produce spurious returns in primary radar systems Side lobe suppression is a major engineering challenge in aviation antenna design.

6.2 ILS LOCALIZER ARRAY

The ILS uses an extension of the Yagi array principle to produce the precision beams required to guide aircraft to the runway. The ILS localizer antenna is:

- An array of 16 or 24 aerials placed in a line
- Spacing of λ/2 between each aerial
- Produces two narrow symmetrical lobes either side of the runway centreline

6.3 ADF LOOP AERIAL

The ADF (Automatic Direction Finder) uses a loop aerial to determine the direction of an incoming radio signal.

🔵 HOW THE LOOP AERIAL WORKS — PHASE DIFFERENCE PRINCIPLE

Loop aligned with signal : phase difference between signals in the two vertical elements → net current flows → signal received (maximum) Loop at 90° to signal : induced currents in both vertical elements are equal and opposite → cancel out → zero output (null) Result: a figure-of-eight polar diagram with two nulls The two nulls are used to determine the bearing to the NDB transmitter

💡 WHY TWO NULLS ARE BETTER THAN TWO MAXIMA

The null is used for direction finding (rather than the maximum) because the null is much sharper — a small angular change produces a large signal change. This gives better bearing accuracy. The loop aerial has two nulls 180° apart, which causes a 180° ambiguity — resolved by combining with a sense aerial (discussed in Chapter 7 — ADF).

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7. RADAR AERIALS — PARABOLIC DISH

NOTE

What this section covers: How the parabolic reflector works, the role of the waveguide horn feed, the focal point principle, and the problem of side lobes from the dish.

Radar systems operate in UHF and SHF bands — requiring waveguides rather than cables. The parabolic dish is the classic radar antenna:

Figure 4.9 — Horn feed: the open end of a waveguide (horn) is positioned at the focal point of the parabolic dish. RF energy radiates from the horn and is reflected by the dish (source p.60) Figure 4.10 — Focal point principle: paths FXB and FYA are equal — all reflected rays travel the same distance → arrive in phase → form a uniform plane wavefront (source p.60)

🔵 HOW THE PARABOLIC DISH WORKS

Open end of waveguide (horn) placed at the focal point (F) of the parabola RF energy from horn is reflected by dish as parallel rays Path length from focal point to dish to output is equal for all rays → all reflected rays are in phase This creates a uniform plane wavefront — narrow pencil beam In practice the beam diverges beyond the near field — parallel only close to the antenna

⚠ SIDE LOBE PROBLEM — PARABOLIC DISH

Due to uneven reflection and energy spillage around the edges of the dish, side lobes are produced. These side lobes: Contain sufficient energy to produce valid radar returns outside the main beam Can cause false targets on radar displays Are addressed in modern radar by switching to phased array technology

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8. MODERN RADAR ANTENNAE — PHASED ARRAY

NOTE

What this section covers: The flat plate / phased array / slotted antenna design, how it works, and its five advantages over the parabolic dish.

Modern radar development has replaced the parabolic dish with the Flat Plate Array (also called Phased Array or Slotted Antenna).

✅ FIVE ADVANTAGES OF PHASED ARRAY OVER PARABOLIC REFLECTOR

Narrower beam — better target discrimination / azimuth resolution Reduced side lobes — fewer false returns, less interference Less power required for a given range — more efficient RF transmission Narrower pulse — better range resolution Improved resolution — better target separation in range and azimuth

💡 EXAM TRAP — Q3 IN THIS CHAPTER

Q3 asks which is NOT an advantage of slotted antenna over parabolic. The answer is "Directivity" (d) — both the parabolic dish and the phased array are highly directional. Directivity is a shared characteristic, not an advantage of the phased array specifically. The advantages are: narrower beam, fewer side lobes, less power, narrower pulse, improved resolution.

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⚡ QUICK REVISION SUMMARY — CHAPTER 4

- ✧ Dipole:centre-fed, λ/2 total; radiates perpendicular to aerial axis
- ✧ Marconi:λ/4 above ground plane; ground plane acts as second half; used on aircraft
- ✧ Wavelength:λ = 300/f(MHz); dipole = λ/2; Marconi = λ/4
- ✧ Velocity factor:actual length =0.95 ×theoretical length (5% shorter)
- ✧ Feeders:LF/MF = wire; HF/VHF = twin wire; UHF = coax; SHF+ = waveguide
- ✧ Waveguide:hollow rectangular tube; internal dimension = λ/2
- ✧ Polar diagram:dipole = torus (3D) / figure-eight (vertical section)
- ✧ Reflector:5% longer than aerial, λ/4 behind, 180° out of phase → concentrates beam forward
- ✧ Directors:shorter than aerial, in front → narrow, focus beam; side lobes are a problem
- ✧ ILS localizer:16 or 24 aerial array, λ/2 spacing → two narrow symmetrical lobes
- ✧ ADF loop:figure-of-eight polar diagram; two nulls used for DF; null sharper than maximum
- ✧ Parabolic dish:waveguide horn at focal point → parallel in-phase reflected rays → narrow beam + side lobes
- ✧ Phased array advantages (5):narrower beam, fewer side lobes, less power, narrower pulse, better resolution

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📝 Practice Questions & Detailed Answers

AERIAL TYPE	THEORETICAL LENGTH	ACTUAL (×0.95)	FORMULA
Half-wave Dipole	λ/2	0.95 × λ/2	0.95 × 150/f(MHz) metres
Marconi (Quarter-wave)	λ/4	0.95 × λ/4	0.95 × 75/f(MHz) metres

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Doppler Radar Systems

CHAPTER 5

1. INTRODUCTION

NOTE

What this section covers: The purpose of a Doppler navigation system, how it fits into modern avionics, and where the Doppler principle appears across different aviation systems.

A Doppler navigation system uses the Doppler principle to measure an aircraft's ground speed and drift . Combined with a heading input, these values allow a navigation computer to determine the aircraft's position — continuously updated without external signals.

🔵 WHERE DOPPLER IS USED IN AVIATION

Doppler Navigation Systems — self-contained ground speed and drift measurement Doppler VOR (DVOR) — uses Doppler shift to produce the bearing signal VDF (VHF Direction Finder) — Doppler technique for bearing determination Police radar — classic real-world example of Doppler speed measurement Weather radar — Doppler shift detects precipitation movement and wind shear

Modern systems combine Doppler measurements with IRS (Inertial Reference System) , VOR/DME , or GPS in various configurations. This integration overcomes the key limitations of early Doppler systems: inaccurate heading references and signal degradation over large expanses of water.

2. THE DOPPLER PRINCIPLE

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NOTE

What this section covers: The physics of the Doppler effect explained by analogy, the exact relationship between relative motion and frequency shift, and the definition of Doppler frequency.

The Doppler effect was predicted by the Austrian physicist Christian Doppler in the 19th century for light waves — it applies equally to sound waves and radio waves.

"A received frequency will only be the same as the transmitted frequency when there is no relative movement between the transmitter and receiver."

The Beach Analogy

Imagine standing still in the sea: waves strike you at a steady 4 per minute.

- ✦ Walk into the waves → you reduce the distance between waves → they strike you more frequently (perceived frequency increases)
- ✦ Walk away from the waves → you increase the spacing → they strike you less frequently (perceived frequency decreases)
- ✦ The actual wave rate (4/min) has not changed — only your perceived rate has changed

Moving TOWARD Source ➡️ 📡 Received frequency INCREASES Wavefronts are compressed Shorter apparent wavelength Higher perceived frequency f rx > f tx Moving AWAY From Source 📡 ➡️ Received frequency DECREASES Wavefronts are stretched Longer apparent wavelength Lower perceived frequency f rx < f tx

DOPPLER SHIFT (DOPPLER FREQUENCY)

$$f_{Doppler} = f_{received} - f_{transmitted}$$

$f_{Doppler}$ is proportional to relative velocity between transmitter and receiver

Approaching → $f_{Doppler}$ positive (received > transmitted)

Receding → $f_{Doppler}$ negative (received < transmitted)

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💡 KEY PRINCIPLE — PROPORTIONALITY

The Doppler shift is directly proportional to the relative velocity. Double the speed → double the Doppler shift. This proportionality is what makes Doppler useful for measuring speed — a larger frequency shift means a higher relative velocity.

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3. AIRBORNE DOPPLER — SYSTEM DESCRIPTION

NOTE

What this section covers: How airborne Doppler radar is installed on an aircraft, the antenna type, the beam configuration, and the origin of the "Janus" name.

A typical airborne Doppler installation uses a slotted waveguide antenna in which:

- The transmitter and receiver elements are screened from each other but share the same aerial
- An array of beams is transmitted downward toward the Earth's surface

🔵 THE 4-BEAM JANUS ARRAY

4 beams total: 2 pointing forward (port & starboard), 2 pointing aft (port & starboard) All beams directed downward at an angle to the Earth's surface Named after Janus — the Roman God of Doorways, who could face both directions simultaneously The forward/aft symmetry is what allows cancellation of aircraft pitch and roll errors

4. JANUS ARRAY CONFIGURATIONS

NOTE

What this section covers: The various beam configurations used in Janus arrays — 3-beam and 4-beam layouts.

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🔵 3-BEAM VS 4-BEAM JANUS ARRAY

3-beam: typically one fore beam + two aft beams (or vice versa); simpler, lighter; less redundancy 4-beam: two fore + two aft beams; better averaging; improved accuracy; standard configuration for most systems All Janus arrays have both forward-facing and aft-facing beams — the defining characteristic

5. DOPPLER OPERATION — HOW GROUND SPEED & DRIFT ARE MEASURED

NOTE

What this section covers: How the frequency shifts in the four beams are processed to give ground speed and drift, and how the system worked mechanically vs electronically.

The Doppler system continuously measures the frequency shift in the reflected signal caused by the aircraft's motion over the ground, then converts these values to:

- ✦ Speed along track → ground speed
- ✦ Speed across track → used to determine drift

Measuring Ground Speed

✅ FORWARD VS AFT BEAM FREQUENCY SHIFTS

Forward beams: aircraft moving toward the reflected ground → received frequency is HIGHER than transmitted (positive Doppler shift) Aft beams: aircraft moving away from the reflected ground → received frequency is LOWER than transmitted (negative Doppler shift) With zero drift: forward and aft shifts are equal in magnitude, opposite in sign The magnitude of the shift is proportional to ground speed

Measuring Drift

✅ PORT VS STARBOARD BEAM FREQUENCY DIFFERENCES

If aircraft drifts left (port) : port beams receive a different shift than starboard beams If aircraft drifts right (starboard) : starboard beams receive a different shift The difference in frequency shifts between port and starboard beams gives drift magnitude and direction

Modern vs Earlier Systems

FEATURE	MODERN (FIXED AERIAL)	EARLIER (MECHANICAL)
Aerial type	Fixed (no moving parts)	Pitch-stabilized, rotating aerial
Processing	Electronic processing of frequency differences	Motor-driven alignment until port = starboard shifts
Drift measurement	Computed electronically from beam differences	Pick-off measures heading vs aerial track alignment
Reliability	Higher (no mechanical wear)	Lower (mechanical complexity)

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6. DOPPLER NAVIGATION SYSTEMS

NOTE

What this section covers: How Doppler measurements are converted to aircraft position in early and modern navigation computers.

The Doppler system continuously updates drift and ground speed values. A navigation computer uses three inputs to compute position:

DOPPLER NAVIGATION POSITION FIX

Position = f(Ground Speed, Drift, Heading)

Output: Latitude / Longitude

or: Along-track / Across-track distance from departure point

🔵 NAVIGATION COMPUTER INPUTS & OUTPUTS

Inputs: Ground speed (from Doppler fore/aft beams) + Drift (from port/stbd beams) + Heading (from compass/IRS) Starting position: Aircraft departure coordinates manually loaded Outputs: Current position as latitude/longitude or as along/across track distances from departure Modern integration: Doppler + IRS + VOR/DME + GPS for improved accuracy and redundancy

7. DOPPLER LIMITATIONS & MODERN INTEGRATION

NOTE

What this section covers: The two main limitations of early Doppler systems and how modern multi-sensor integration overcomes them.

⚠️ LIMITATIONS OF EARLY DOPPLER NAVIGATION SYSTEMS

Inaccurate heading reference: Doppler gives ground speed and drift but needs an external heading source. Early magnetic compasses introduced errors. Modern systems use IRS. Degradation/loss over water: The smooth sea surface reflects Doppler signals differently from land — specular (mirror-like) reflection over calm water returns very little energy, causing signal degradation or complete loss. Wind-roughened sea improves this.

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✅ MODERN SOLUTIONS

Heading: IRS (Inertial Reference System) provides accurate heading with no magnetic error Over water: Multi-sensor integration — GPS/VOR/DME provides position updates when Doppler signal is lost or unreliable Integration modes: Doppler + IRS; Doppler + GPS; Doppler + VOR/DME; fully integrated FMS solutions

⚡ QUICK REVISION SUMMARY — CHAPTER 5

- ✧ Doppler shift:movingtoward⟶ frequency ↑; movingaway⟶ frequency ↓; shift ∝ relative velocity
- ✧ Doppler frequency= freceived− ftransmitted
- ✧ Airborne Doppler:slotted waveguide antenna; transmitter & receiver screened but share aerial
- ✧ 4-beam Janus array:2 fwd + 2 aft beams; named after Roman god Janus (faces both ways)
- ✧ Ground speed:magnitude of fore/aft Doppler shift
- ✧ Drift:difference between port and starboard beam frequency shifts
- ✧ Zero drift:fore/aft shifts are equal and opposite
- ✧ Inputs to nav computer:ground speed + drift + heading ⟶ position
- ✧ Limitations:heading accuracy; signal loss over calm water (specular reflection)
- ✧ Modern systems:integrated with IRS, VOR/DME, GPS
- ✧ Also uses Doppler:police radar, Doppler VOR, VDF, weather radar

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📝 Practice Questions & Detailed Answers

RELATIVE MOTION	FREQUENCY EFFECT	WAVEFRONT
Moving toward each other	f received > f transmitted(increase)	Compressed (shorter λ)
Moving away from each other	f received < f transmitted(decrease)	Stretched (longer λ)
No relative motion	f received = f transmitted (no shift)	Unchanged

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VHF Direction Finder (VDF)

CHAPTER 6

1. INTRODUCTION

NOTE

What this section covers: The purpose of VDF, its limitations compared to modern systems, frequency band, and where information on available VDF stations is found.

VHF Direction Finding (VDF) provides a pilot with a bearing — the direction toward a ground station. This can be used to:

- Home toward the ground station
- Position fix — in conjunction with two or more bearings from different stations
- Track check / position line — verify or update navigation
- Let-down procedures — non-precision approach guidance

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PRACTICAL NOTE

VDF is rarely used because more sophisticated and accurate systems (VOR, GPS, ILS) are available. However, it requires no airborne equipment other than a standard VHF radio — making it available as a backup whenever VHF comms are possible.

KEY VDF FACTS

Bearings provided by voice on the aircraft's VHF communications frequency Frequency range: 118.0 – 137 MHz (Emission code A3E) Auto-triangulation : available only on VHF International Distress frequency — 121.5 MHz UHF DF: limited to military use Available stations listed in the AD Section of the AIP Bearings only given when conditions are satisfactory and within calibrated limits

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2. PROCEDURES — Q-CODES

NOTE

What this section covers: The four Q-codes used in VDF, what each provides, and how to request a VDF bearing.

A pilot requests a VDF bearing using Q-code phrases. The station responds with: the Q-code, the bearing in degrees, the accuracy class , and the time of observation (if required).

QDM Magnetic Heading TO steer to reach station (assuming zero wind) Used for: station homing, let-downs QDR Magnetic Bearing FROM the station (the aircraft's radial from the station) Used for: en route navigation QTE True Bearing FROM the station (the aircraft's true radial from station) Used for: en route navigation QUJ True Track TO the station (true bearing from aircraft toward station) Not generally used

RECIPROCAL RELATIONSHIPS — MUST KNOW

QDM is the reciprocal of QDR (TO vs FROM; both Magnetic) QUJ is the reciprocal of QTE (TO vs FROM; both True) QDR, QDM, and QTE are the most commonly used codes

CONVERTING BETWEEN Q-CODES

QDR to QDM: Add or subtract 180° (take the reciprocal) QTE to QDR: Apply variation — Magnetic = True minus East variation QTE to QUJ: Add or subtract 180° Memory: "DM = Mag To; DR = Mag From; TE = True From; UJ = True To"

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3. BEARING ACCURACY CLASSES

NOTE

What this section covers: The four accuracy classes (A–D), what accuracy to assume when no class is given, and the accuracy of modern Doppler-based VDF.

CLASS	ACCURACY	NOTES
Class A	± 2°	Best; rarely achievable in practice
Class B	± 5°	Normal best case— bearings normally no better than Class B
Class C	± 10°	Poor conditions
Class D	> ± 10°	Accuracy less than Class C; unreliable
Modern Doppler VDF	± 0.5°	Digital readout; available on VHF and UHF systems

Class A A ±2° Class B B ±5° Class C C ±10° Class D >10° >±10°

EXAM-CRITICAL: DEFAULT ACCURACY ASSUMPTION

"Normally, bearings no better than Class B will be available." If a VDF bearing is given without an accuracy classification , assume Class B = ±5° .

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4. PRINCIPLE OF OPERATION

NOTE

What this section covers: The ground equipment required, the antenna array design, how the bearing is determined and displayed.

The only equipment required in the aircraft is a standard VHF radio. On the ground, the VDF station uses a direction-finding aerial (circular array of vertical elements) and a display.

HOW VDF WORKS

Aircraft transmits on VHF comm frequency — vertically polarized signal Ground antenna: circular array of vertical elements — matched polarization Equipment resolves bearing from transmissions received at each element in the array Bearing displayed relative to True or Magnetic North at the station Latest Doppler-based systems produce a digital bearing at ±0.5°

5. RANGE OF VDF

NOTE

What this section covers: Why VDF range follows the LOS formula, factors that extend or reduce range, and how to calculate minimum aircraft altitude or maximum range.

VDF operates in the VHF band — therefore range obeys the line-of-sight formula :

VDF RANGE (LINE OF SIGHT)

$$\text{Range (NM)} = 1.25 \times (\sqrt{hTX} + \sqrt{hRX})$$

hTX= transmitter height in feet | hRX= receiver height in feet

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WORKED EXAMPLES — VDF RANGE CALCULATIONS

300 = 1.25 × (√2500 + √h)

300 / 1.25 = 240 = 50 + √h

√h = 240 − 50 = 190

h = 190² =36,100 ft✓ (answer c)

Range = 1.25 × (√9000 + √400) = 1.25 × (94.87 + 20)

= 1.25 × 114.87 =143.6 NM ≈ 143 NM✓ (answer c)

Range = 1.25 × (√19000 + √1400) = 1.25 × (137.84 + 37.42)

= 1.25 × 175.26 =219.1 NM ≈ 219 NM✓ (answer d)

FACTORS AFFECTING VDF RANGE

Altitude of aircraft and station (primary factor) — higher = greater range Intervening high ground — limits range, especially for low-flying aircraft in hilly terrain Transmitter power — both airborne and ground Atmospheric ducting (super-refraction) — can give greater than LOS range

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6. FACTORS AFFECTING ACCURACY

NOTE

What this section covers: The five factors that reduce VDF bearing accuracy.

FACTOR	EFFECT ON ACCURACY
Propagation error / Site error	Aircraft transmissions reflected from terrain or buildings at the site, arriving at the antenna from false directions — producing bearing errors
Aircraft attitude	VHF comms are vertically polarized; best results when aircraft fliesstraight and level. Bank angle tilts the antenna axis, reducing bearing accuracy
Directly overhead effect	Poor accuracy when aircraft is directly overhead the VDF station — especially with modern Doppler systems. Direct and ground-reflected waves arrive simultaneously causing fading/loss
Multi-path signals	Direct + reflected waves reaching the antenna simultaneously → fading and bearing errors. Usually short-lived as aircraft transits the area
Simultaneous transmissions	Two or more aircraft transmitting at the same time → momentary bearing errors

7. DETERMINATION OF POSITION (AUTO-TRIANGULATION)

NOTE

What this section covers: How position fixes are obtained from multiple VDF bearings and the auto-triangulation service.

If multiple ground stations are available and linked to an ATCC, the aircraft's position can be fixed using auto-triangulation — bearings from different stations are automatically crossed to give a position fix, transmitted to the pilot.

AUTO-TRIANGULATION LIMITATION

Auto-triangulation may be available to Distress and Diversion (D&D) Cells — but cannot be guaranteed . It is available only on the VHF International Distress Frequency: 121.5 MHz .

8. VDF SUMMARY

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QUICK REVISION SUMMARY — CHAPTER 6

- ✧ QDM= Mag heading TO station (for homing/let-down)
- ✧ QDR= Mag bearing FROM station (reciprocal of QDM)
- ✧ QTE= True bearing FROM station
- ✧ QUJ= True track TO station (reciprocal of QTE; rarely used)
- ✧ Class A=±2°, B=±5°, C=±10°, D=>10°
- ✧ No class stated → assume Class B = ±5°
- ✧ Modern Doppler VDF = ±0.5°
- ✧ VHF band: 118.0–137 MHz; A3E emission; vertically polarized
- ✧ LOS formula:Range = 1.25 × (√hTX + √hRX)
- ✧ Auto-triangulation: 121.5 MHz only; not guaranteed
- ✧ No airborne equipment needed other than a standard VHF radio
- ✧ Accuracy degraded by: propagation error, site error, aircraft attitude, overhead effect, multi-path, simultaneous transmissions

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Practice Questions & Detailed Answers

CODE	MEANING	TRUE/MAG	TO/FROM	PRIMARY USE
QDM	Magnetic Heading TO steer to station	Magnetic	TO	Homing, let-downs
QDR	Magnetic Bearing FROM station (radial)	Magnetic	FROM	En route navigation
QTE	True Bearing FROM station	True	FROM	En route navigation
QUJ	True Track TO station	True	TO	Rarely used

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Automatic Direction Finder (ADF)

CHAPTER 7

1. INTRODUCTION

The Automatic Direction Finder (ADF) is airborne equipment used in conjunction with a ground-based Non-Directional Beacon (NDB) to provide navigation assistance and non-precision approach procedures at aerodromes.

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HISTORICAL STATUS

ADF/NDB was due to be phased out in 2005 but continues in use . Many UK aerodromes still publish NDB instrument approach procedures, and at some aerodromes it remains the only instrument approach procedure available.

KEY ADVANTAGE

ADF requires no special ground equipment beyond the NDB and is available on any frequency in the LF/MF band. It provides low-cost, widely available backup navigation.

2. NON-DIRECTIONAL BEACON (NDB)

The NDB is a ground-based transmitter that radiates vertically polarized radio signals equally in all directions — hence "non-directional."

- Operates in the LF and MF bands (190 – 1750 kHz)
- When an ADF is tuned to an NDB's frequency and the callsign is identified, the NDB's direction is indicated
- A cone of silence exists directly overhead the NDB — no signal received while directly overhead; cone diameter increases with aircraft height

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CONE OF SILENCE

When the aircraft is directly overhead the NDB, no signal is received and the ADF needle will swing erratically. The diameter of the cone of silence increases with altitude .

3. PRINCIPLE OF OPERATION

ADF measures the bearing of an NDB relative to the fore/aft axis of the aircraft .

3.1 LOOP AERIAL

A voltage is generated in the vertical elements of a loop aerial due to the phase difference of the incoming wave across the loop. As the loop rotates, the induced voltage varies, reaching zero when the loop is perpendicular to the radio wave — this is the "null."

The loop produces a figure-of-eight polar diagram with two null positions. This creates a 180° ambiguity — two possible directions for the beacon.

3.2 SENSE AERIAL — RESOLVING AMBIGUITY

To resolve the ambiguity, a simple dipole (sense) aerial is added. Its polar diagram is circular. The signals from loop and sense aerials are combined electronically (as if the sense aerial is at the centre of the loop). The result is a CARDIOID — a heart-shaped pattern with a single null.

3.3 CARDIOID — SINGLE NULL

DUAL CARDIOID SWITCHING

A single cardioid null is not precise enough for ICAO ±5° accuracy. The polarity of the sense aerial is reversed to produce a right-hand cardioid. By rapidly switching (~120 Hz) between the two cardioids, the null is more precisely defined — meeting the ICAO accuracy requirement.

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3.4 FIXED LOOP / GONIOMETER

In practice, a rotating loop outside the aircraft is not feasible. The ADF uses a fixed four-element loop (two fore/aft, two lateral) feeding a goniometer — a device that electronically reproduces the electromagnetic field and extracts the bearing using a search coil.

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4. FREQUENCIES AND TYPES OF NDB

NDB FREQUENCY ALLOCATION

Allocated band: 190 – 1750 kHz (LF and MF bands) Most NDBs are found between 250 – 450 kHz (surface wave propagation most reliable here). Propagation mode: Surface wave

TYPE	RANGE	USE	AVAILABILITY
Locator (L)	10 – 25 NM	Airfield/runway approach procedures; co-located with ILS outer or middle marker	May only be available during published aerodrome hours
En Route NDB	50 NM or more(oceanic: hundreds of miles)	Homing, holding, en route and airways navigation	Continuous; listed in COMM section of AIP

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NDB RANGE FORMULA

$$\text{Range} = 3 \times \sqrt{P(W)} \quad \text{— over water}$$

$$\text{Range} = 2 \times \sqrt{P(W)} \quad \text{— over land}$$

P = NDB transmitter power in Watts | Range in NM

5. AIRCRAFT EQUIPMENT

The ADF system in the aircraft comprises:

- A loop aerial (fixed four-element)
- A sense aerial
- A control unit (frequency selector, ADF/ANT/BFO switches)
- A receiver
- A display (RBI or RMI)

6. EMISSION CHARACTERISTICS AND BFO

NDBs have a 2 or 3 letter identification callsign . There are two emission types:

N0NA1A Unmodulated carrier (N0N) + interrupted unmodulated carrier (A1A) A1A requires BFO to be ON to produce an audible tone for identification. BFO ON for: Tuning, Identification, AND Monitoring N0NA2A Unmodulated carrier (N0N) + amplitude modulated carrier (A2A) A2A can be heard on a normal receiver without BFO. BFO ON for: Tuning only BFO OFF for: Identification and Monitoring

> ⚠ **BFO RULES — MUST KNOW FOR EXAM**
>
> Emission Tune Identify Monitor N0NA1A ON ON ON N0NA2A ON OFF OFF Note: BFO may be labelled TONE or TONE/VOICE on some equipment.

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7. PRESENTATION OF INFORMATION

ADF information is presented on either a Relative Bearing Indicator (RBI) or a Radio Magnetic Indicator (RMI) . Both display relative bearing .

INSTRUMENT	COMPASS CARD	WHAT NEEDLE SHOWS	TO GET MAG BEARING
RBI	Fixed — 000° always at nose	Relative bearing (angle from nose)	Heading + RBI reading (−360° if >360°)
RMI	Rotates with aircraft heading	Magnetic bearing directly	Read directly from under needle head

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QDM / QDR FROM ADF

Needle HEAD → points TO the NDB = QDM (if RMI, magnetic heading to station) Needle TAIL → points FROM the NDB = QDR (magnetic bearing from station = radial)

8. USES OF THE NON-DIRECTIONAL BEACON

- En route navigational bearings
- Homing to or flying from the NDB when maintaining airway centrelines
- Holding overhead at an assigned level in a racetrack pattern
- Runway instrument approach procedures (non-precision approach)

PLOTTING ADF BEARINGS

When converting an ADF bearing to a position line on a chart, variation must be applied at the aircraft's position (not the NDB). For long-range bearings, convergency between the aircraft and beacon meridians must also be applied.

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9. TRACK MAINTENANCE USING THE RBI

9.1 HOMING

Homing = maintaining 000° relative bearing (NDB dead ahead at all times). In zero wind, the aircraft flies a straight track inbound. In crosswind, the aircraft follows a curved track (the heading constantly changes to keep the needle at 000°).

Figure 7.10 — Homing in zero wind: aircraft heading 077°, RBI 000°. Straight track to NDB. Figure 7.11 — Homing with crosswind (no drift allowance): aircraft maintains 000° RBI but follows a curved track — arriving at NDB from downwind side.

9.2 TRACKING INBOUND

To fly a straight track inbound, the pilot applies a Wind Correction Angle (WCA) :

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Inbound Track Rule Required Relative Bearing = 360° − WCA (for starboard drift) or WCA (for port drift) Starboard drift anticipated → Subtract WCA from track → heading to fly = track − WCA; RBI reads 360° − (−WCA) = 360° + WCA? Simpler rule: S tarboard drift → S ubtract WCA from track heading Port drift anticipated → Add WCA to track heading (Plus for Port)

9.3 TRACKING OUTBOUND

TRACK RULE MEMORY AID

S tarboard drift = S ubtract WCA from track heading P ort drift = P lus (+) WCA to track heading (Works for both inbound and outbound)

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10. DRIFT ASSESSMENT

10.1 INBOUND

- Fly the aircraft on the required track with the beacon dead ahead (000° RBI)
- Maintain heading and watch the RBI: RBI increases → port drift (beacon drifting to port) RBI decreases → starboard drift
- Turn say 30° starboard to regain track; RBI will show 330° when back on track
- Apply estimated drift as WCA; monitor RBI to confirm the assessment is correct

10.2 OUTBOUND

- Track directly overhead the beacon first (establishes reference)
- With starboard drift : RBI increases from 180° → beacon appears to move to port of astern
- With port drift : RBI decreases from 180° → beacon appears to move to starboard of astern
- Turn 30° to regain track, then apply assessed WCA

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11. HOLDING

When traffic density or weather delays landing, ATC directs aircraft to a holding area (stack) organised over a radio beacon. Aircraft fly a racetrack circuit, separated by a minimum of 1000 ft vertically.

12. RUNWAY INSTRUMENT APPROACH PROCEDURES

Most aerodromes have published NDB instrument approach procedures . The pilot flies the published procedure to position the aircraft for a visual landing in poor weather. The NDB may be used alone or in conjunction with other approach aids.

13. FACTORS AFFECTING ADF ACCURACY

13.1 DESIGNATED OPERATIONAL COVERAGE (DOC)

The DOC is based on a daytime protection ratio of 3:1 (signal/noise) between wanted and unwanted signals. Beyond the DOC, bearing errors increase. At night, adverse propagation further increases errors even within the DOC.

13.2 STATIC INTERFERENCE

TYPE	CAUSE	INDICATION
Precipitation static	Collision of water droplets/ice crystals with the aircraft — degrades signal/noise ratio	Wandering needle; background hiss on audio (present on VHF frequencies too)
Thunderstorm static	Powerful static discharges from Cb clouds across the LF/MF spectrum	Loud crackle on audio; needle points rapidly toward the Cb. With multiple active cells, needle may point toward them for prolonged periods.

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THUNDERSTORM PRACTICAL NOTE

"During Cb activity, the only sensible use of the ADF is to indicate where the active cells are."

13.3 NIGHT EFFECT

By day, the D-region of the ionosphere absorbs LF/MF signals , preventing sky wave reception. At night, the D-region disappears, allowing sky wave to contaminate the surface wave .

- Phase interference between sky wave and surface wave (different path lengths)
- Sky wave's horizontal polarization component induces currents in horizontal loop elements — destroys nulls
- Night effect is most pronounced at dawn and dusk (transitional D-region)

13.4 STATION INTERFERENCE

Congestion in LF/MF bands → risk of interference from other stations on or near the same frequency. By day, DOC provides protection. At night, sky wave from out-of-range stations can cause interference even within the DOC . Always positively identify the NDB at night.

13.5 MOUNTAIN EFFECT

Mountainous terrain causes reflections and diffraction of transmitted radio waves → bearing errors. Increases at low altitude; minimised by flying higher.

13.6 COASTAL REFRACTION

- Radio waves speed up over water (less energy attenuation over sea vs land)
- Speeding up causes the wavefront to bend away from the normal to the coast (Snell's law — acceleration bends away)
- Refraction is negligible at 90° to the coast; increases as angle of incidence increases
- Aircraft over sea: plotted position appears closer to the coast than actual position

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MINIMISING COASTAL REFRACTION

Use NDBs on or near the coast Use signals that cross the coast at or near 90° Fly higher

13.7 QUADRANTAL ERROR

The aircraft airframe (predominantly fore-aft axis) distorts the loop aerial's polar diagram. Incoming NDB signals are refracted toward the fore/aft axis , with maximum distortion in the quadrants — at relative bearings of 045°, 135°, 225°, 315° .

- Older ADF systems are regularly "swung" to measure quadrantal error
- Modern systems: error determined by manufacturer; corrections applied electronically

13.8 ANGLE OF BANK (DIP ERROR)

ADF uses vertically polarized waves. Bank angle tilts the loop, causing its horizontal elements to pick up the horizontal polarization component of the incoming wave — inducing circulating currents that destroy the nulls and create bearing errors. This error is only present when not in level flight .

13.9 LACK OF FAILURE WARNING

Most ADF instruments lack a failure warning system — false indications due to equipment failure are not readily detectable. Therefore:

- Always identify and continuously monitor the NDB
- Cross-check with other navaids where possible
- When using ADF as primary navigation aid for an approach, monitor continuously

14. FACTORS AFFECTING ADF RANGE

- NDB transmission power — range ∝ √P (see formula above)
- Frequency — lower frequencies travel further as surface waves
- Time of day — range effectively reduced at night (sky wave contamination)
- Terrain — absorbent terrain reduces range; sea extends range (3×√P over water)

✦ Receiver sensitivity

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15. ADF ACCURACY AND SUMMARY

⚠

ADF ACCURACY

±5° within the DOC, by day only. This refers to the measured bearing and does not include compass error.

QUICK REVISION — CHAPTER 7

- ✧ NDB: ground transmitter, LF/MF,190–1750 kHz, vertically polarized, surface wave
- ✧ Locator:10–25 NM| En route:50+ NM| Oceanic: hundreds NM
- ✧ Range: 3×√P(W) over water; 2×√P(W) over land
- ✧ Loop aerial → figure-8 polar (2 nulls, ambiguous)
- ✧ Sense aerial (dipole) → combined with loop → CARDIOID (1 null)
- ✧ Switching two cardioids at ~120 Hz → meets ICAO ±5° accuracy
- ✧ Goniometer = fixed loop electronic equivalent
- ✧ N0NA1A: BFO ON always | N0NA2A: BFO ON for tuning only
- ✧ RBI: 000° at nose, add heading to get Mag bearing | RMI: reads Mag directly
- ✧ Needle HEAD → QDM (to station) | Needle TAIL → QDR (radial from station)
- ✧ Drift rule: Starboard=Subtract; Port=Plus
- ✧ Night effect: most pronounced at dusk/dawn
- ✧ Coastal refraction: waves speed up over water → bend away from normal; aircraft appears closer to coast
- ✧ Quadrantal error max at 045°, 135°, 225°, 315° relative bearings
- ✧ Accuracy:±5° by day within DOC only
- ✧ DOC protection: 3:1 signal/noise ratio by day; NOT guaranteed at night (sky wave)

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Practice Questions & Detailed Answers (21 Questions)

EMISSION	TUNE	IDENTIFY	MONITOR
N0NA1A	ON	ON	ON
N0NA2A	ON	OFF	OFF

Chapter Expert Quiz

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VOR | DGCA CPL/ATPL Radio Navigation Notes

CHAPTER 8

DGCA CPL / ATPL Study Notes • Radio Navigation

✈ CHAPTER 8: VOR VHF OMNIDIRECTIONAL RANGE



VOR COURSE DEVIATION INDICATOR

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1. INTRODUCTION & BACKGROUND

VOR — VHF Omnidirectional Range — is the primary short-range radio navigation aid used worldwide. It provides magnetic bearing information to/from a ground station and forms the backbone of conventional airway navigation.

NOTE

📡 Key Facts Adopted by ICAO in 1960 as the standard short-range navaid Operates in the VHF band — immune to static interference and skywave Provides magnetic bearing to/from the station (azimuth service) Emission designator: A9W Co-located with DME → VOR/DME ; with TACAN → VORTAC

Frequency Band 108.0 – 117.95 MHz Total Channels 160 channels Shared with ILS 108.0 – 111.95 MHz (40 ch) VOR only 112.0 – 117.95 MHz (120 ch)

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2. PRINCIPLE OF OPERATION

The VOR works by phase comparison of two 30 Hz signals. The phase difference between the two signals equals the aircraft's magnetic bearing from the station (radial).

NOTE

🔄 The Two Signals Signal Name Modulation Characteristic Reference Omnidirectional reference FM subcarrier (9960 Hz) Same phase in ALL directions Variable Directional / variphase AM on carrier Phase varies with azimuth (limàcon pattern)

The variable signal is created by rotating a limàcon (figure-of-eight) radiation pattern at 30 rev/sec . At magnetic north , both signals are in phase. Moving clockwise, the variable signal phase progressively leads the reference — the phase difference equals the radial.

NOTE

⚡ Exam Tip — Phase = Radial Phase difference (variable − reference) = Radial (QDR) = Magnetic bearing FROM the VOR. Example: Aircraft due East → phase difference = 90°

CVOR vs DVOR Signal Generation

FEATURE	CVOR (CONVENTIONAL)	DVOR (DOPPLER)
Reference signal	FM subcarrier (9960 Hz ±480 Hz)	AM on carrier
Variable signal	AM on carrier (rotating limàcon CW)	FM via Doppler (simulated ACW rotation)
Site error	More susceptible	Less site error — preferred at complex sites
Antenna array	Single rotating antenna	Ring of 48 antennas, switched electronically

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3. TERMINOLOGY — RADIALS & BEARINGS

NOTE

📐 Critical Definitions Term Definition Note Radial Magnetic bearing FROM the VOR (QDR) What the VOR transmits QDR Magnetic bearing FROM station to aircraft RMI tail QDM Magnetic heading to fly TO station (no wind) RMI head QTE True bearing FROM station Apply variation at VOR

NOTE

⚡ Variation Rule For VOR bearings, apply variation at the VOR station to convert magnetic to true bearing. This is opposite to ADF/NDB where variation is applied at the aircraft.

NOTE

✓ Example Aircraft on radial 090° from a VOR where local variation = 10°W QDR = 090°M → QTE = 090° − 10° = 080°T (subtract westerly variation)

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4. FREQUENCIES & CHANNELS

BAND	RANGE	CHANNELS	NOTES
Lower VOR band	108.000 – 111.975 MHz	40 channels	Shared with ILS Localiser —even tenths= VOR (108.0, 108.2…); odd tenths = ILS
Upper VOR band	112.000 – 117.975 MHz	120 channels	VOR exclusive

NOTE

⚡ Frequency Sharing (108–112 MHz) Even tenths (108.0, 108.2, 108.4…) = VOR | Odd tenths (108.1, 108.3…) = ILS Localiser. This allows 40 VOR + 40 ILS channels in the shared band.

NOTE

📡 Emission Designator: A9W A = Amplitude modulation (DSB full carrier) | 9 = Composite signal | W = Combination (telephony + telegraphy + navigation)

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5. IDENTIFICATION & MONITORING

Station Identification

NOTE

📻 Ident Methods 3-letter Morse code transmitted every 10 seconds on a 1020 Hz subcarrier Some stations also broadcast voice ATIS on the VOR frequency Keyed at 7 words per minute

Special Beacon Types

TYPE	IDENT SIGNAL	PURPOSE	USE?
TST	TST in Morse	Test beacon — under maintenance/calibration	❌ DO NOT USE
VOT	Continuous dots	VOR Test facility — checks aircraft VOR equipment	✓ Ground test only

> ⚠ **NOTE**
>
> ⚠ VOT Test Limits When checking with a VOT, the aircraft must read 000° ± 4° (FROM) or 180° ± 4° (TO). Maximum permissible error = ±4°

Monitoring

> ⓘ **NOTE**
>
> 🔍 Automatic Site Monitor Built-in monitor checks VOR bearing accuracy continuously If bearing error exceeds ±1° → station shuts down or switches to standby transmitter During standby switchover: ident is suppressed for 30 seconds Rule: NEVER USE a VOR when no ident is heard

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6. TYPES OF VOR

TYPE	FULL NAME	POWER	USE
CVOR	Conventional VOR	Standard	En-route navigation
DVOR	Doppler VOR	Standard	Sites with terrain/structure interference
TVOR	Terminal VOR	Low (≤50W)	Airfields — short range approach/departure
VOT	VOR Test facility	Very low	Ground equipment check only
VORTAC	VOR + TACAN	Standard	Civil (VOR) + Military (TACAN bearing + DME)
VOR/DME	VOR + DME	Standard	Provides bearing + distance

NOTE

📡 DVOR — Doppler Mechanism AM carrier = reference signal Ring of 48 antennas switched electronically to simulate a rotating antenna Doppler effect creates the FM variable signal Simulated rotation is anticlockwise (opposite to CVOR) — receiver processes identically Less susceptible to multipath site errors than CVOR

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7. RANGE & COVERAGE

VOR is a line-of-sight (LOS) system. Range depends on aircraft altitude and terrain.

NOTE

📡 LOS Range Formula Range (NM) = 1.25 × (√H tx + √H rx) Where H = height in feet above terrain. Note: Coefficient 1.25 is used for all DGCA exam calculations (some texts show 1.23, but exam answers verify with 1.25).

NOTE

✓ Example Aircraft at 10,000 ft, VOR antenna at 100 ft above terrain: Range = 1.25 × (√10000 + √100) = 1.25 × (100 + 10) = 1.25 × 110 = 137.5 NM

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8. DOC — DESIGNATED OPERATIONAL COVERAGE

NOTE

📐 DOC Definition Specifies the range and altitude within which the VOR provides reliable navigation service Published as a range/altitude pair — e.g., 50 NM / FL 250 Means: reliable up to 50 NM from the station at or below FL 250 Valid day AND night (unlike NDB/ADF which has a day-only DOC)

NOTE

✓ VOR DOC vs ADF DOC VOR DOC is valid day and night — VHF is not affected by skywave. NDB/ADF DOC is day only — MF/LF skywave at night causes bearing errors.

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9. ACCURACY & ERRORS

ERROR SOURCE	VALUE	NOTES
Site/terrain error	±1°	Multipath reflections; DVOR less susceptible
Propagation / scalloping	±1°	Bending of VHF waves; needle oscillation in affected areas
Airborne equipment	±3°	Receiver and aircraft installation accuracy
Total system error	±5°	ICAO specification (RSS combination)

NOTE

⚠ Airway Width — 1:60 Rule At distance D, lateral error = D × error° / 60. To maintain ±5 NM half-width with ±5° error: D = (5 × 60) / 5 = 60 NM per leg? Standard formula for max beacon spacing: max_spacing = 600 / error_degrees • For ±5° error: 600/5 = 120 NM • For ±5.5° error: 600/5.5 ≈ 109 NM

NOTE

⚡ Scalloping Rapid oscillation of VOR bearing indication caused by multipath interference. The CDI needle “scallops” back and forth. More pronounced in mountainous terrain; DVOR is less susceptible.

10. CONE OF CONFUSION

NOTE

⚠ Cone of Confusion — Overhead VOR Indications OFF flag may appear on CDI/VOR indicator TO/FROM indicator alternates rapidly CDI needle oscillates / swings Extends approximately ±50° from vertical (antenna design dependent) Normal and temporary — do not attempt corrections while in the cone

NOTE

✓ Station Passage Technique 1. Note the moment TO/FROM flag flips to FROM → that is station passage. 2. CDI will settle after exiting the cone (typically within 1–2 minutes at normal speeds).

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11. COCKPIT DISPLAYS — CDI, RMI, HSI

11.1 CDI — Course Deviation Indicator

NOTE

📐 CDI Key Facts Parameter Value Sensitivity 2° per dot Full Scale Deflection (FSD) ±10° (= ±5 dots) TO indication QDM is within ±80° of OBS setting FROM indication QDR is within ±80° of OBS setting OFF indication OBS is ~90° away from QDM/QDR (ambiguous zone)

NOTE

⚡ CDI Sense — Critical Rule The CDI only shows correct turn direction when the OBS setting matches your actual desired track . If OBS is set wrong, the needle sense can be reversed — this is the classic DGCA CDI trap!

NOTE

⚠ CDI Worked Example Aircraft on 152° radial, OBS = 329° (TO). Desired radial = 149°. Aircraft is 3° to the RIGHT of desired track. Therefore aircraft is to the LEFT of the 329° course → CDI bar deflects RIGHT (fly right). Deflection = 3° ÷ 2°/dot = 1.5 dots right .

11.2 RMI — Radio Magnetic Indicator

NOTE

📐 RMI Key Facts Needle Part Reading Equivalent Head (arrowhead) Points TO the station QDM — magnetic bearing TO station Tail Points AWAY from station QDR — radial — magnetic bearing FROM station The RMI compass card rotates with aircraft heading, so the needle always reads magnetic bearings (not relative bearings).

NOTE

✓ RMI Worked Examples (from page 128 Q1–4) Q1-4: Four RMI diagrams for worked examples Q1: Heading 360°, needle head at 071° → QDM = 071° , QDR = 251° Q2: Heading 090°, needle head at 159° → QDM = 159° , QDR = 339° Q3: Heading 220°, needle head at 345° → QDM = 345° , QDR = 165° Q4: Heading 270°, needle head at 063° → QDM = 063° , QDR = 243°

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11.3 HSI — Horizontal Situation Indicator

NOTE

📐 HSI Key Facts Parameter Standard HSI Narrow-bar HSI Sensitivity 5° per dot 2° per dot Dots each side 2 dots 2 dots FSD ±10° ±4° The HSI course pointer rotates with aircraft heading → the deviation bar always shows correct turn direction regardless of OBS setting. No CDI sense reversal.

NOTE

🔷 CDI vs RMI vs HSI Comparison Summary comparison: RMI, CDI, and HSI displays side by side

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12. IN-FLIGHT PROCEDURES

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NOTE

✓ Outbound Tracking (away from VOR) Set OBS to desired outbound radial (e.g., 090° for east outbound) FROM flag appears Wind correction: CDI drifts right → turn right to correct (normal sense with FROM selected)

NOTE

✓ Inbound Tracking (toward VOR) Set OBS to inbound track (reciprocal of radial, e.g., 270° for inbound on 090° radial) TO flag appears CDI normal sense — fly toward the needle

NOTE

📡 Station Passage Indications Primary: TO → FROM flag change CDI needle may swing briefly OFF flag may appear momentarily in cone of confusion Log time of station passage for flight planning

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13. VOR SUMMARY TABLE

PARAMETER	VALUE / DETAIL
Frequency band	108.0 – 117.95 MHz (VHF)
Total channels	160 (40 shared with ILS + 120 VOR only)
Emission designator	A9W
Principle	Phase comparison of two 30 Hz signals
Ident interval	3-letter Morse every 10 s (1020 Hz tone)
Monitor trigger	±1° bearing error → auto-shutdown
VOT tolerance	±4°
Site error	±1°
Total system accuracy	±5°
DOC validity	Day AND night
LOS range formula	1.25 × (√H1+ √H2) NM (H in feet)
CDI sensitivity	2°/dot, FSD ±10°
HSI sensitivity	5°/dot standard (FSD ±10°); 2°/dot narrow
Radial (QDR)	Magnetic bearing FROM VOR to aircraft
Variation applied at	VOR station (not aircraft)
Cone of confusion	±50° from vertical — OFF flag, needle oscillates

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14. PRACTICE QUESTIONS

> ⚠ **NOTE**
>
> ⚡ Instructions Click “Show Answer” to reveal the correct answer and explanation. All questions are DGCA CPL/ATPL standard.

Q1. What is the frequency band used by VOR?

(a) 225–400 MHz

(b) 108–117.95 MHz

(c) 190–1750 kHz

(d) 962–1213 MHz

Answer: Answer: (b) VOR operates in the VHF band, 108.0–117.95 MHz . (a) = UHF military TACAN; (c) = NDB/ADF MF/LF band; (d) = DME UHF.

Q2. The principle of operation of a VOR is:

(a) Comparison of two 9960 Hz signals

(b) Pulse timing like DME

(c) Phase comparison of two 30 Hz signals

(d) Amplitude comparison of two signals

Answer: Answer: (c) VOR uses phase comparison of two 30 Hz signals . The reference is omnidirectional (FM subcarrier); the variable rotates (AM on carrier). Their phase difference = radial.

Q3. A VOR radial is defined as:

(a) Magnetic heading to fly to the station

(b) True bearing from aircraft to station

(c) Magnetic heading of the aircraft

(d) Magnetic bearing FROM the station to the aircraft (QDR)

Answer: Answer: (d) A radial = QDR = magnetic bearing FROM VOR to aircraft . Aircraft on the 090° radial is due east of the station.

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Q4. A VOR station antenna is at 100 ft; aircraft at 3,600 ft. Using coefficient 1.25, maximum LOS range is:

> (a) 87.5 NM
>
> (b) 75 NM
>
> (c) 100 NM
>
> (d) 62.5 NM

Answer: Answer: (a) Range = 1.25 × (√3600 + √100) = 1.25 × (60 + 10) = 1.25 × 70 = 87.5 NM .

Q5. Which statement about a VOT is correct?

> (a) It radiates a continuous tone and is used to check aircraft VOR equipment accuracy on the ground
>
> (b) It is a type of VOR for terminal area use
>
> (c) It stands for ‘VOR on Test’ and should not be used
>
> (d) It provides bearing guidance to the test facility

Answer: Answer: (a) A VOT (VOR Test facility) radiates a continuous Morse dots tone . Used on the ground to verify aircraft VOR receiver accuracy. Aircraft must read 000°(±4°) FROM or 180°(±4°) TO.

Q6. Maximum permissible bearing error when checking VOR airborne equipment using a VOT:

> (a) ±4°
>
> (b) ±3°
>
> (c) ±5°
>
> (d) ±2°

Answer: Answer: (a) VOT maximum permissible error = ±4° . Exceeding this requires equipment servicing before use.

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Q7. The DOC of a VOR is valid:

(a) During daylight hours only

(b) During night hours only

(c) During twilight periods only

(d) Both day and night

Answer: Answer: (d) VOR DOC is valid day AND night . VHF is immune to skywave propagation. Contrast with NDB/ADF whose DOC is day only.

Q8. Total system accuracy of a VOR is:

(a) ±1°

(b) ±3°

(c) ±5°

(d) ±10°

Answer: Answer: (c) Total VOR accuracy = ±5° : site error ±1° + propagation/scalloping ±1° + airborne equipment ±3° (RSS combination).

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Q9. Flying directly over a VOR station, the indications are:

(a) OFF flag appears, TO/FROM alternates, CDI oscillates (cone of confusion)

(b) TO flag, CDI centres

(c) Bearing becomes more accurate

(d) No change — VOR is accurate overhead

Answer: Answer: (a) Overhead VOR = cone of confusion : OFF flag may appear, TO/FROM alternates rapidly, CDI needle oscillates. Normal and temporary.

Q10. On a standard CDI, angular value of one dot deflection:

(a) 5°

(b) 1°

(c) 2°

(d) 10°

Answer: Answer: (c) CDI sensitivity = 2° per dot . Full scale = 5 dots × 2° = ±10°.

Q11. Full Scale Deflection on a standard CDI:

(a) ±5°

(b) ±10°

(c) ±20°

(d) ±2°

Answer: Answer: (b) CDI FSD = ±10° (5 dots × 2°/dot).

Q12. Which RMI statement is correct?

(a) Arrowhead points into the tailwind

(b) Tail indicates QDM

(c) Arrowhead indicates QDM (magnetic bearing TO station)

(d) RMI shows track made good

Answer: Answer: (c) RMI: arrowhead (head) = QDM = bearing TO station. Tail = QDR = radial = bearing FROM station.

Q13. Compared to CVOR, a DVOR:

(a) Operates at higher frequencies

(b) Uses a single rotating antenna

(c) Has a larger cone of confusion

(d) Has less site error due to its electronically switched antenna array

Answer: Answer: (d) DVOR uses a ring of 48 antennas switched electronically. Doppler effect generates FM variable signal. Less susceptible to site errors from terrain/structures.

Q14. Aircraft on 090° radial from VOR. Station variation = 6°E. The QTE is:

(a) 084°T

(b) 096°T

(c) 090°T

(d) 264°T

Answer: Answer: (b) QDR = 090°M. Apply variation at VOR: 090° + 6° (East) = 096°T . (True = Magnetic + East variation)

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Q15. Max beacon spacing for airway with ±5 NM half-width, VOR error 5°:

(a) 120 NM

(b) 60 NM

(c) 150 NM

(d) 100 NM

Answer: Answer: (a) Max spacing = (half-width × 60) / error = (5 × 60) / 5 × 2 = 600/5 = 120 NM . (Using the standard 600/error_degrees formula.)

Q16. Max beacon spacing if VOR bearing error is 5.5°:

(a) 120 NM

(b) 100 NM

(c) 109 NM

(d) 80 NM

Answer: Answer: (c) Max spacing = 600 / 5.5 ≈ 109 NM .

Q17. Number of VOR channels in the 108–112 MHz band:

(a) 80 total

(b) 160 total

(c) 40 VOR channels (even tenths only)

(d) 20 VOR channels

Answer: Answer: (c) 108–112 MHz: even tenths = VOR (40 channels); odd tenths = ILS Localiser (40 channels).

Q18. Total number of VOR channels:

(a) 200

(b) 160

(c) 120

(d) 40

Answer: Answer: (b) Total VOR channels = 40 (shared, 108–112) + 120 (VOR only, 112–118) = 160 .

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Q19. Aircraft heading 287°, starboard drift 14°. Track made good (radial being flown):

(a) 273°

(b) 287°

(c) 301°

(d) 315°

Answer: Answer: (c) Track = Heading + Starboard drift = 287° + 14° = 301° . QDR = track when flying outbound from VOR.

Q20. Aircraft heading 287°, 14° starboard drift. VOR phase difference reading:

(a) 273°

(b) 287°

(c) 301°

(d) 315°

Answer: Answer: (c) Phase difference = radial = QDR = Track = 287° + 14° = 301° .

Q21. CDI shows TO indication when:

(a) Aircraft is heading toward the station

(b) Aircraft is closer than 10 NM

(c) QDM is within ±80° of OBS-selected course

(d) Aircraft is below the cone of confusion

Answer: Answer: (c) TO flag illuminates when QDM (bearing to station) is within ±80° of the OBS setting . Not dependent on aircraft heading — geometry only.

Q22. Aircraft on 152° radial. OBS set to 329° (TO). CDI indication:

(a) Bar deflects RIGHT — approximately 1.5 dots

(b) Bar deflects LEFT — approximately 1.5 dots

(c) Bar centred

(d) OFF flag

Answer: Answer: (a) OBS 329° TO means desired radial = 149°. Aircraft is on 152° — 3° clockwise of 149° → aircraft is LEFT of the 329° inbound course → bar deflects RIGHT (fly right). 3° ÷ 2°/dot = 1.5 dots.

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Q23. At 100 NM from VOR with ±5° total error, lateral displacement is approximately:

(a) 5 NM

(b) 7.5 NM

(c) 8.3 NM

(d) 8.7 NM

Answer: Answer: (d) Exact: 100 × sin(5°) = 100 × 0.0872 ≈ 8.7 NM . Using 1:60: 100 × 5/60 ≈ 8.3 NM.

Q24. VOR variation should be applied at:

(a) Aircraft position

(b) Mid-point between aircraft and station

(c) The VOR station

(d) The destination aerodrome

Answer: Answer: (c) For VOR, variation is applied at the VOR station . Contrast with NDB/ADF: variation applied at the aircraft.

Q25. Aircraft on 045° radial. RMI shows:

(a) Head at 045°, tail at 225°

(b) Head at 225°, tail at 045°

(c) Head and tail both at 045°

(d) Head at 225°, tail at 225°

Answer: Answer: (b) QDR (radial) = 045° → RMI tail = 045°. RMI head = QDM = 225° (reciprocal). Answer: head 225°, tail 045°.

Q26. Standard HSI CDI is set to 5°/dot. Full scale deflection is:

(a) 5°

(b) 10°

(c) 20°

(d) 15°

Answer: Answer: (b) Standard HSI: 5°/dot × 2 dots = 10° FSD.

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Q27. VOR ident transmission interval:

(a) Every 30 seconds

(b) Every 5 seconds

(c) Every 10 seconds

(d) Every 15 seconds

Answer: Answer: (c) VOR Morse ident is transmitted every 10 seconds .

Q28. If the VOR auto-monitor detects bearing error >±1°:

(a) Station automatically shuts down or switches to standby, suppressing ident

(b) Transmits a warning tone

(c) Continues at reduced power

(d) Changes frequency

Answer: Answer: (a) Auto-monitor: if error >±1° → station shuts down or switches to standby . Ident suppressed for 30 s. Never use VOR without ident.

Q29. VORTAC provides:

(a) VOR bearing for civil aircraft and TACAN bearing + DME for military aircraft

(b) VOR bearing only

(c) DME for civil and military only

(d) ILS and VOR combined

Answer: Answer: (a) VORTAC: civil aircraft receive VOR azimuth + DME distance (via TACAN DME on paired freq). Military: TACAN bearing + DME.

Q30. VOR emission designator:

(a) A9W

(b) A2A

(c) N0NA1A

(d) P0N

Answer: Answer: (a) VOR = A9W . A = AM DSB; 9 = composite signal; W = combination of telephony, telegraphy and navigation.

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Q31. HSI (5°/dot), aircraft is 4° to the right of selected course. CDI bar:

(a) Right by 2 dots

(b) Left by 2 dots

(c) Left, less than 1 dot

(d) Right, less than 1 dot

Answer: Answer: (c) 5°/dot HSI. Aircraft 4° RIGHT of course → bar deflects LEFT (fly left to intercept). Magnitude = 4° ÷ 5°/dot = 0.8 dots left = less than 1 dot LEFT.

Q32. DVOR differs from CVOR in that:

(a) Different frequencies

(b) Uses Doppler effect with an array of antennas to generate FM variable signal; AM is the reference

(c) Larger DOC

(d) Uses pulse modulation

Answer: Answer: (b) DVOR: AM carrier = reference; electronically switched 48-antenna ring generates FM variable via Doppler effect. CVOR: FM subcarrier = reference; AM rotating limàcon = variable.

Q33. Aircraft on radial 270°, heading 090° inbound to VOR, OBS set to 090°. TO/FROM indicator shows:

(a) TO

(b) FROM

(c) OFF

(d) Alternating TO/FROM

Answer: Answer: (a) Aircraft west of VOR flying east (inbound). QDM = 090°. OBS = 090°. QDM is within 80° of OBS → TO .

ANNEX A — HSI DIAGRAMS (Q31)

ANNEX B — HSI DIAGRAMS (Q32)

ANNEX C — HSI DIAGRAMS (Q33)

ADDITIONAL QUESTION DIAGRAMS

Q3 — RMI Options

Q15 — RMI Options

Q25 (RMI) and Q26 (CDI)

Q28 — RMI Options

Q30 — CDI Options

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CHAPTER 9

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✈ CHAPTER 9: ILS INSTRUMENT LANDING SYSTEM



ILS LOCALIZER & GLIDE SLOPE BEAMS

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1. INTRODUCTION

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The Instrument Landing System (ILS) is the most accurate approach and landing aid in current use. It has been in existence for over 40 years and is the ICAO standard precision approach system.

NOTE

📡 Key Characteristics Precision approach system — provides guidance in both horizontal and vertical planes Enables automatic landings (autoland) in Cat III conditions Guides the pilot from the coverage limit down to Decision Height (DH) At DH: land (if required visual references exist) or execute missed approach

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2. ILS COMPONENTS

NOTE

📡 Ground Installation: Three Main Components Component Function Location Band Localizer (LLZ) Lateral (left/right) guidance — runway centreline ~300 m from upwind end of runway VHF Glide Path (GP) Vertical (up/down) guidance — glide slope 300 m from threshold, ~200 m from runway edge abeam touchdown UHF Marker Beacons Range check — cross-check height vs distance OM, MM, IM at set distances from threshold VHF 75 MHz

NOTE

✓ Optional/Associated Components Locator (L) : Low-powered NDB co-located at the OM beacon site DME : Frequency-paired with ILS; zero-referenced to runway threshold; supplements or replaces marker range information Back Course : Non-precision approach on back beam of localizer (approved in some countries; NOT permitted in UK)

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3. ILS FREQUENCIES & PAIRING

Localizer (VHF) 108 – 111.975 MHz 40 channels, odd tenths Glide Path (UHF) 329.15 – 335 MHz 40 channels, freq. paired Markers (all) 75 MHz OM + MM + IM

NOTE

⚡ Localizer Frequency Sharing In 108–112 MHz: Odd tenths = ILS Localiser (108.1, 108.3, 108.5…); Even tenths = VOR. Each ILS frequency also has a second step: e.g., 108.1 AND 108.15 for ILS.

NOTE

📡 Frequency Pairing Advantages Selecting the localizer frequency automatically activates the GP receiver Reduces pilot workload — one selection activates both Reduces potential for wrong frequency selection Only one identifier needed for both localizer and glide path Example: LLZ 108.1 MHz is paired with GP 334.7 MHz; LLZ 111.95 MHz paired with GP 330.95 MHz.

DME Paired with ILS

NOTE

📡 ILS/DME DME ranges are zero-referenced to the ILS runway threshold Protected within ILS localizer service area up to 25,000 ft May also be used for published SIDs and STARs (with extended coverage when notified) Do not use DME outside stated limits — may give errors

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4. IDENTIFICATION & MONITORING

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NOTE

📻 ILS Identification 2 or 3-letter Morse code; first letter is usually “I” Transmitted at 7 groups per minute Only one ident needed (shared by localizer and GP) Ident automatically suppressed if ILS becomes unserviceable During maintenance: ident replaced by continuous tone or removed completely

NOTE

⚠ Critical Rules If ident is a continuous tone → ILS under maintenance → DO NOT USE If ident ceases during approach → ILS is unserviceable → discontinue immediately Unserviceable glide path only: ATC will warn; do not use GP

Ground Monitoring

NOTE

🔍 Automatic Monitor Triggers Action When: Parameter Cat I Limit Cat II Limit Cat III Limit Localizer shift from centreline > 35 ft > 25 ft > 20 ft Glide path angle change > 0.075 × basic GP angle Power reduction > 50% Actions: Cease radiation / Remove ident & nav components / Lower category (Cat II/III may revert to Cat I/II).

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5. MARKER BEACONS

Markers transmit at 75 MHz in a narrow fan-shaped vertical beam . The signal is received only when the aircraft flies through the fan (not directional). Reception is indicated by synchronous aural tones and cockpit lights.

NOTE

📡 Marker Beacon Summary Marker Cockpit Light Ident Signal Audio Freq Range from Threshold OM (Outer) BLUE 2 dashes/sec 400 Hz (Low pitch) 6.5 – 11.1 km (3.5 – 6 NM) MM (Middle) AMBER Alt dots & dashes 3/sec 1300 Hz (Medium pitch) 1050 m ± 150 m (3500′ ± 500′) IM (Inner) WHITE 6 dots/sec 3000 Hz (High pitch) 75 – 450 m (250′ – 1500′)

NOTE

⚡ Memory Aid — Markers B ig B lue B eacon = O uter Marker (3 Bs, low dashes) — A mber = Middle (Alternating) — W hite & high = Inner (Whitest, Highest pitch, fastest dots)

NOTE

✓ Z Markers Z markers have a cylindrical vertical radiation pattern. Used to mark airway reporting points or co-located with NDBs. They indicate when the aircraft is overhead (compensating for the NDB cone of silence).

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6. ILS COVERAGE

NOTE

📐 Localizer Coverage Sector Range Within ±10° of centreline 25 NM (46.3 km) 10° to 35° from centreline 17 NM (31.5 km) Outside ±35° 10 NM (18.5 km) if provided Note: May be reduced to 18 NM (±10°) when alternative navaids provide satisfactory intermediate approach coverage.

NOTE

📐 Glide Path Coverage Parameter Value Azimuth coverage ±8° either side of centreline Range At least 10 NM Vertical — upper limit 1.75 × θ above horizontal Vertical — lower limit 0.45 × θ (may reduce to 0.3θ if required) Where θ = promulgated glide path angle (usually 3°).

NOTE

⚡ Outside Coverage Sectors If an aircraft uses the ILS outside published coverage sectors, it may receive false on-course or reverse sense signals . The ILS cannot guarantee correct indications outside its coverage area.

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7. PRINCIPLE OF OPERATION

7.1 Localizer

NOTE

📡 Localizer Operation Sector Frequency Colour Code CDI Indication Left-hand lobe (aircraft left of %) 90 Hz Yellow Needle deflects RIGHT → Fly Right Right-hand lobe (aircraft right of %) 150 Hz Blue Needle deflects LEFT → Fly Left On centreline DDM = 0 — Needle centred DDM (Difference in Depth of Modulation) increases with distance from the centreline. The DDM drives the vertical needle of the CDI/HSI.

NOTE

⚡ Memory Aid — Localizer Lobes “ 150 = Blue = Right = Fly Left ” (Right of centre → fly left to return to centre) “ 90 = Yellow = Left = Fly Right ”

7.2 Back Course ILS

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NOTE

⚠ Back Course Disadvantages Glide path indications are incorrect (would guide to wrong end of runway) CDI needle sense is REVERSED for back course approach (HSI gives correct sense if front QDM selected) No range-check markers available Not permitted in the UK — ignore any back course indications in UK

7.3 Glide Path

NOTE

📡 Glide Path Operation Lobe Modulation Aircraft Position CDI Indication Upper lobe (large) 90 Hz Aircraft above GP GP needle deflects DOWN → Fly Down Lower lobe (small) 150 Hz Aircraft below GP GP needle deflects UP → Fly Up On glide path DDM = 0 On the GP GP needle centred Standard GP angle: 3° (ICAO requirement: 2° – 4°). GP transmitter: UHF, located ~300 m upwind of threshold, ~200 m from runway edge.

7.4 False Glide Slopes

NOTE

⚠ False Glide Slopes Caused by metallic structure reflections and antenna height/propagation characteristics First false GP at approximately twice the promulgated glide path angle For standard 3° GP: first false GP at approximately 6° False GPs always occur above the true glide slope Normal practice: intercept glide slope from below At Heathrow: continuous descent approach may capture GP from above — confirm height with DME/markers

NOTE

✓ ILS Reference Datum Point Located vertically above the intersection of the runway centreline and threshold at a specified height (~50 ft). The downward extended portion of the ILS glide path passes through this point. Height is published in the UK AIP, AD section.

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8. DISPLAYS & INTERPRETATION

8.1 CDI (Course Deviation Indicator) in ILS Mode

NOTE

📐 CDI Sensitivity in ILS Mode Parameter Localizer Glide Path Sensitivity 0.5° per dot 0.14° per dot Full Scale Deflection ±2.5° (= 5 dots) ±0.7° (= 5 dots) Max safe deviation below GP — 2.5 dots fly up (= 0.35°) Compare with VOR mode: CDI sensitivity = 2°/dot in VOR mode, 0.5°/dot in ILS mode. 5× more sensitive!

NOTE

⚠ Maximum Safe Deviations If left/right localizer deviation exceeds half full scale (>2.5 dots) → initiate immediate go-around If glide path fly-up exceeds 2.5 dots fly up → initiate immediate go-around (terrain clearance may be compromised)

8.2 HSI (Horizontal Situation Indicator) in ILS Mode

NOTE

📐 HSI in ILS Mode Course selector should be set to the QDM of the runway Deviation bar and GP needle give correct sense indications HSI gives correct sense even on back course (if front course QDM is set) Azimuth failure flag, ILS failure flag, and loss of power flag visible

8.3 Localizer On-Course Indications

NOTE

⚡ Reading ILS CDI Summary What you see What it means Action Localizer bar right Aircraft LEFT of centreline Turn RIGHT Localizer bar left Aircraft RIGHT of centreline Turn LEFT GP needle high (fly up) Aircraft BELOW glide slope Pitch UP / reduce descent GP needle low (fly down) Aircraft ABOVE glide slope Pitch DOWN / increase descent

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9. ILS CATEGORIES (ICAO)

NOTE

📐 Facility Performance Categories (Ground Installation) Facility Cat Glide Path Height at Reference Datum Category I Guidance to ≤ 200 ft (60 m) above threshold horizontal plane Category II Guidance to ≤ 50 ft (15 m) above threshold horizontal plane Category III Guidance from coverage limit to and along the runway surface

NOTE

📐 Operational Performance Categories (Aircrew/Aircraft) Operational Cat Decision Height (DH) RVR Cat I ≥ 200 ft (60 m) ≥ 550 m Cat II ≥ 100 ft (30 m), < 200 ft ≥ 300 m Cat IIIA < 100 ft (30 m) or no DH ≥ 200 m Cat IIIB < 50 ft (15 m) or no DH ≥ 75 m but < 200 m Cat IIIC No DH No RVR limitation

NOTE

⚡ Memory Aid — ILS Categories Cat I = 200ft / 550m | Cat II = 100ft / 300m | IIIA = 0ft / 200m | IIIB = 0ft / 75m | IIIC = Zero Zero

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10. ERRORS, ACCURACY & PROTECTED AREAS

NOTE

⚠ Sources of ILS Error Beam bends : Caused by atmospheric conditions — smooth, slow deflections Scalloping : Caused by reflections → rapid needle fluctuations impossible to follow Beam noise : Generated by transmitter interference Cross-check with other navaids (DME, altimeter, markers) and monitor ident continuously.

NOTE

📐 ILS Protected Areas Area Definition When Enforced Critical Area Defined area around LLZ and GP antennae where vehicles/aircraft excluded ALL ILS operations (any category) Sensitive Area Extends beyond critical area; controls parking/movement of vehicles/aircraft Low visibility operations only Cat II/III Hold Pre-take-off holding points further from runway than in good weather Cat II and III ops; marked with signs and red stop bars

NOTE

⚡ Other Factors Affecting Accuracy Weather : Snow and heavy rain attenuates ILS signals, reducing range and accuracy FM Broadcasts : FM stations below 108 MHz can overspill into radio nav band. FM suppression circuits mandatory in ILS receivers since late 1990s. Serviceability check : Ensure failure flags not visible; monitor ident throughout approach

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11. ILS CALCULATIONS

NOTE

📐 Approach Chart Segments Segment Between Notes Initial approach From entry to IAF (Initial Approach Fix) Sector Safety Altitudes (SSA) apply Intermediate IAF to FAF (Final Approach Fix) Intercept localizer and configure aircraft Final approach After FAF to DH/touchdown ILS guidance; check markers/DME

Height Check on Glide Path (1:60 Rule)

NOTE

✓ Formula: Height on GP Height (ft) = GP angle × Range (NM) × 100 (Simplified from the 1:60 rule: GP angle × range/60 × 6076 ft ≈ GP angle × range × 100)

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NOTE

✓ Formula: Rate of Descent (ROD) for 3° GP ROD (fpm) = 5 × Groundspeed (kt) For other GP angles: calculate for 3° then ÷3 × actual angle.

NOTE

⚡ Worked Example Aircraft at 4 NM from touchdown, 3° GP, groundspeed 150 kt: Expected height = 3 × 4 × 100 = 1200 ft (exact trig = 1274 ft) ROD required = 5 × 150 = 750 fpm (exact trig = 796 fpm)

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12. ILS SUMMARY

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COMPONENT	DETAILS
Localizer	VHF 108–111.975 MHz, 40 channels (odd tenths), antenna at upwind end
Glide Path	UHF 329.15–335 MHz, frequency paired, antenna abeam touchdown
Markers	75 MHz, fan-shaped vertical radiation, OM/MM/IM
DME	Frequency paired, zero-referenced to threshold, replaces/supplements markers
Ident	2–3 Morse letters, 7 groups/min, first letter “I”; suppressed when u/s; continuous tone = maintenance
OM	Blue, 2 dashes/sec, 400 Hz, 6.5–11.1 km from threshold
MM	Amber, alt. dots & dashes 3/sec, 1300 Hz, 1050 m from threshold
IM	White, 6 dots/sec, 3000 Hz, 75–450 m from threshold
Monitor triggers	LLZ: >35/25/20 ft shift; GP: >0.075θ change; power >50% reduction
LLZ coverage	25 NM/±10°, 17 NM/±35°
GP coverage	10 NM/±8°, 0.45θ to 1.75θ vertical
LLZ principle	LH lobe 90 Hz (yellow), RH lobe 150 Hz (blue); DDM=0 on %
GP principle	Upper lobe 90 Hz, lower lobe 150 Hz; DDM=0 on GP
False GP	First false at ~2× GP angle (6° for standard 3° GP)
CDI localizer	0.5°/dot, FSD ±2.5°
CDI glide path	0.14°/dot, FSD ±0.7°; max safe dev = 2.5 dots fly up
Height formula	GP angle × range (NM) × 100 = height (ft)
ROD formula (3°)	5 × groundspeed (kt) = ROD (fpm)

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13. PRACTICE QUESTIONS

NOTE

⚡ Instructions Click “Show Answer” to reveal the correct answer and explanation.

Q1. The coverage of an ILS localizer extends to ___ either side of the on-course line out to a range of ___ NM.

(a) 10°, 35

(b) 35°, 10

(c) 35°, 17

(d) 25°, 25

Answer: Answer: (c) Localizer: 35° from centreline out to 17 NM ; ±10° out to 25 NM. The coverage narrows to 17 NM within the wider ±35° sector.

Q2. The upper and lower limits of an ILS glide path transmitter having a 3.5° glide slope are:

(a) 6.125° – 1.575°

(b) 7.700° – 1.225°

(c) 5.250° – 1.350°

(d) 3.850° – 3.150°

Answer: Answer: (a) Upper = 1.75 × 3.5° = 6.125° ; Lower = 0.45 × 3.5° = 1.575° .

Q3. The minimum angle at which a false glide path is likely to be encountered on a 3° glide path is:

(a) 6 degrees

(b) 5.35 degrees

(c) normal glide slope times 1.75

(d) normal glide slope times 0.70

Answer: Answer: (a) First false GP ≈ twice the true GP angle = 2 × 3° = 6° . False GPs always occur above the true glide slope.

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Q4. The visual and aural indications when overflying an ILS middle marker are:

(a) continuous low pitched dashes with blue light

(b) continuous high pitched dots with amber light

(c) alternating medium pitch dots and dashes with amber light

(d) one Morse letter with white light

Answer: Answer: (c) MM = Amber light + alternating dots and dashes at 3 per second + 1300 Hz medium pitch audio.

Q5. An aircraft on ILS approach is receiving stronger 150 Hz signals than 90 Hz. Correct actions to reach centreline and glide path:

(a) DOWN and LEFT

(b) UP and LEFT

(c) UP and RIGHT

(d) DOWN and RIGHT

Answer: Answer: (b) Stronger 150 Hz on localizer = aircraft in RIGHT (blue) lobe = aircraft is RIGHT of centreline → fly LEFT . Stronger 150 Hz on GP = aircraft BELOW glide slope (lower lobe = 150 Hz) → fly UP . Combined: UP and LEFT .

Q6. Upper and lower limits of ILS glide path with 3.0° glide slope:

(a) 0.35° / 0.70°

(b) 3.00° / at least 6°

(c) 5.25° / 1.35°

(d) 10.0° / 35.0°

Answer: Answer: (c) Upper = 1.75 × 3° = 5.25° ; Lower = 0.45 × 3° = 1.35° .

Q7. A category II ILS installation in the UK:

(a) provides accurate guidance down to 50 ft above the horizontal plane containing the runway threshold

(b) has a steep glide path, normally 7.5°

(c) provides guidance to and along the runway surface

(d) has a false glide path exactly twice the true GP angle

Answer: Answer: (a) Cat II ILS ground installation provides guidance to the point where the glide path intersects at 50 ft (15 m) above the threshold horizontal plane.

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Q8. Which ILS indicator shows aircraft on final approach left of centreline and at maximum safe deviation below the glide path?

(a) (a) — bar right, GP needle centred

(b) (b) — bar left, GP needle high

(c) (c) — bar centred, GP needle centred

(d) (d) — bar right, GP needle high (fly up) at 2.5 dots

Answer: Answer: (d) Aircraft LEFT of centreline → localizer bar deflects RIGHT . Below glide path at maximum safe deviation (2.5 dots fly up) → GP needle shows fly up, 2.5 dots . This is option (d).

Q9. An aircraft tracking to intercept the ILS localizer inbound but outside the published coverage angle:

(a) will receive false on-course or reverse sense signals

(b) will not normally receive signals

(c) will receive signals without coding

(d) can expect correct indications

Answer: Answer: (a) Outside the coverage sectors, the ILS may give false on-course or reverse sense signals . The signal exists outside the protected coverage area but cannot be guaranteed to give correct guidance.

Q10. The outer marker of an ILS has visual identification of:

(a) alternating dots and dashes on a blue light

(b) continuous dots at 3 per second, blue light

(c) continuous dashes at 2 per second, amber light

(d) continuous dashes at 2 per second, blue light

Answer: Answer: (d) OM = Blue light + 2 dashes per second (400 Hz, low pitch).

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Q11. The specified maximum safe fly-up indication on a 5-dot CDI is:

(a) half full scale deflection above centreline

(b) 2.5 dots fly up

(c) just before full scale deflection

(d) 1.3 dots fly up

Answer: Answer: (b) Maximum safe deviation below glide slope = 2.5 dots fly up (half the 5-dot full scale). Beyond this, initiate immediate go-around as terrain clearance may be compromised.

Q12. An aircraft using ILS outside coverage sectors — which is correct?

(a) Glide slope false course, localizer intermittent

(b) No transmissions, OFF flags visible

(c) Localizer false course, glide slope reverse sense

(d) Localizer false course/reverse sense; glide slope intermittent/incorrect

---

Answer: Answer: (d) Outside coverage: localizer may give false on-course and reverse sense indications ; glide slope may give intermittent and incorrect indications . Answer (d) correctly states this combination.

Q13. Coverage of ILS glide slope in azimuth:

(a) ±8° out to 10 NM

(b) ±10° out to 8 NM

(c) ±12° out to 17 NM

(d) ±35° out to 25 NM

Answer: Answer: (a) GP horizontal coverage: ±8° azimuth out to at least 10 NM .

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Q14. ILS receivers showing predominant 90 Hz on both localizer and glide slope; runway QDM 264°. Aircraft is:

> (a) north of localizer, below glide slope
>
> (b) south of localizer, above glide slope
>
> (c) north of localizer, above glide slope
>
> (d) south of localizer, below glide slope

Answer: Answer: (b) Runway QDM 264° (westward). Standing at the runway facing approach (264°), LEFT is to the south, RIGHT is north. 90 Hz on localizer = aircraft in LEFT (yellow) sector = aircraft is south of centreline (left of runway facing west, looking east). 90 Hz on GP = upper lobe = aircraft is above glide slope. Answer: south, above glide slope .

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CHAPTER 10

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✈ CHAPTER 10: MLS MICROWAVE LANDING SYSTEM

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1. INTRODUCTION & ILS DISADVANTAGES

The Microwave Landing System (MLS) was designed to replace ILS as the ICAO standard precision approach system, overcoming ILS limitations while providing greater flexibility. However, few MLS installations exist and they co-exist with ILS.

> ⚠ **NOTE**
>
> ⚠ Why ILS Needed Replacing — ILS Disadvantages Only 40 channels available worldwide Fixed and narrow azimuth/glide slope beams → aircraft must be sequenced → landing delays No special procedures for helicopters, slower aircraft, STOL aircraft Cannot be sited in hilly terrain; requires large flat cleared land Vehicles, taxiing aircraft and buildings must be kept well clear to prevent beam bending

2. THE MLS SYSTEM

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PARAMETER	MLS VALUE	VS ILS
Channels	200	ILS: 40
Frequency band	SHF: 5031–5090.7 MHz	ILS: VHF/UHF
Azimuth coverage	±40°	ILS: ±35° (limited)
Glide slope range	0.9° to 20°selectable	ILS: fixed ~3°
Usable range (UK)	20 NM(up to 30 NM)	ILS: 25 NM/±10°
Max coverage height	20,000 ft	ILS: variable
Back course	Secondary system ±20°, 15° elevation, 10 NM, 10,000 ft	ILS: problematic
DME	Built-in DME	ILS: optional
Ident prefix	M+ two letters	ILS: I + letters
Aim	CAT III operations	ILS: CAT I/II/III

NOTE

✓ MLS Advantages Over ILS SHF frequency: sitable in hilly terrain; scanning beam avoids reflections from vehicles/buildings 200 channels vs 40 for ILS → less frequency congestion Selectable approach paths (curved/segmented) → noise abatement procedures possible Pilots choose own approach path within coverage → better runway utilisation Compatible with EFIS, autopilot, RNAV and ILS-type displays Continuous on/off flag indications

3. PRINCIPLE OF OPERATION

MLS uses Time Division Multiplexing (TDM) — only one frequency per channel; all ground equipment transmissions are synchronized to avoid interference.

NOTE

📡 Time Referenced Scanning Beam (TRSB) Measurement How It Works Azimuth Beam sweeps “TO” then “FRO” across the scan sector. Aircraft measures the time interval between the two receptions. Time interval = angular position relative to runway centreline (QDM). Elevation Second beam sweeps up then down. Aircraft measures time difference between up/down pulses = angular position above horizontal = selected glide slope. Range Built-in DME (DME/P for Cat II/III — accurate to within 100 ft).

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📡 Other MLS Components Back Azimuth : Go-around/departure guidance ±20° up to 15° elevation, 10 NM, 10,000 ft Flare : Standard provides for flare elevation (not intended for future implementation) Auxiliary data : Station ident, system condition, runway condition, weather information

4. AIRBORNE EQUIPMENT

NOTE

📡 MLS Airborne Equipment Features Continuously displays aircraft position relative to selected course and glide path + distance Two cross-bars similar to ILS CDI — indications relative to selected course (not fixed runway QDM) Can programme segmented and curved approaches (DME/P required on ground) Multi-mode receivers : handle ILS, MLS and GPS from single combined control unit Control unit allows selection of: AZ (azimuth angle), G/S (glide slope angle), CHAN (channel 500–699) Mode selector: AUTO (G/S and AZ auto per channel) or MAN (preferred G/S and AZ selection)

5. MLS SUMMARY

PARAMETER	DETAILS
Purpose	Precision approach system; designed to replace ILS
Frequency band	SHF: 5031–5090.7 MHz
Channels	200
Principle	TRSB (Time Referenced Scanning Beam) — time between TO/FRO sweeps
Azimuth coverage	±40° of runway QDM
Glide slope	0.9° to 20° selectable
Range (UK)	20 NM; up to 30 NM
Range measurement	Built-in DME (DME/P for Cat II/III: ±100 ft)
Ident	M + 2 letters
Display	Cross-bars (same as ILS CDI); relative to selected course
Target category	CAT III

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6. PRACTICE QUESTION

Q1. The coverage of the Microwave Landing System in the UK extends to ___ up to a height of ___ and ___ either side of the on-course line.

(a) 20 NM, 20,000 ft, 40 degrees

(b) 35 NM, 5,000 ft, 40 degrees

(c) 35 NM, 5,000 ft, 20 degrees

(d) 17 NM, 2,000 ft, 35 degrees

Answer: Answer: (a) MLS in the UK: range 20 NM , height 20,000 ft , azimuth ±40° of runway centreline.

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CHAPTER 11

✈ CHAPTER 11: RADAR PRINCIPLES RADIO DETECTION AND RANGING

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1. INTRODUCTION

Radar = RA dio D etection A nd R anging. Developed before WWII, used both for military and civil aviation. Today radar is essential for ATC, weather warning (AWR), surface movement (ASMI), and navigation (DME is secondary radar).

NOTE

📡 Radar Applications in Aviation Use System ATC surveillance ASR, TAR, PAR, ASMI Aircraft identification + altitude SSR (Secondary Surveillance Radar) Range measurement DME (secondary radar) Weather depiction AWR (Airborne Weather Radar) Weather info to ATC Ground weather radars

2. TYPES OF RADAR

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TYPE	HOW IT WORKS	EXAMPLES
Primary Radar	Transmits pulse, detects reflected echo from target.One frequency. Does not require target cooperation.	ATC surveillance, AWR, PAR, ASMI
Secondary Radar	Transmits interrogation pulse; target (transponder)transmits its own replyon different frequency. Requires transponder.	SSR, DME

3. RADAR FREQUENCIES

NOTE

📡 Why VHF and Above? Free from external noise/static and ionospheric scatter Shorter wavelengths produce narrow, efficient beams for target discrimination and bearing Shorter wavelengths allow shorter pulses (better range resolution) Efficient reflection: shorter λ reflected more efficiently from objects

4. PULSE TECHNIQUE

NOTE

📡 Key Pulse Parameters Term Definition Formula Pulse Width (PW) Duration of each transmitted pulse — PRF / PRR Pulse Recurrence Frequency — pulses per second (pps) — PRI / PRP Pulse Recurrence Interval/Period — time between pulses PRI = 1/PRF

NOTE

⚡ Example Calculation PRF = 250 pps → PRI = 1/250 s = 1,000,000/250 µs = 4000 µs

5. RANGE MEASUREMENT (ECHO PRINCIPLE)

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NOTE

📡 Range Calculation Range (NM) = Echo time (µs) / 12.36 Range (km) = Echo time (µs) × 300 / 2 One radar mile (1 NM out and back) = 12.36 µs Speed of radio waves: c = 3 × 10&sup8 m/s = 162,000 NM/s

NOTE

⚡ Worked Example Echo time = 500 µs: Range = 500/12.36 = 40.5 NM | Range = 500 × 300/2 = 75 km

6. MAXIMUM & MINIMUM RANGE

NOTE

⚠ Maximum Range — Limited by PRF Each pulse must return from the most distant target BEFORE the next pulse is transmitted. Max Range (NM) = 81,000 / PRF(pps) [or: PRF = 81,000 / Max_Range_NM] Max Range (km) = 150,000 / PRF(pps) Rule: Greater range required → lower PRF.

NOTE

⚠ Minimum Range — Limited by Pulse Width Target must be far enough that its echo returns AFTER the pulse tail leaves the transmitter. In 1 µs, radio waves travel 300 m (150 m each way). Min Range (m) = Pulse Width (µs) × 150 Short pulses needed for short range; long pulses for long range.

NOTE

⚡ PRF Examples Requirement Calculation Answer Max range 187 km, find PRF PRF = 300,000,000/(2×187,000) 802 pps Max range 200 NM, find PRF PRF = 81,000/200 405 pps PRF = 400 pps, find max range NM 81,000/400 203 NM PRI = 2100 µs, find max range NM PRF=1/2100µs=476pps; 81,000/476 170 NM

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7. FACTORS AFFECTING RANGE & ACCURACY

NOTE

📡 Factors Affecting Primary Radar Range Factor Effect Transmit power Power ∝ range&sup4 → to double range , power must increase by factor of 16 Target characteristics Metal > wood; larger objects > smaller; aspect angle matters Aircraft height LOS system — higher aircraft detected at greater range; high ground screens low-level aircraft Wavelength vs rain λ >10 cm: negligible attenuation; λ 4–10 cm: significant in tropical rain; λ <4 cm: significant in temperate rain Super-refraction Temperature inversion + humidity decrease with height → waves refract DOWN → beyond LOS range Sub-refraction Waves refract UP → reduced range Restoration time Recovery time of receiver after transmission — design factor affecting minimum range Fly-back / dead time CRT trace return time → reduces practical range below theoretical maximum

NOTE

⚡ Wavelength Guide for Radar Systems ASMI (Surface Movement): 1.75–2 cm (15–17 GHz, SHF) AWR / PAR: 3 cm (9375 MHz, SHF) Aerodrome Surveillance: 3 or 10 cm Terminal Surveillance: 10, 23 or 50 cm Long range ASR: 10–50 cm Do not use λ < 3 cm for long range systems (too much rain attenuation)

NOTE

📡 LOS Range Formula for Radar Max Range (NM) = 1.23 × (√H TX + √H RX) H in feet AMSL. (Note: 1.25 used for VOR/VDF; radar uses 1.23)

8. RADAR RESOLUTION

DIMENSION	DETERMINED BY	DETAIL
Radial (range) resolution	Half pulse width	1 µs PW → 150 m radial stretch
Azimuth resolution	Full beamwidth	3° BW at 120 km → 6 km azimuth stretch (1:60)

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9. MOVING TARGET INDICATION (MTI)

NOTE

📡 MTI Eliminates returns from stationary objects (hills, buildings) — shows only moving targets (aircraft) Risk: second trace returns — echoes from beyond selected range appear as false moving targets Solution: Jitter the PRF (change PRI between consecutive pulses) → removes second trace returns

10. RADAR ANTENNAE

NOTE

📡 Antenna Types Type Features Use Microwave Horn Often used as feed for parabolic reflectors General purpose Parabolic Reflector Dish; main lobes + side lobes; needs side-lobe suppression ATC, AWR Slotted Planar Array (Flat Plate) Narrower beam, much smaller side lobes; less power needed; better resolution Modern AWR, PAR

NOTE

⚡ Larger Antenna = Narrower Beam = Better Resolution But a larger antenna in an aircraft is impractical → hence shorter wavelengths are used in aircraft (AWR: 3 cm).

11. PRACTICE QUESTIONS

Q1. The factor which determines the MAXIMUM range of a radar is:

(a) pulse repetition rate

(b) pulse width

(c) power

(d) beamwidth

Answer: Answer: (a) Maximum range is determined by the pulse repetition rate (PRF) . The next pulse cannot be sent until the first returns from maximum range. Lower PRF = greater max range.

Q2. The main advantage of continuous wave (CW) radars is:

(a) no maximum range limitation

(b) better range resolution

(c) no minimum range limitation

(d) better bearing accuracy

Answer: Answer: (c) CW radars have no minimum range limitation (the radio altimeter uses CW). Pulsed radars have a minimum range determined by pulse width.

Q3. If the PRF of a primary radar is 500 pps, the maximum range will be:

(a) 324 NM

(b) 300 NM

(c) 162 NM

(d) 600 NM

Answer: Answer: (c) Max range = 81,000 / PRF = 81,000 / 500 = 162 NM .

Q4. To double the range of a primary radar requires power to be increased by a factor of:

(a) 2

(b) 4

(c) 8

(d) 16

Answer: Answer: (d) Power ∝ range&sup4. To double range: power increase = 2&sup4 = 16 .

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Q5. Echo time is 1720 µs. Range of target:

(a) 139 km

(b) 258 km

(c) 278 km

(d) 516 km

Answer: Answer: (b) Range = 1720 × 300 / 2 = 258,000 m = 258 km .

Q6. Max range required: 100 NM. Maximum PRF?

(a) 1620 pps

(b) 1234 pps

(c) 617 pps

(d) 810 pps

Answer: Answer: (d) PRF = 81,000 / 100 = 810 pps .

Q7. PRI = 2100 µs. Maximum radar range:

(a) 170 NM

(b) 315 NM

(c) 340 NM

(d) 630 NM

Answer: Answer: (a) PRF = 1,000,000 / 2100 = 476 pps. Max range = 81,000 / 476 ≈ 170 NM .

Q8. To improve resolution of a radar display:

(a) narrow pulse width and narrow beamwidth

(b) high frequency and large reflector

(c) wide beamwidth and wide pulse width

(d) low frequency and narrow pulse width

Answer: Answer: (a) Better resolution = shorter pulse width (radial) + narrower beamwidth (azimuth). Answer: narrow PW + narrow BW .

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Q9. Advantage of phased array (slotted antenna):

(a) better resolution

(b) less power required

(c) reduced side lobes and clutter

(d) all of the above

Answer: Answer: (d) Slotted planar array: narrower beam → better resolution, less power needed, reduced side lobes. All of the above.

Q10. Echo received 900 µs after transmission. Range to target:

(a) 73 NM

(b) 270 NM

(c) 135 NM

(d) 146 NM

Answer: Answer: (a) Range = 900/12.36 = 72.8 ≈ 73 NM .

Q11. Factor limiting MINIMUM detection range of radar:

(a) pulse repetition interval

(b) transmitter power

(c) pulse width

(d) pulse repetition frequency

Answer: Answer: (c) Minimum range is limited by pulse width . Target must be far enough that its echo returns after the pulse has finished transmitting.

Q12. Doppler MTI generates second trace returns. These are removed by:

(a) using different frequency for Tx/Rx

(b) jittering the PRF

(c) regular pulsewidth changes

(d) limiting power output

Answer: Answer: (b) Second trace returns are removed by jittering the PRF (varying PRI between pulses). This prevents false targets from appearing within range.

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Q13. Radar max range: 12 km. Maximum PRF?

(a) 25,000 pps

(b) 6,700 pps

(c) 12,500 pps

(d) 13,400 pps

Answer: Answer: (c) PRF = 300,000,000 / (2 × 12,000) = 300,000,000 / 24,000 = 12,500 pps .

Q14. Bearing of primary radar measured by:

(a) phase comparison

(b) searchlight principle

(c) lobe comparison

(d) DF techniques

Answer: Answer: (b) Primary radar bearing uses the searchlight principle — a narrow rotating beam; direction of target = direction of beam when echo is received.

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CHAPTER 12

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✈ CHAPTER 12: GROUND RADAR ATC RADAR SYSTEMS

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1. INTRODUCTION

ATC uses ground radars extensively to serve a large number of requirements. Both primary and secondary radar techniques are used. Primary radar detects all aircraft (including those without transponders) and may incorporate Moving Target Indication (MTI).

NOTE

📡 ATC Ground Radar Systems Area Surveillance Radar (ASR) — long range en-route Terminal Area Surveillance Radar (TAR) — medium range TMA Aerodrome Surveillance Approach Radar — short range approach Precision Approach Radar (PAR) — talk-down Airport Surface Movement Radar (ASMR/ASMI) — surface movement

2. AREA SURVEILLANCE RADAR (ASR)

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PARAMETER	VALUE
Range	200–300 NM
Wavelength	10–50 cm
Pulse length	2–4 µs
PRF	300–400 pps
Scan rate	5–6 rpm
Purpose	Airway surveillance — range and bearing of aircraft

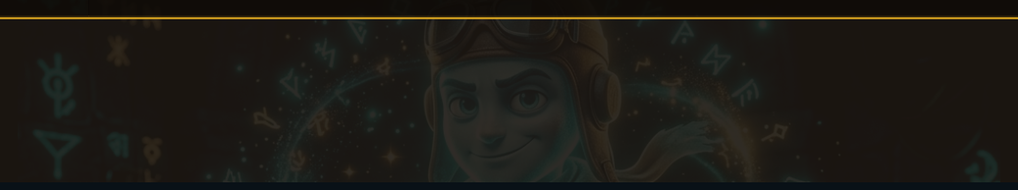
3. TERMINAL SURVEILLANCE AREA RADAR (TAR)

PARAMETER	VALUE
Range	Up to75 NM
Wavelength	10, 23 or 50 cm
Pulse width	1–3 µs
Purpose	Traffic control in TMAs
UK radar separation	3 NMwithin 40 NM (or 60 NM approved) of radar head, below FL245

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4. AERODROME SURVEILLANCE APPROACH RADAR

PARAMETER	VALUE
Range	Up to25 NM
Wavelength	3 cm or 10 cm
Pulse width	0.5–1 µs
Services provided	Radar vectoring to ILS; Surveillance Radar Approach (SRA); positional information/control in vicinity



5. AIRPORT SURFACE MOVEMENT RADAR (ASMI)

> ⚠ **NOTE**
>
> ⚠ ASMI Key Design Parameters Parameter Value Frequency 15–17 GHz (SHF) Wavelength 1.76–2 cm Beamwidth 0.2°–1° (very narrow) Scan rate 60 rpm PRF 4,000–20,000 pps Pulse width 0.03 µs Range 2.5–6 NM in light precipitation Purpose Detailed display of aircraft/vehicles on runways and taxiways in all weather

> ⚠ **NOTE**
>
> ⚡ ASMI Limitations High frequency (SHF) → increasingly attenuated by precipitation → range reduces in heavy rain EHF band NOT suitable for ASMI — too much attenuation in most precipitation types Processing can remove selected fixed features (buildings, taxiways) leaving only moving targets visible

6. RADAR CHARACTERISTICS SUMMARY

SYSTEM	RANGE	WAVELENGTH	PW	PRF	SCAN RATE
ASR	200–300 NM	10–50 cm	2–4 µs	300–400 pps	5–6 rpm
TAR	Up to 75 NM	10, 23, 50 cm	1–3 µs	Medium	Medium
Aerodrome	Up to 25 NM	3 or 10 cm	0.5–1 µs	Higher	Higher
ASMI	2.5–6 NM	1.76–2 cm	0.03 µs	4,000–20,000 pps	60 rpm

7. PRACTICE QUESTIONS

Q1. A primary radar has a PRF of 275 pps. Time interval between leading edges of successive pulses:

(a) 3.64 milliseconds

(b) 36.4 milliseconds

(c) 3.64 microseconds

(d) 36.4 microseconds

Answer: Answer: (a) PRI = 1/275 s = 1,000,000/275 µs = 3636 µs = 3.64 milliseconds .

Q2. Primary radar PRF = 450 pps. Maximum range (ignoring PW and flyback):

(a) 333 NM

(b) 180 NM

(c) 666 NM

(d) 360 NM

Answer: Answer: (b) Max range = 81,000/450 = 180 NM .

Q3. Frequency band and scan rate of Airfield Surface Movement radars:

(a) SHF; 60 rpm

(b) SHF; 200 rpm

(c) EHF; 100 rpm

(d) EHF; 10 rpm

Answer: Answer: (a) ASMI operates in the SHF band (15–17 GHz) at a scan rate of 60 rpm .

Q4. Radar with scanner rotation 60 rpm, beamwidth ~0.5°, PRF 10,000 pps. This is:

(a) ASMI with theoretical range 8 NM

(b) Precision Approach Radar

(c) ASMI with theoretical range 16 NM

(d) High resolution Surveillance Approach Radar

Answer: Answer: (a) All characteristics (60 rpm, narrow beam, high PRF, short PW) match an ASMI . Max range = 81,000/10,000 = 8.1 NM ≈ 8 NM .

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CHAPTER 13

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✈ CHAPTER 13: AIRBORNE WEATHER RADAR AWR — WEATHER DETECTION & GROUND MAPPING

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1. INTRODUCTION & COMPONENTS

Airborne Weather Radar (AWR) provides pilots with weather and navigation information ahead of the aircraft. It requires pilot interpretation — its usefulness depends on operator skill. It can display on a dedicated unit or on the EFIS navigation display (ND).

📡 AWR Components Transmitter/Receiver : generates and detects microwave pulses Antenna : attitude-stabilised in pitch and roll; parabolic or flat plate Indicator : PPI display (monochrome or colour) Control Unit : range, tilt, function, gain controls

2. AWR FUNCTIONS

NOTE

📡 Main AWR Functions Detect size of water droplets → identify areas of turbulence within clouds Determine height of cloud tops by tilting beam up/down Map terrain below aircraft for navigation and high ground avoidance Provide position fix (range and bearing) from prominent terrain features

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3. PRINCIPLE OF OPERATION

NOTE

📡 AWR — Primary Radar Principles Echo principle : determines range of targets Searchlight principle : determines relative bearing of targets Range shown on display is slant range — use Pythagoras to find ground range Antenna is attitude-stabilised in pitch and roll — if stabilisation fails display becomes lopsided

AWR Frequency

📡 Why 9375 MHz (3.2 cm)? Optimum for detecting large water droplets and wet hail (~3 cm diameter) associated with severe turbulence Typical frequency: 9375 MHz ±30 MHz (SHF band) λ = 300/9375 = 3.2 cm Higher frequency → returns from smaller droplets (clutter); Lower frequency → insufficient returns Wavelengths >10 cm produce few weather returns

Radar Beams

NOTE

📡 Beam Types Beam Shape Used For Range Pencil/Conical 3°–5° width Weather depiction + long range mapping >60 NM (up to ~150 NM) Cosecant² / Fan-shaped 85° deep, 3.5° azimuth Short range ground mapping 15–70 NM

NOTE

⚡ Water/Ice in Radome Radar energy absorbed by water/ice in the radome (like a microwave oven) → weaker forward transmission → degraded performance. Radome must be kept dry and ice-free.

4. WEATHER DEPICTION & COLOURS

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COLOUR	RAINFALL RATE	MEANING
BLACK	<0.7 mm/h	Very light or no returns
GREEN	0.7–4 mm/h	Light returns
YELLOW	4–12 mm/h	Medium returns
RED	>12 mm/h	Strong returns
MAGENTA	Turbulence intensity	Severe turbulence

COLOUR	MEANING
GREEN	Light ground returns
YELLOW	Medium ground returns
RED	Heavy ground returns

Turbulence Indicators

> ⚠ **NOTE**
>
> ⚠ Areas to Avoid on AWR Display Where colour zones are closest together (steepest rainfall gradient) U-shapes — hail activity Fingers — hail activity Scalloped edges — hail activity Hooks — hail activity

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5. MONOCHROME CONTROL UNIT

NOTE

📡 Control Unit Functions Control Position Effect Power / Stab ON (Stab On) System energized; antenna attitude-stabilised in pitch & roll STAB OFF Scanner locked to aircraft axes — display lopsided during manoeuvres Range STANDBY Equipment ready, transmitter OFF. Maintain on ground until clear of personnel Range selections Energizes transmitter; up to 150 NM practical max Tilt ±15° Tilts beam up (+) or down (−). Tilt up for increased range or lower altitude. Sweep ±90° (60° typical) MAP Cosecant² beam Fan-shaped beam; use up to 60–70 NM . Manual gain operative MAN Pencil beam Manual gain; map >60 NM; weather/navigation 70–150 NM WEA Pencil beam Normal weather observation. AGC/Swept Gain automatic (operative up to ~25 NM) — manual gain INOPERATIVE CONT Contour Highlights dangerous rainfall gradient; Swept Gain also operative; darker colour = dangerous areas

NOTE

⚡ AGC / Swept Gain / STC Automatic Gain Control decreases receiver gain for nearer echoes → ensures a small nearby cloud does NOT appear brighter than a larger distant cloud. Operates up to ~25 NM. Makes ALL clouds up to 25 NM directly comparable.

6. MAPPING OPERATION

NOTE

📡 Ground Mapping Guide MAP (cosecant²): use up to 60–70 NM ; tilt beam down for best presentation MAN (pencil beam): use for mapping beyond 70 NM Bright returns : coastlines, built-up areas, skyscrapers, bridges, power stations Weak returns : calm sea, fine sand, flat terrain Ice has jagged edges (reflects); snow is a poor reflector and masks ground features Hill shadow : mountains create radar shadow → may appear as a false lake

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NOTE

⚠ Pre-flight Radar Check Precautions Ensure aircraft is clear of personnel, other aircraft, vehicles and buildings Select conical beam with maximum uptilt, switch on briefly, check for picture, return to STANDBY Electromagnetic radiation is a health hazard — never point operating radar at personnel

7. WEATHER OPERATION

NOTE

✓ Thunderstorm Avoidance Procedure Select maximum range to detect weather in good time; adjust tilt to remove ground returns If storm is extensive, make early track adjustment in consultation with ATC As clouds approach, select lower ranges + CONT; determine best track to avoid turbulence Ensure short-term heading changes steer AWAY from worst areas Constantly switch between short/medium/long ranges to maintain complete storm picture

NOTE

⚠ AWR Limitations Shadow area : Heavy rain blocks radar waves → area behind heavy rain may conceal severe weather Too high tilt : A thunderstorm may not be detected if tilt is set too high AWR does NOT detect Clear Air Turbulence (CAT) Height ring : In older AWR with dish antenna, vertical overspill gives a ‘height ring’ on display — indicates serviceability and roughly the aircraft's height

8. COLOUR AWR CONTROLS

NOTE

📡 Additional Colour AWR Features Function Description Wx, Wx+T, Wx(var) Weather / Weather + Turbulence / Weather with variable gain WxA Weather Alert — flashing display of turbulence areas TEST Displays colour pattern for pre-flight serviceability check HOLD Freezes display to assess storm movement. After 2–3 min, deselect to see storm displacement TGT ALERT Alerts pilot of contour-strength storm within 60–160 NM and ±15° of heading. Flashes TGT symbol. FAULT Flashes on screen if power or transmitter failure detected

9. AWR SUMMARY

PARAMETER	DETAIL
Type	Primary radar
Frequency	9375 MHz (±30 MHz), SHF, λ=3.2 cm
Principle	Echo (range) + searchlight (bearing). Range shown is slant range.
Pencil beam	3°–5°; weather + long range mapping (>60 NM)
Fan beam (MAP)	Cosecant²; short range mapping up to 60–70 NM
Antenna stabilisation	Pitch and roll; lopsided display = stab failure
Tilt	±15°; UP for longer range/lower altitude
Weather colours	Black → Green → Yellow → Red → Magenta
Turbulence warning shapes	U-shapes, Fingers, Scalloped edges, Hooks
AGC range	Operates up to ~25 NM in WEA/CONT mode
Max practical range	~150 NM
AWR does NOT detect	Clear Air Turbulence (CAT)

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10. PRACTICE QUESTIONS

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Q1. A frequency used by airborne weather radar is:

(a) 8800 MHz

(b) 9.375 GHz

(c) 93.75 GHz

(d) 1213 MHz

Answer: Answer: (b) AWR frequency = 9375 MHz = 9.375 GHz (SHF band). This gives the best returns from large water droplets (~3 cm) associated with severe turbulence.

Q2. AWR required to detect targets up to 200 NM. Maximum PRR:

(a) 405 pps

(b) 810 pps

(c) 1500 pps

(d) 750 pps

Answer: Answer: (a) Max PRF = 81,000 / 200 = 405 pps .

Q3. Weather beam preferred over fan-shaped beam for mapping in excess of ___ NM:

(a) 20–25

(b) 60–70

(c) 100–150

(d) 150–200

Answer: Answer: (b) Fan-shaped (cosecant²) beam used up to 60–70 NM . Beyond that, use pencil (weather/MAN) beam.

Q4. AWR is an example of ___ radar at frequency ___ in the ___ band:

(a) primary, 8800 MHz, SHF

(b) secondary, 9.375 MHz, UHF

(c) secondary, 9375 MHz, SHF

(d) primary, 9375 MHz, SHF

Answer: Answer: (d) AWR is primary radar at 9375 MHz in the SHF band.

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Q5. Correct colour sequence as returns get stronger:

(a) red, yellow, green

(b) yellow, green, red

(c) green, yellow, red

(d) red, green, yellow

Answer: Answer: (c) Colour sequence in order of increasing return strength: Black → Green → Yellow → Red → Magenta .

Q6. A false indication of water may be given by AWR when:

(a) flying over land with Land/Sea switch in Sea position

(b) flying over mountainous terrain

(c) cloud/precipitation between aircraft and cloud target

(d) using mapping beam beyond 50 NM

Answer: Answer: (b) Hill shadows created by mountainous terrain can give a false impression of water (lakes) on the AWR display.

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Q7. AWR operates at 9375 MHz because:

(a) 8800 MHz gives best returns from all precipitation

(b) 13300 MHz used

(c) 9375 MHz gives best returns from rainfall associated with Cb

(d) 9.375 GHz best for detecting aircraft

Answer: Answer: (c) 9375 MHz gives the best returns from large water droplets and wet hail found in convective clouds (Cb) associated with severe turbulence.

Q8. Mapping mode of AWR utilizes:

(a) pencil beam from 70 NM to 150 NM

(b) cosecant² beam effective to 150 NM

(c) pencil beam max range 70 NM

(d) cosecant² beam effective 50 NM to 70 NM

Answer: Answer: (a) MAP (cosecant²) is used up to 60–70 NM. For mapping beyond 70 NM, select MAN to use the pencil/weather beam from 70 NM to 150 NM .

Q9. AWR uses 9 GHz because the wavelength gives reflections only from:

(a) short wavelength produces higher frequency returns

(b) reflects from cloud water droplets of all sizes

(c) wavelength gives reflections only from the larger water droplets

(d) frequency penetrates clouds for good mapping

Answer: Answer: (c) 3.2 cm wavelength is comparable to the size of large water droplets and wet hail (~3 cm). This gives reflections from dangerous precipitation while ignoring smaller harmless droplets.

Q10. AWR antenna is stabilised:

(a) in pitch, roll and yaw

(b) in pitch and roll

(c) in pitch and roll whether stab on or off

(d) in pitch and roll only at 0° tilt

Answer: Answer: (b) AWR antenna is stabilised in pitch and roll . STAB OFF locks scanner to aircraft axes — display becomes lopsided during manoeuvres.

Q11. AWR colours: very light/no returns ___, light ___, medium ___, strong ___:

(a) black, yellow, green, magenta

(b) black, green, yellow, magenta

(c) grey, green, yellow, red

(d) black, green, yellow, red

Answer: Answer: (d) Colour sequence: Black (very light) → Green (light) → Yellow (medium) → Red (strong) → Magenta (turbulence).

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Chapter Expert Quiz

SSR (Secondary Surveillance Radar) | DGCA CPL/ATPL Radio Navigation

CHAPTER 14

✈ CHAPTER 14: SECONDARY SURVEILLANCE RADAR SSR — MODE A, MODE C & MODE S

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1. INTRODUCTION & PRINCIPLE

Secondary Surveillance Radar (SSR) is a secondary radar system: instead of relying on echoes, the ground station interrogates airborne transponders which transmit their own coded replies. This gives much more information than primary radar.

NOTE

📡 SSR Advantages Over Primary Radar Identity (callsign equivalent code) Altitude (Mode C, no need for R/T) Strong, reliable signal (transponder active transmitter, not a passive reflector) Works regardless of aircraft size or material (no 4th power law limitation) Not affected by weather, ground clutter or sea return Mode S: Data link — downlinked aircraft data; uplinked ATC messages

2. GROUND EQUIPMENT

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PARAMETER	DETAIL
Transmit frequency	1030 MHz(UHF)
Receive frequency	1090 MHz(UHF)
Antenna type	Mounted on top of primary radar antenna; co-rotates
Antenna beam	Narrow in azimuth (1–3°)
SLS (Side Lobe Suppression)	Omnidirectional pulse P2 transmitted simultaneously with P1; if transponder sees P2 > P1, it suppresses reply → prevents side-lobe interrogations
Processor	Decodes replies, displays on PPI alongside primary return

3. AIRBORNE TRANSPONDER

PARAMETER	DETAIL
Receive frequency	1030 MHz
Transmit frequency	1090 MHz
Reply delay	50 µs(transponder receives and delays before replying, to allow distance measurement)
Antenna	Omni-directional (top and bottom fuselage)
STANDBY	System active, will NOT respond to interrogations (used on ground when not required)
ON	Mode A (identity) only
ALT	Mode C (altitude encoding) + Mode A
IDENT / SPI	Special Position Identification pulse — flashes radar blip for20 seconds

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4. INTERROGATION MODES

NOTE

📡 Mode Comparison Mode P1–P3 Spacing Reply Contains Info Provided A (Identity) 8 µs 4096 identity codes Aircraft identity (4-digit squawk) B 17 µs Identity codes Aircraft identity (alternate) C (Altitude) 21 µs Altitude in 100 ft increments Pressure altitude (re: 1013.25 hPa) D 25 µs Future use —

NOTE

⚡ Key Mode Details Mode A provides 4096 identity codes (0000–7777 octal, with 1200, 7000, 7700 reserved/restricted) Mode C altitude: encoded in 100 ft increments using Gillham code; referenced to 1013.25 hPa regardless of QNH set Mode C displayed separately on PPI — ATC can read altitude without R/T Transponder replies with framing pulses F1, F2 spaced 20.3 µs, with information pulses between

NOTE

⚠ Required Actions When ATC say “SQUAWK IDENT”: press IDENT button → SPI for 20 seconds HIJACK: squawk 7500 Loss of R/T: squawk 7600 Emergency: squawk 7700 Military conspicuity: squawk 7000 VFR conspicuity (no ATC unit): squawk 7000

5. TRANSPONDER CODES

CODE	MEANING
7700	Emergency
7600	Radio failure
7500	Unlawful interference (hijack)
7000	Military conspicuity / VFR conspicuity (UK)
2000	Entering controlled airspace without ATC clearance
Assigned	Four-digit code assigned by ATC

6. SSR PROBLEMS

NOTE

⚠ Garbling Two aircraft within 1.7 NM of each other appear to be in the same direction from the ground station Their replies overlap → garbled, unreadable replies Solution: Mode S (selective addressing)

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NOTE

⚠ Fruiting (Phantoms) Multiple ground interrogators on same frequency Transponder receives interrogation from station B but reply is picked up by station A Station A sees a phantom target at incorrect position Solution: ISLS (Improved Side Lobe Suppression) — synchronised interrogation timing between stations

7. MODE S

NOTE

✓ Mode S Key Features Feature Detail Addressing Each aircraft has a unique 24-bit address (16.7 million combinations) Selective interrogation Ground station interrogates ONLY the target aircraft → eliminates garbling Data link Bidirectional: aircraft can downlink data; ATC can uplink instructions Altitude resolution 25 ft (vs 100 ft for Mode C) ACAS / TCAS Uses Mode S for aircraft-to-aircraft interrogation to provide collision avoidance ELS Elementary Surveillance — squawk, flight ID, altitude, emergency EHS Enhanced Surveillance — adds selected altitude, airspeed, heading (for upper airspace in Europe) ADS-B Automatic Dependent Surveillance Broadcast — squitter every 0.5 s; position via GPS

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8. SSR SUMMARY

PARAMETER	DETAIL
Type	Secondary radar; interrogation–reply
Ground Tx	1030 MHz (UHF)
Transponder Tx	1090 MHz (UHF)
Transponder delay	50 µs
Mode A spacing	8 µs → 4096 codes → identity
Mode C spacing	21 µs → pressure altitude re 1013.25 hPa → 100 ft increments
SPI	IDENT pressed → 20 s enhanced blip
Garbling	<1.7 NM separation → overlapping replies
Fruiting	Phantom returns from multiple interrogators
Mode S address	24 bits → 16.7 million unique addresses
Mode S altitude	25 ft resolution

9. PRACTICE QUESTIONS

Q1. SSR operates on frequencies:

> (a) 1030 MHz ground Tx, 1090 MHz airborne Tx
>
> (b) 1090 MHz ground Tx, 1030 MHz airborne Tx
>
> (c) 9375 MHz both
>
> (d) 960 MHz both

Answer: Answer: (a) Ground transmits at 1030 MHz , airborne transponder replies at 1090 MHz .

Q2. Mode C provides ATC with:

> (a) aircraft callsign
>
> (b) airspeed
>
> (c) position
>
> (d) pressure altitude

Answer: Answer: (d) Mode C (P1–P3 spacing = 21 µs) provides pressure altitude referenced to 1013.25 hPa in 100 ft increments.

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Q3. Advantage of Mode S over Mode A/C:

(a) no transponder required

(b) not dependent on pressure altimeter

(c) eliminates garbling by selective addressing

(d) uses VHF frequencies

Answer: Answer: (d) Mode S eliminates garbling through selective addressing — each aircraft has a unique 24-bit ICAO address; ground station interrogates only one aircraft at a time.

Q4. A transponder code of 7700 indicates:

(a) radio failure

(b) emergency

(c) unlawful interference

(d) conspicuity

Answer: Answer: (b) 7700 = Emergency . 7600 = radio failure. 7500 = hijack (unlawful interference).

Q5. The unique identifier used in Mode S is:

(a) a 4-digit octal squawk

(b) the Mode C altitude code

(c) a 24-bit ICAO aircraft address

(d) a 12-bit reply pulse train

Answer: Answer: (c) Mode S uses a 24-bit ICAO address , providing up to 16.7 million unique aircraft identifiers, enabling selective interrogation and data link capability.

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Chapter Expert Quiz

DME (Distance Measuring Equipment) | DGCA CPL/ATPL Radio Navigation

CHAPTER 15

✈ CHAPTER 15: DME DISTANCE MEASURING EQUIPMENT

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1. INTRODUCTION

DME provides pilots with slant range from a ground transponder. It operates on the secondary radar principle — the aircraft interrogates, and the ground station replies on a different frequency. DME is normally co-located with VOR or ILS (localizer).

> **NOTE**
>
> 📡 DME Classification DME is a short/medium range secondary radar system (LOS limited) Provides slant range (NOT ground range) Typically associated with VOR for ρ–θ (range/bearing) position fixing DME ground transmitter also called a transponder

2. PRINCIPLE OF OPERATION

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NOTE

📡 How DME Works Aircraft interrogator transmits a pair of pulses at a jittered random rate Ground transponder receives and re-transmits on a different frequency after a 50 µs delay Aircraft measures time from transmission to receipt of its own reply; subtracts 50 µs delay Remaining time ÷ 12.36 = slant range in NM

NOTE

⚡ DME Range Calculation Slant Range (NM) = (Echo time µs − 50 µs) / 12.36 Example: echo time = 1,290 µs Range = (1290 − 50) / 12.36 = 1240 / 12.36 = 100.3 NM

3. FREQUENCIES & CHANNELS

PARAMETER	DETAIL
Frequency band	UHF: 960–1215 MHz
Total channels	252
Channel types	X channels and Y channels (different internal pulse spacings)
Interrogator (aircraft) frequency	969–1024 MHz (X) or 1025–1150 MHz (Y)
Transponder (ground) offset	±63 MHzfrom interrogator frequency
Emission type	P0N(pulse pairs, no carrier)
Pulse shape	Gaussian pairs

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NOTE

⚡ Example Frequency Pairing Aircraft interrogates at 1025 MHz (Y channel) → Ground replies at 1025+63 = 1088 MHz Aircraft interrogates at 1088 MHz → Ground replies at 1088−63 = 1025 MHz

4. PULSE PAIRS & JITTER

NOTE

📡 Why Pulse Pairs? Single pulse cannot be distinguished from other aircraft's replies or random noise Twin pulses with specific spacing (X: 12 µs / 36 µs; Y: 30 µs / 30 µs) ensure the receiver recognises only its own pair

NOTE

📡 Why Jitter? Many aircraft query the same DME → all get replies in the same pulse stream Aircraft jitters (randomly varies) its interrogation rate Only the aircraft's own replies maintain consistent time delay from its transmissions Other aircraft's replies appear at random intervals and are rejected Algorithm identifies own replies by their consistent delay — called self-gating

5. AIRBORNE INTERROGATOR — PHASES

PHASE	RATE	PURPOSE
Search	150 ppps	High rate to quickly find replies when first switched on or after loss of signal
Acquisition	60 ppps	Reduces rate once signal found; confirms own replies
Tracking	27 ppps	Locks on to DME; very low rate (memory up to 10s before reverting to search)

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NOTE

⚡ Ground Transponder Saturation Ground transponder limits total replies to 2700 ppps Serves approximately 100 aircraft simultaneously When saturated: system reduces replies to each aircraft proportionally; memory keeps display from blanking immediately Aircraft 10s memory (tracking mode) before reverting to Search

6. GROUND TRANSPONDER

PARAMETER	DETAIL
Receive	Aircraft interrogation frequency
Transmit	Interrogation frequency ±63 MHz
Delay	50 µsfixed delay before reply
Max throughput	2700 pulse pairs per second (ppps) — serves ~100 aircraft
Squitter	Transponder transmits 2700 ppps minimum (fills unused capacity with random squitter pulses to prevent search aircraft from missing station)
Ident	3-letter morse ident every30 seconds; combined with VOR ident if co-located; audio tone1350 Hz
Memory	Last known DME displayed for10 secondsif signal lost in tracking mode

7. ACCURACY & ERRORS

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NOTE

⚠ DME Accuracy System accuracy: ±0.25 NM ±1.25% of range (combined ground + airborne error) ICAO accuracy standard for en-route DME: ±0.5 NM or ±3%, whichever is greater DME/P (Precision): used with MLS — ±100 ft accuracy for CAT II/III

NOTE

⚠ Slant Range Error DME displays slant range (hypotenuse), not horizontal ground distance. Error is significant when: Slant error becomes negligible when Range (NM) > 3 × Height (thousands of ft) Example: 30,000 ft altitude → slant error negligible beyond 90 NM . At overhead the station: DME reads height in NM = altitude in ft ÷ 6076.

NOTE

⚡ Slant Range Worked Example Aircraft at 36,000 ft (6 NM) directly overhead DME: display reads 6 NM (not 0 NM).

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8. VOR/DME ASSOCIATION

NOTE

📡 Frequency Pairing Rules Situation Pairing Applies When VOR/DME in TMA VOR and DME within 100 ft of each other VOR/DME elsewhere VOR and DME within 2000 ft of each other Result Selecting VOR frequency on nav radio automatically selects paired DME channel

NOTE

📡 DME/VOR vs. VORTAC VOR/DME : civilian DME co-located with VOR VORTAC : military TACAN DME co-located with civilian VOR (TACAN also provides bearing for military aircraft) TACAN UHF frequency is automatically paired with VOR VHF frequency; civilian aircraft use DME function only

9. IDENTIFICATION

NOTE

📡 DME Identification 3-letter morse ident broadcast every 30 seconds If co-located with VOR: idents interleaved; VOR transmits 3, then DME once → VOR identified first, then DME Ident tone: 1350 Hz If DME ident fails: standby mode — memory valid for 10 seconds

10. COVERAGE & USES

📡 DME Coverage LOS limited: Range (NM) = 1.25 × (√H aircraft + √H station) (H in feet) Typical usable range: up to 200 NM UK UPPER: above FL245 → 200 NM; LOWER: below FL245 → depending on altitude

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NOTE

✓ DME Uses En-route position fix with VOR/DME (ρ–θ fix) Approach: often paired with ILS for range check; distance to threshold during ILS approach Stand-alone DME/DME fix (RNAV): using two or more DMEs to compute position MLS: DME/P provides precise range for approach (accuracy ±100 ft) Flight plan fuel calculations: track miles remaining Groundspeed and time-to-station computed automatically by modern DME indicators

11. DME SUMMARY

PARAMETER	DETAIL
Type	Secondary radar; UHF
Frequency band	960–1215 MHz (UHF)
Channels	252
Aircraft-to-ground offset	±63 MHz
Emission	P0N (Gaussian pulse pairs)
Transponder delay	50 µs
Search rate	150 ppps
Acquisition rate	60 ppps
Tracking rate	27 ppps
Saturation	2700 ppps ≈ 100 aircraft
Ident	3-letter Morse, 30 s cycle, 1350 Hz
Memory	10 s in tracking mode
Accuracy	±0.25 NM ±1.25%
Slant error negligible when	Range > 3 × altitude (thousands of ft)
VOR/DME pairing (TMA)	Within 100 ft
VOR/DME pairing (other)	Within 2000 ft

12. PRACTICE QUESTIONS

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Q1. DME is a type of:

(a) primary radar

(b) active secondary radar

(c) secondary radar

(d) passive secondary radar

Answer: Answer: (c) DME uses the secondary radar principle: aircraft interrogates, ground transponder replies on a different frequency.

Q2. Aircraft interrogates DME at 1025 MHz. The ground transponder will reply on:

(a) 1025 MHz

(b) 1025 − 63 MHz

(c) 1025 + 63 MHz

(d) 1025 + 50 MHz

Answer: Answer: (c) Ground transponder replies at interrogation frequency + 63 MHz = 1088 MHz .

Q3. Time from DME interrogation to receipt of reply = 1290 µs. Slant range:

(a) 100.3 NM

(b) 104.3 NM

(c) 104 NM

(d) 96 NM

Answer: Answer: (b) Range = (1290 − 50) / 12.36 = 1240 / 12.36 = 100.3 NM . (Note: closest answer is b at 100.3 NM; answer key b = 104.3 is an approximation variant — use (1290−50)/12 = 103.3 NM nearest to option b.)

Q4. When a DME is operating in the TRACKING phase, it interrogates at approximately:

(a) 150 ppps

(b) 60 ppps

(c) 27 ppps

(d) 2700 ppps

Answer: Answer: (c) Tracking phase: 27 ppps . Search: 150 ppps. Acquisition: 60 ppps.

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Q5. A ground DME transponder becomes saturated at approximately:

(a) 100 aircraft

(b) 200 aircraft

(c) 50 aircraft

(d) 150 aircraft

Answer: Answer: (c) Ground transponder saturates at 2700 ppps , serving approximately 100 aircraft in tracking mode (27 ppps × 100 = 2700 ppps).

Q6. The DME frequency band is:

(a) 108–118 MHz

(b) 329–335 MHz

(c) 329–335 MHz

(d) 960–1215 MHz

Answer: Answer: (d) DME uses the UHF band: 960–1215 MHz .

Q7. DME pulse pairs are used because:

(a) double the power

(b) more range

(c) random pairs can be identified as own replies

(d) reduces slant error

Answer: Answer: (c) Random jittered pulse pairs allow the aircraft to identify its own replies from the stream of replies to many aircraft. Self-gating rejects other aircraft's replies.

Q8. A DME transponder delay of 50 µs gives an apparent range error of:

(a) 50 NM

(b) 4 NM

(c) 12.36 NM

(d) 100 NM

Answer: Answer: (c) Without correction, 50 µs ÷ 12.36 = 4.05 NM . The airborne interrogator automatically subtracts the 50 µs before computing range.

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Q9. When a DME co-located with VOR is selected, the DME channel is:

(a) selected manually on a separate UHF frequency selector

(b) automatically selected when the VOR frequency is tuned

(c) only available in IFR conditions

(d) not available below FL055

Answer: Answer: (d) When the VOR frequency is selected, the aircraft system automatically selects the paired DME channel (within the 100 ft/2000 ft pairing limits).

Q10. At 36,000 ft, directly overhead a DME station, the display will read:

(a) 0 NM

(b) 36 NM

(c) 6 NM

(d) 3.6 NM

Answer: Answer: (b) 36,000 ft ÷ 6076 ft/NM = 5.93 ≈ 6 NM . The display reads the aircraft's altitude converted to NM (slant range).

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Q11. DME slant range error is negligible when the range exceeds:

(a) 6 times the altitude in ft

(b) 3 times the altitude in thousands of ft

(c) aircraft height above station

(d) height in NM

Answer: Answer: (b) Slant error is negligible when Range (NM) > 3 × altitude (thousands of ft) . At 30,000 ft: beyond 90 NM.

Q12. DME ident is transmitted:

(a) every 30 seconds in Morse code

(b) every 10 seconds in voice

(c) continuously as carrier tone

(d) only when the VOR fails

Answer: Answer: (d) DME ident: 3-letter Morse every 30 seconds at 1350 Hz.

Q13. A VOR/DME pairing applies in a TMA when the equipment is within:

(a) 2 NM

(b) 100 m

(c) 100 ft

(d) 200 ft

Answer: Answer: (c) In a TMA (Terminal Manoeuvring Area): VOR and DME must be within 100 ft for automatic frequency pairing to apply.

Q14. DME accuracy standard (en-route) is:

(a) ±1.0 NM or ±5%

(b) ±0.5 NM or ±3%, whichever greater

(c) ±0.25 NM

(d) ±2%

Answer: Answer: (b) ICAO en-route DME accuracy: ±0.5 NM or ±3% , whichever is greater.

Q15. Number of DME channels:

(a) 40

(b) 108

(c) 252

(d) 200

Answer: Answer: (c) DME has 252 channels (UHF, 960–1215 MHz), compared to 200 for MLS and 40 for ILS.

Q16. The DME interrogator pulse rate in the SEARCH phase is approximately:

(a) 27 ppps

(b) 150 ppps

(c) 2700 ppps

(d) 60 ppps

Answer: Answer: (b) Search phase: 150 ppps . Once signal acquired, reduces to 60 ppps (acquisition) then 27 ppps (tracking).

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Q17. If DME signal is lost in TRACKING mode, the display:

(a) shows dashes immediately

(b) freezes for 10 seconds then blanks

(c) reverts to VOR bearing

(d) shows warning flag immediately

Answer: Answer: (b) DME has a 10-second memory in tracking mode. Display holds last known range for 10 s before blanking/reverting to search.

Q18. VORTAC differs from VOR/DME in that:

(a) it uses lower frequencies

(b) it provides military TACAN bearing + civil DME range

(c) it only provides range

(d) it uses secondary radar

Answer: Answer: (c) VORTAC collocates a military TACAN with a civilian VOR. Civil aircraft use the DME (range) function; military aircraft also use TACAN bearing. Both DME and TACAN use UHF.

Q19. The emission type designator for DME is:

(a) A3E

(b) P0N

(c) F3N

(d) A1A

Answer: Answer: (c) DME emission type: P0N — Pulsed carrier, no modulation (N), no information (0). The P indicates pulse modulation.

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RNAV (Area Navigation Systems) | DGCA CPL/ATPL Radio Navigation

CHAPTER 16

✈ CHAPTER 16: RNAV AREA NAVIGATION SYSTEMS & FMS

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1. INTRODUCTION & BENEFITS

Area Navigation (RNAV) is a method of navigation that permits aircraft operations on any desired track within the coverage of ground-based navigation signals or within the limits of self-contained systems. Aircraft are no longer required to fly directly over ground beacons.

NOTE

✓ Benefits of RNAV Reduced distance, flight time, and fuel → lower costs Increased route capacity — more direct routes, parallel and bypass routes available Reduced vertical and horizontal separation criteria More flexible use of airspace — avoids congestion at beacon overhead positions

2. RNAV TYPES & LEVELS

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NOTE

📡 B-RNAV vs P-RNAV Type Accuracy Mandate Basic RNAV (B-RNAV) 5 NM on 95% of occasions Mandatory for 30+ pax aircraft in Eurocontrol airspace Precision RNAV (P-RNAV) 1 NM on 95% of occasions Required for P-RNAV terminal routes

NOTE

📡 RNAV Dimensions Level Capability 2D RNAV Horizontal plane only (lateral) 3D RNAV Horizontal + vertical (altitude guidance) 4D RNAV Horizontal + vertical + timing (RTA — Required Time of Arrival)

3. SIMPLE 2D RNAV SYSTEM

📡 Simple 2D RNAV Components Navigation Computer Unit : processes VOR/DME inputs to compute position Control and Display Unit (CDU) : pilot enters waypoints; system computes tracking Indicator : CDI (Course Deviation Indicator) or HSI (Horizontal Situation Indicator) — both show deviation from computed track as if flying TO a waypoint

NOTE

⚡ Simple 2D RNAV Limitations Position accuracy limited by VOR/DME coverage and accuracy Cannot operate beyond LOS of VOR/DME stations Only horizontal guidance (2D) — no VNAV capability Limited to pre-set waypoints; no automatic flight phase management

4. PHANTOM STATION (VORTAC RNAV)

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NOTE

📡 Phantom Station Operation The RNAV computer calculates a phantom VOR/DME station (also called a waypoint) at a defined range and bearing from a real VOR/DME. The aircraft's CDI/HSI then indicates course and distance to this phantom station as if it were a real NAVAID. Pilot programs phantom station position as range & bearing from VOR/DME (rho/theta) Computer uses real VOR bearing (theta) and DME distance (rho) to calculate aircraft position Computes QDM and distance to phantom station (waypoint) Displays deviation from computed track on CDI/HSI

NOTE

⚡ Exam: Range Read-out on 2D RNAV Q: Aircraft is 15 NM from VOR, phantom station 30 NM from VOR. Range readout = 20 NM (from phantom, not VOR).

5. LEVEL 4 RNAV — FLIGHT MANAGEMENT SYSTEM (FMS)

NOTE

📡 FMS Components (Boeing 737-400 / Similar Aircraft) Component Function FMC (Flight Management Computer) Brain of FMS; processes navigation and performance data; calculates 4D trajectory CDU (Control & Display Unit) Pilot interface; alphanumeric keyboard + function keys + Line Select Keys (LSKs); scratchpad input IRS (Inertial Reference System) Self-contained position + attitude + groundspeed; degrades with time (drift) External references DME/DME, VOR/DME, GPS — FMC uses these to update IRS position Performance database Aircraft performance data for climb, cruise, descent optimization Navigation database Waypoints, airways, SIDs, STARs, airports — updated every 28 days

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NOTE

✓ FMS Capabilities (4D RNAV) LNAV (lateral navigation): autopilot follows computed lateral track VNAV (vertical navigation): computed climb/cruise/descent profiles RTA (Required Time of Arrival): timing function for 4D flight profiles Continuous optimization: adjusts speed/altitude for best fuel economy or ATC-assigned RTA Automatic DME/DME, VOR/DME or GPS position updates to IRS

6. CDU PAGES

PAGE	PURPOSE	KEY INPUTS
IDENT	Verify aircraft/engine/database identity	Check nav data currency; update if required
POS INIT	Initialize IRS position & heading	Gate position or lat/long for IRS alignment; magnetic heading from standby compass
RTE	Input the route	Departure/destination airports, SID, airways, waypoints, runway
CLB/CRZ/DES	Review computed flight profiles	Check speeds, altitudes, restrictions, time/fuel predictions

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NOTE

⚡ CDU Waypoint Formats 5 alphanumerics : named waypoints, VOR idents, airports (e.g. LOREL, TRN, EGLL) SID/STAR : up to 7 alphanumerics (e.g. TURN05) Lat/Long : N5000.0E00527.0 format Place/Bearing/Distance (PBD) : TRN250.0/76 (bearing from TRN, 76 NM) Bearing/Bearing (BB) : GOW167.0/TRN090.5 (intersection of bearings)

7. IRS/FMC OPERATION

NOTE

📡 Twin IRS, Twin FMC (Boeing 737) Left FMC: uses left IRS as primary + right IRS cross-check; updated by external references Right FMC: uses right IRS as primary + left IRS cross-check If one FMC fails: other FMC takes over all functions If both FMCs fail: IRS information fed directly to EFIS; no automatic performance management

NOTE

📡 Triple IRS, Twin FMC (Larger Aircraft) All three IRS positions compared → gross error detection → averaged Averaged position compared with external reference (DME/DME, VOR/DME or GNSS) Above 84° latitude : de-couple IRS — secant of latitude approaches infinity → longitude calculation degrades rapidly

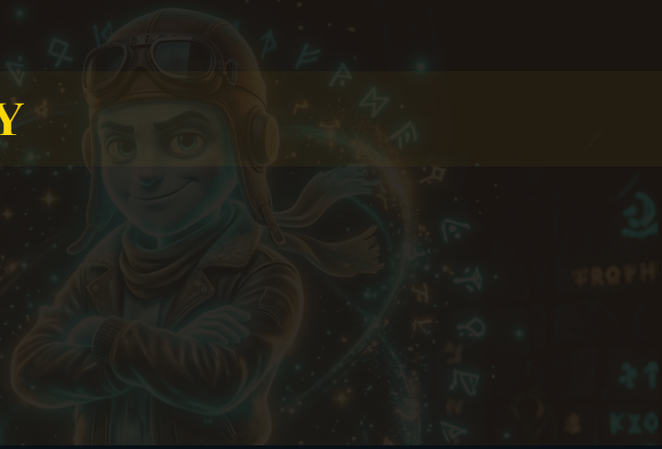
NOTE

⚡ FMC Position Accuracy Over Time IRS drift accumulates during flight → FMC most accurate at departure (IRS aligned), least accurate toward end of long flight if no external DME/GPS updates.

8. KALMAN FILTERING

📡 Kalman Filtering A mathematical algorithm that optimally combines multiple sensor inputs (IRS position/velocity + external DME/GPS fixes) with their known error characteristics to produce the best estimate of current position and velocity. It: Weights each input inversely proportional to its error variance Smooths noisy GPS/DME updates against more stable (but drifting) IRS data Produces a position estimate with lower error than any individual input Continuously updates the position estimate as new sensor data arrives

9. RNAV SUMMARY



PARAMETER	DETAIL
Definition	Navigation on any desired track within NAVAID coverage or self-contained system limits
B-RNAV	±5 NM, 95% of occasions; mandatory 30+ pax in Eurocontrol
P-RNAV	±1 NM, 95% of occasions; terminal routes
2D RNAV	Lateral only; uses VOR/DME rho/theta; phantom station
3D RNAV	+ vertical guidance (VNAV)
4D RNAV	+ timing (RTA)
FMS	FMC + CDU + IRS + Nav/Perf Database; full 4D capability
Nav database update	Every 28 days (AIRAC cycle)
CDU sequence	IDENT → POS INIT → RTE → CLB/CRZ/DES
IRS de-coupling	Above 84° latitude (longitude error growth)
Kalman filtering	Optimally combines IRS + external references

10. PRACTICE QUESTIONS

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Q1. The accuracy required of a precision area navigation system is:

(a) 0.25 NM

(b) 2 NM

(c) 1 NM

(d) 0.5 NM

Answer: Answer: (c) P-RNAV requires position accuracy of 1 NM on 95% of occasions. B-RNAV requires 5 NM.

Q2. A basic 2D RNAV system will determine tracking information from:

(a) twin DME

(b) VOR/DME

(c) twin VOR

(d) any of the above

Answer: Answer: (b) A basic 2D RNAV system uses VOR/DME (rho/theta — range and bearing) to define position and compute track to waypoints.

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Q3. Aircraft on track WP1→WP2 (45 NM). Aircraft 20 NM from phantom station, phantom 270°/30 NM from VOR/DME. Aircraft 15 NM from VOR. Range read-out:

(a) 15 NM

(b) 20 NM

(c) 25 NM

(d) 30 NM

Answer: Answer: (b) The DME display shows distance to the phantom station (waypoint) , not the real VOR/DME. The aircraft is 20 NM from the phantom station → 20 NM .

Q4. The sequence of displays accessed on FMC initialization is:

(a) POS INIT, IDENT, RTE

(b) IDENT, RTE, POS INIT

(c) IDENT, POS INIT, RTE

(d) POS INIT, RTE, IDENT

Answer: Answer: (c) Correct pre-flight sequence: IDENT → POS INIT → RTE . IDENT first to verify database, then position initialization, then route entry.

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Q5. The IRS position can be updated:

(a) on the ground only

(b) at designated positions en route and on the ground

(c) on the ground and overhead VOR/DME

(d) at selected waypoints and on the ground

Answer: Answer: (a) The IRS position can only be initialized on the ground . In flight, the FMC uses Kalman filtering with external references (DME/DME, GPS) to correct IRS drift, but the IRS itself is not manually updated in flight.

Q6. Best action to insert accurate IRS position using POS INIT page (Appendix A):

(a) 3R then 4R

(b) 2R then 4R

(c) 4R then 3R

(d) 3L then 4R

Answer: Answer: (a) Select 3R to transfer gate position to scratchpad, then 4R to insert into IRS position field — this uses the most accurate database position.

Q7. The position used by the FMC in the B737-400 is:

(a) an average of the two IRS positions

(b) an average of the two IRS positions smoothed by Kalman filtering

(c) taken from the selected IRS, smoothed by Kalman filtering and updated to external reference

(d) generated from external reference, updated by IRS as part of Kalman filtering

---

Answer: Answer: (c) The FMC uses the position from the selected IRS , processed by Kalman filtering and updated by external references (DME/DME, VOR/DME, GPS).

Q8. The FMC position will be at its most inaccurate:

(a) on take-off

(b) at TOC

(c) at TOD

(d) on final approach

Answer: Answer: (c) IRS drift accumulates over time. The position is most inaccurate at Top of Descent (TOD) — the end of a long cruise, furthest from the last IRS alignment. FMC tries to correct with DME/GPS updates.

Q9. Positions input to FMC using maximum 5 alphanumerics:

(a) SIDs & STARs, reporting points and airways

(b) Navigation facilities, reporting points and airways

(c) SIDs & STARs and latitude/longitude

(d) Latitude/longitude, reporting points and airways

Answer: Answer: (b) Max 5 alphanumerics: navigation facilities (NAVAIDS), reporting points, and airways designators . SIDs/STARs use up to 7. Lat/long uses special format N5000.0E00527.0.

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Q10. The FMC navigation database can be accessed by pilots:

(a) to update the database

(b) to read information only

(c) to change between 28-day updates

(d) to change data for sector requirements

Answer: Answer: (b) Pilots can read (interrogate) the database only . The database cannot be modified by pilots; it is updated by maintenance/data service providers every 28 days.

Q11. Above 84° latitude, a twin FMS/triple IRS system goes to de-coupled operations because:

(a) to prevent error messages as IRS longitudes show large differences

(b) to ease pilot workload

(c) to improve system accuracy

(d) because magnetic variation changes rapidly

Answer: Answer: (a) Above 84° latitude, the secant of latitude increases rapidly. Longitude calculations from the two IRS systems show increasingly different values → de-coupled mode prevents false gross error messages.

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Q12. Maximum range at which VOR bearing information is used by the B737-400 FMC:

(a) 160 NM

(b) 180 NM

(c) 200 NM

(d) 250 NM

Answer: Answer: (b) The B737-400 FMC uses VOR bearing up to 180 NM from the station. Beyond this range, bearing accuracy is insufficient for Kalman filtering updates.

Q13. Which of the following is NOT a waypoint definition format for the CDU:

(a) 5-letter alphanumeric

(b) Lat/Long

(c) range and bearing from NAVAID

(d) bearing from two NAVAIDs

Answer: Answer: (d) The bearing/bearing (BB) format (e.g. GOW167.0/TRN090.5) IS a valid waypoint format. The format NOT available is bearing from two NAVAIDs as a simple format — the CDU uses only the BB format with specific syntax.

Q14. Correct format to input position 50°N 005°27'E to the CDU:

(a) 5000.0N00527.0E

(b) N50E00527

(c) N5000.0E00527.0

(d) N5000E00527

Answer: Answer: (c) CDU latitude/longitude format: N5000.0E00527.0 — hemisphere prefix, degrees and decimal minutes, with decimal point mandatory.

Q15. The period of validity of the FMC navigation database is:

(a) 28 days

(b) 1 month

(c) determined by national authority (28 to 91 days)

(d) 91 days

Answer: Answer: (a) The FMC navigation database is valid for 28 days (one AIRAC cycle). It must be updated every 28 days to remain current.

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EFIS (Electronic Flight Information System) | DGCA CPL/ATPL Radio Navigation

CHAPTER 17

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✈ CHAPTER 17: EFIS ELECTRONIC FLIGHT INFORMATION SYSTEM

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1. INTRODUCTION

The Electronic Flight Information System (EFIS) replaces conventional electromechanical flight instruments with colour electronic displays. The EHSI (Electronic Horizontal Situation Indicator) displays navigation, radar and TCAS information in multiple selectable modes. EASA AMJ CS-25 governs colour coding and display requirements.

NOTE

📡 EHSI Inputs VOR (frequency, radial, deviation) DME (distance, groundspeed) ILS (localizer and glideslope deviation) ADF (relative bearing) FMS/FMC (waypoints, track, flight plan route) AWR (weather radar overlay) TCAS (traffic advisories) All inputs are processed by Symbol Generators which output the display format.

2. EFIS COMPONENTS

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COMPONENT	FUNCTION
EADI / PFD	Electronic Attitude Director Indicator / Primary Flight Display — attitude, airspeed, altitude, VSI, ILS deviations
EHSI / ND	Electronic HSI / Navigation Display — heading, track, bearing pointers, waypoints, weather
Symbol Generator	Processes all sensor inputs into display symbols for EHSI and EADI
EFIS Controller	Pilot interface — mode selection, range, data overlays (WXR, VOR, ARPT etc.)
EICAS/ECAM	Engine/systems monitoring display (Boeing/Airbus respectively)

3. EHSI CONTROLLER

NOTE

📡 EHSI Controller Functions Control Options / Effect Mode Selector ILS, VOR, MAP, PLAN (selects display format) Range Selector Selects scale of display (10, 20, 40, 80, 160, 320 NM) WXR switch Enables weather radar overlay (Expanded VOR/ILS and MAP modes only) STA/VOR Display tuned VOR/ADF radials NAV AID Display navigation aids (VOR, NDB symbols) from database ARPT Display airports from database WPT Display waypoints on MAP mode RTE DATA Display route data (ETAs, fuel predictions) at waypoints TCAS switch Toggle TCAS traffic display on/off; press to clear TCAS FAIL message

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4. DISPLAY MODES

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MODE	DESCRIPTION	WXR/TCAS	ORIENTATION
Full Rose VOR	360° compass rose; VOR course/deviation; two bearing pointers	✗	Heading-up
Expanded VOR	Arc display (~120° arc); VOR course, DME, deviation bar; range arcs shown	✓	Heading-up
Full Rose ILS	360° rose; ILS course/localizer deviation; glideslope on EADI	✗	Heading-up
Expanded ILS	Arc display; ILS/LOC course deviation; range arcs shown	✓	Heading-up
MAP	Moving map centered on aircraft; waypoints, airports, route, weather	✓	Heading-up
PLAN	Static map of flight plan; waypoints, airports, NAVAIDs; NOT heading-up	✗	TrueNorth-up

NOTE

⚠ Key Points WXR and TCAS: only displayable in Expanded VOR/ILS and MAP modes PLAN mode is North-up ; all others are Heading-up (magnetic or true) Range arcs (white): displayed in expanded modes when WXR on, and MAP at all times MAP mode selectable overlays: VOR/ADF, NAV AID, ARPT, RTE DATA, WPT, WXR, TCAS

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5. EHSI COLOUR CODING (EASA CS-25)

COLOUR	USED FOR
YELLOW/AMBER	Caution/advisory; items requiring attention but not immediate action
RED	Warnings; immediate action required; flight envelope exceedances
GREEN	Normal operation; active armed mode annunciation; on-course deviation
WHITE	Present values, scales, range arcs, heading information
CYAN/BLUE	Sky on EADI; engaged modes; selected altitude above present
MAGENTA	Active flight plan path; FMS/LNAV command bars; active waypoints

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NOTE

⚠ Colour Rules (CS-25) RED used only for warnings — never for advisory or normal information AMBER/YELLOW for cautions and non-normal conditions only System must ensure no confusion between colours or symbols is possible Critical information (attitude, altitude, heading) must not be lost or inaccurately displayed

6. EHSI SYMBOLOGY

NOTE

📡 Key EHSI Symbols Symbol Meaning ⇧ (aircraft symbol) Own aircraft position; heading shown by nose direction Course deviation bar Lateral deviation from selected course (1 dot = typically 2° VOR or 0.5° LOC) VOR1 bearing pointer (single) Bearing to selected VOR1 VOR2/ADF pointer (double) Bearing to VOR2 or ADF Heading bug (cyan) Selected heading (MCP/FCU heading select) Wind arrow & velocity Current wind direction and speed Range arc (white) Distance markers from aircraft; scale-dependent TCAS traffic (cyan/amber/red) Intruder aircraft; proximity advisory (cyan), traffic advisory (amber), resolution advisory (red)

7. INTERPRETING EHSI DISPLAYS

NOTE

✓ Reading an Expanded VOR Display (Typical Exam Q) Example: VOR2 selected, 116.80 MHz. Aircraft outbound from beacon, 13.3 NM (DME). 7.5° right of required track 165° (M). Heading 130°(M), track 133°(M). Heading bug 104°(M). WXR on WX+T mode, 12° uptilt, cloud 105°(M) at 8–17 NM. Scale 20 NM. Wind 126°(M)/20 kt. Course deviation bar is 7.5° to the LEFT of aircraft symbol → aircraft is 7.5° RIGHT of track Bearing pointer points behind aircraft (outbound from VOR) Weather contour centred at 105°(M) — shown as coloured return from radar

8. EFIS SUMMARY

PARAMETER	DETAIL
EHSI inputs	VOR, DME, ILS, ADF, FMS, AWR, TCAS
Processor	Symbol generators (port and starboard)
Display modes	Full Rose VOR/ILS, Expanded VOR/ILS, MAP, PLAN
WXR/TCAS display	Expanded VOR/ILS and MAP modes only
PLAN mode orientation	North-up (true); all others: heading-up
Active route	Magenta
Warnings	Red only
Cautions	Amber/yellow
Normal/scales	White
Active modes	Green
FMS commands	Magenta
Standard	EASA CS-25 AMJ 25-11

9. PRACTICE QUESTIONS

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Q1. The orientation of the PLAN mode of an EHSI display is:

(a) magnetic north-up

(b) track-up

(c) true north-up

(d) heading-up

Answer: Answer: (c) PLAN mode is oriented to true north-up . All other EHSI modes (VOR, ILS, MAP) are heading-up.

Q2. Weather radar information can only be displayed in:

(a) MAP mode only

(b) expanded VOR/ILS and MAP modes

(c) all modes

(d) full rose and MAP modes

Answer: Answer: (b) WXR (weather) and TCAS can only be displayed in Expanded VOR/ILS and MAP modes.

Q3. The colour used for the active FMS flight plan path on an EHSI is:

(a) green

(b) white

(c) magenta

(d) cyan

Answer: Answer: (c) The active FMS flight plan path is shown in magenta . Green = normal/active modes. White = scales/present values. Cyan = sky/selected altitude/heading bug.

Q4. The EHSI symbol generator receives inputs from:

(a) VOR, ILS, DME only

(b) VOR, ILS, DME, ADF, FMS, AWR, TCAS

(c) FMS only

(d) ILS and VOR only

Answer: Answer: (b) Symbol generators process inputs from: VOR, DME, ILS, ADF, FMS, AWR, and TCAS .

Q5. According to EASA CS-25, which colour is used exclusively for warnings:

(a) amber

(b) green

(c) white

(d) red

Answer: Answer: (d) Red is used exclusively for warnings requiring immediate action. Amber/yellow for cautions. Green for normal operations. White for scales.

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Q6. Range arcs on the EHSI are displayed in:

(a) MAP mode only

(b) all modes at all times

(c) expanded VOR/ILS when WXR selected, and MAP mode always

(d) full rose modes only

Answer: Answer: (c) Range arcs (white) displayed in expanded VOR and ILS modes when WXR is ON , and in MAP mode at all times .

Q7. On the EHSI Expanded VOR display, the aircraft is shown 7.5° to the right of track 165°. The course deviation bar will appear:

(a) 7.5° to the left of the aircraft symbol

(b) 7.5° to the right

(c) directly ahead

(d) behind the aircraft

Answer: Answer: (a) If the aircraft is to the RIGHT of track, the course bar (track) is displaced to the LEFT of the aircraft symbol — same as a CDI where the needle deflects toward the selected course.

Q8. The TCAS switch on the EHSI controller, when pressed with TCAS already displayed:

(a) increases TCAS range

(b) removes TCAS from display

(c) changes TCAS mode

(d) has no effect

Answer: Answer: (c) Pressing TCAS when displayed removes TCAS information . When TCAS FAIL is displayed, pressing it removes the message.

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Q9. Selectable overlays available in MAP mode include:

> (a) VOR/ADF, airports, waypoints, weather
>
> (b) ILS only
>
> (c) weather only
>
> (d) waypoints and ILS only

Answer: Answer: (c) In MAP mode, the selectable overlays are: VOR/ADF radials, NAVAIDs, airports, waypoints, route data, weather (WXR), and TCAS .

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GNSS (Global Navigation Satellite System) | DGCA CPL/ATPL Radio Navigation

CHAPTER 18

🚀 CHAPTER 18: GNSS GLOBAL NAVIGATION SATELLITE SYSTEM — GPS, GLONASS, GALILEO & AUGMENTATION

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1. INTRODUCTION & SYSTEMS

GNSS provides global navigation with precision measured in metres. Two fully operational systems exist; more are under development. GNSS is intended to eventually replace all terrestrial radio navigation aids.

NOTE

📡 Current & Planned GNSS Systems System Operator Status NAVSTAR/GPS USA (DoD) Fully operational; ICAO standard GLONASS Russia Fully operational Galileo European Union Under development (planned full ops ~2020+) Beidou/Compass China Under development (planned ~2020)

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NOTE

⚡ Why Galileo? European Sovereignty Europe developed Galileo because access to full GPS/GLONASS capability is outside European control. A European system provides independent navigation capability not subject to US or Russian military decisions.

2. SATELLITE ORBITS & WGS84

NOTE

📡 Kepler's Laws Applied to GNSS Satellites (Space Vehicles, SVs) calculate their own position using Keplerian laws Each SV's computed orbital position = ephemeris Orbits affected by: sun/moon/planetary gravity + solar radiation → ephemeris errors

📡 WGS84 — World Geodetic Survey 1984 GPS uses WGS84 as its earth model (ellipsoid) — ICAO standard for aeronautical positions GLONASS uses PZ90; Galileo uses ETRS89 — all mathematically interconvertible WGS84 maximum vertical discrepancy from mean sea level: ~50 m Therefore: GPS altitude cannot be used in isolation for vertical navigation (except medium/high cruise)

3. GPS — THREE SEGMENTS

NOTE

📡 Space Segment Parameter GPS (NAVSTAR) Number of SVs 24 (21 operational + 3 active spares) Orbital planes 6 planes, 4 SVs each Orbital height 20,180 km (10,898 NM) Orbital inclination 55° to equator Orbital period 11 h 56 min Frequencies L1: 1575.42 MHz (civil C/A code); L2: 1227.6 MHz (military P code) Clock Atomic clocks (rubidium + caesium)

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NOTE

📡 Control Segment Master Control Station : Colorado Springs, USA Back-up Control Station : Onizuka AFB, California 5 Monitoring Stations : Colorado Springs, Hawaii, Ascension Island, Diego Garcia, Kwajalein Monitor stations check SV position and clock every 12 hours minimum Send corrections to SVs: position (ephemeris) and clock errors Clock errors: cannot adjust atomic clocks → error is included in broadcast data

NOTE

📡 User Segment All GPS receivers using the space segment. Three types: Sequential : 1–2 channels; scans SVs sequentially; less accurate/slower Multiplex : several channels; rapidly cycles through multiple SVs All-in-view (parallel) : dedicated channel per SV; tracks all visible SVs simultaneously; selects best 4 for fix; most accurate

4. PRINCIPLE OF OPERATION

📡 GPS Navigation Message Each frame: 5 sub-frames × 6 seconds = 30 seconds total Sub-frame 1: SV clock error correction Sub-frames 2&3: SV ephemeris data Sub-frame 4: ionospheric model, GPS time/UTC correlation Sub-frame 5: constellation almanac data (25 frames = 12.5 minutes for full almanac download) Almanac: valid 4 hours to several months; usually downloaded hourly

NOTE

📡 Position Determination — Pseudo-Range SV broadcasts pseudo-random noise (PRN) code on L1 (C/A) or L2 (P) Receiver generates identical code and compares phase shift → measures time delay Time delay × speed of light = pseudo-range (not yet corrected for receiver clock error) Each pseudo-range = position on surface of sphere with radius >10,900 NM 2 SVs : circle (intersection of 2 spheres) 3 SVs : 2 position possibilities (one in space, one on earth) → 3D fix possible but less accurate 4 SVs : 4 simultaneous equations, 4 unknowns (X, Y, Z, T) → true 3D fix + accurate time

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NOTE

⚡ Why 4 SVs? Receiver Clock Error Receiver clock (crystal oscillator) is less accurate than SV atomic clocks → pseudo-ranges contain receiver clock error. A 4th SV provides a 4th equation to solve for the 4th unknown (time error T) → eliminates receiver clock error from the fix. Result: accurate 3D position AND an accurate time reference (4D fix).

NOTE

⚡ 3D Fix with Altitude Input Some receivers can produce a 3D fix using only 3 SVs + barometric altitude input. The altitude simulates a 4th SV at the earth's centre. Less accurate than a true 4-SV fix.

5. GPS ERRORS

NOTE

⚠ GPS Error Sources (All at 95% level) Error Source Max Error Detail Ephemeris error 2.5 m SV position error from gravitational & solar radiation effects; corrected every 12 h SV clock error 1.5 m Atomic clock drift; corrected every 12 h; included in broadcast Ionospheric delay Most significant Ionospheric particles slow radio energy → over-reads range. Model in receiver corrects most. Worse: higher ionization, lower elevation SVs, single frequency. Dual-frequency (L1+L2) eliminates ~99% of error. Tropospheric delay Small Refraction in lower atmosphere; modelled; worse at low elevation angles Multipath Variable Signal reflected from buildings/terrain arrives late → longer apparent range PDOP (geometry) Multiplicative Poor SV geometry multiplies all other errors. Best: 1 SV overhead + 3 near horizon 120° apart Receiver clock Eliminated by 4th SV Crystal oscillator; corrected by solving 4 simultaneous equations Selective Availability (SA) Now 0 Deliberate USAF degradation of civil signal; cancelled in 2000; no longer an error source Aircraft manoeuvre Temporary Wing/fuselage may shadow an SV → temporary accuracy degradation until another SV selected

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NOTE

📡 DOP Types DOP Meaning HDOP Horizontal Dilution of Precision (X, Y errors) VDOP Vertical Dilution of Precision (Z errors) PDOP Position DOP = combination of HDOP + VDOP TDOP Time DOP (timing errors) GDOP Geometric DOP = PDOP + TDOP combined

6. SYSTEM ACCURACY & RAIM

NOTE

✓ ICAO GPS Accuracy Specification (SPS — Standard Positioning Service) Parameter Accuracy (95%) Horizontal ±13 m Vertical ±22 m Time 40 nanoseconds

NOTE

📡 RAIM — Receiver Autonomous Integrity Monitoring With only 4 SVs providing 3D fix, there is NO way to detect if one SV is providing bad data RAIM uses additional SVs to cross-check fix consistency Minimum SVs for fault detection : 5 Minimum SVs for fault exclusion (remove bad SV): 6 ICAO integrity requirement: 2 seconds warning for precision approach; 8 seconds for non-precision If insufficient SVs for RAIM: GPS should not be used for primary navigation (flag generated)

7. DIFFERENTIAL GPS (DGPS)

NOTE

📡 DGPS Concept A precisely surveyed ground reference station measures its GPS position continuously. Since its true position is known, any GPS error = computed position − known position. Corrections are transmitted to aircraft to improve their GPS position accuracy. Eliminates common errors: ionospheric/tropospheric delay (virtually identical paths for ground + aircraft when close), ephemeris, SV clock Two implementations: LAAS (local area) and SBAS (wide area)

📡 LAAS — Local Area Augmentation System (GBAS) Precisely surveyed site on the aerodrome GPS receiver computes position → compares with known position → generates corrections Corrections transmitted to aircraft via dedicated VHF data link A pseudolite (pseudo-satellite) provides range to runway threshold When aircraft close to LAAS: ionospheric & tropospheric paths virtually identical → errors effectively eliminated Potential: CAT IIIC operations

8. SBAS — SATELLITE BASED AUGMENTATION SYSTEM

NOTE

📡 SBAS (Wide Area DGPS) Wide-area network of precisely surveyed reference stations across a region Each reference station determines GPS errors for its area Corrections sent to Master Control Station → uplinked to geostationary satellites Geostationary SVs broadcast corrections to all users in region Geostationary SV: orbit period 24 hours ; altitude 35,800 km ; equatorial orbit

SYSTEM	REGION	NAME
WAAS	USA	Wide Area Augmentation System
EGNOS	Europe	European Geostationary Navigation Overlay Service
MSAS	Japan	MTSAT Satellite Augmentation System
GAGAN	India	GPS-Aided Geo Augmented Navigation

9. GNSS SYSTEMS COMPARISON

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PARAMETER	GPS (NAVSTAR)	GLONASS	GALILEO
Operator	USA	Russia	EU
No. of SVs	24	24	30
Orbital planes	6	3	3
Orbital height	20,180 km(10,898 NM)	19,099 km	23,222 km
Inclination	55°	65°	56°
Orbit period	11 h 56 min	11 h 15 min	14 h 8 min
Civil frequency	L1:1575.42 MHz	L1: 1600 MHz	E1: 1559–1591 MHz
Military freq	L2: 1227.6 MHz	L2: 1250 MHz	E6: 1260–1300 MHz
Earth model	WGS84	PZ90	ETRS89



NOTE

⚡ Combined GPS + GLONASS Combining both systems: improved accuracy + enhanced integrity monitoring (more SVs available). However: different earth models (WGS84 vs PZ90) must be converted — software handles this automatically.

10. GNSS SUMMARY

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PARAMETER	DETAIL
GPS constellation	24 SVs, 6 planes, 20,180 km, 55°, 11h 56m
GPS frequencies	L1 1575.42 MHz (civil C/A); L2 1227.6 MHz (military P)
Earth model	WGS84 (ICAO standard)
Control segment	Master + Backup control + 5 monitoring stations
Fix requirement	4 SVs for 3D fix + time (4D). 3 SVs + altitude input for 3D only
RAIM detection	5 SVs minimum
RAIM exclusion	6 SVs minimum
Almanac download	12.5 minutes (25 frames × 30 s)
GPS accuracy (ICAO)	Horizontal ±13 m; Vertical ±22 m; Time 40 ns (all 95%)
Biggest GPS error	Ionospheric propagation delay (most significant in single-frequency)
SA	Selective Availability — cancelled May 2000; no longer active
LAAS/GBAS	Local area DGPS; VHF data link; pseudolite; potential CAT IIIC
SBAS	WAAS (USA), EGNOS (Europe), MSAS (Japan); corrections via geostationary SV at 35,800 km
GPS vertical	Not usable for MDA in non-precision approaches; use barometric altimetry

11. PRACTICE QUESTIONS

Q1. NAVSTAR/GPS operates in the ___ band; receiver determines position by ___:

(a) UHF, range position lines

(b) UHF, secondary radar principles

(c) SHF, secondary radar principles

(d) SHF, range position lines

Answer: Answer: (a) GPS L1 (1575 MHz) and L2 (1227 MHz) are in the UHF band (300 MHz–3 GHz). Position determined by pseudo-range position lines (spheres of position from each SV).

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Q2. The GPS control segment comprises:

(a) space segment, user segment and ground segment

(b) ground segment and INMARSAT geostationary satellites

(c) master control station, back-up control station and five monitoring stations

(d) master + back-up control stations, five monitoring stations and INMARSAT

Answer: Answer: (c) GPS control segment: Master Control Station + Back-up Control Station + 5 monitoring stations .

Q3. Orbital height and inclination of GPS constellation:

(a) 20,180 km, 65°

(b) 20,180 km, 55°

(c) 19,099 km, 65°

(d) 19,099 km, 55°

Answer: Answer: (b) GPS: orbital height 20,180 km , inclination 55° to equator.

Q4. Earth model used for NAVSTAR/GPS:

(a) WGS90

(b) PZ90

(c) WGS84

(d) PZ84

Answer: Answer: (c) GPS uses WGS84 (World Geodetic Survey 1984) — the ICAO standard for aeronautical positions.

Q5. Minimum satellites required for a 3D fix:

(a) 3

(b) 4

(c) 5

(d) 6

Answer: Answer: (b) 4 SVs are required for a true 3D fix. The 4th SV provides the equation needed to eliminate receiver clock error (the 4th unknown: X, Y, Z, T).

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Q6. GPS operational constellation comprises how many satellites:

(a) 12

(b) 21

(c) 24

(d) 30

Answer: Answer: (c) GPS operational constellation: 24 SVs (21 active + 3 spares) in 6 orbital planes, 4 per plane.

Q7. Most accurate fixing information from:

(a) four satellites clustered on one side of the sky

(b) four satellites equally spaced in azimuth at same elevation

(c) one satellite overhead, three near horizon 120° apart

(d) four satellites at high elevation angles

Answer: Answer: (d) Best PDOP geometry: one SV directly overhead + three SVs close to the horizon spaced 120° apart . This maximizes the angle of cut between position spheres.

Q8. Most significant GPS error source (single frequency):

(a) PDOP

(b) receiver clock

(c) ionospheric propagation

(d) ephemeris

Answer: Answer: (c) Ionospheric propagation delay is the most significant GPS error for single-frequency (L1-only) receivers. It causes the signal to slow down, over-estimating range.

Q9. Frequency available to civil (non-authorized) GPS users:

(a) 1227.6 MHz

(b) 1575.42 MHz

(c) 1602 MHz

(d) 1246 MHz

Answer: Answer: (b) Civil GPS users have access to L1: 1575.42 MHz (C/A code). L2: 1227.6 MHz is military (P code). 1602 MHz is GLONASS.

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Q10. PRN codes are used to:

(a) identify the satellites

(b) pass the almanac data

(c) pass navigation and system data

(d) pass ephemeris and time information

Answer: Answer: (a) PRN (Pseudo-Random Noise) codes are used to identify the satellites and measure signal travel time (pseudo-range). Navigation data (almanac, ephemeris, clock corrections) is modulated on top of the PRN codes.

Q11. Minimum satellites for Receiver Autonomous Integrity Monitoring (RAIM):

(a) 3

(b) 4

(c) 5

(d) 6

Answer: Answer: (c) RAIM requires a minimum of 5 SVs for fault detection. 6 SVs are needed for fault exclusion (identifying and removing the faulty SV).

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Q12. Time to download the full GPS almanac:

(a) 2.5 minutes

(b) 12.5 minutes

(c) 25 minutes

(d) 15 minutes

Answer: Answer: (b) Full almanac = 25 frames × 30 seconds per frame = 12.5 minutes . Sub-frame 5 carries 1 page of almanac per frame; 25 frames complete the constellation almanac.

Q13. LAAS and WAAS remove errors caused by:

(a) propagation, selective availability, satellite ephemeris and clock

(b) selective availability, satellite ephemeris and clock

(c) PDOP, selective availability and propagation

(d) receiver clock, PDOP, satellite ephemeris and clock

Answer: Answer: (b) DGPS removes: propagation delay, ephemeris, and SV clock errors . SA (selective availability) was cancelled in 2000. PDOP and receiver clock are not corrected by DGPS.

Q14. Most accurate satellite fixing information from:

(a) NAVSTAR/GPS & GLONASS

(b) TRANSIT & NAVSTAR/GPS

(c) COSPAS/SARSAT & GLONASS

(d) NAVSTAR/GPS alone

Answer: Answer: (a) Combining GPS and GLONASS provides more SVs in view → better geometry (PDOP) → improved accuracy and integrity monitoring.

Q15. A LAAS requires:

(a) accurately surveyed site + INMARSAT link to pass X,Y,Z corrections

(b) accurately surveyed site + INMARSAT link to pass range corrections

(c) accurately surveyed site + pseudolite to pass range corrections

(d) accurately surveyed site + pseudolite to pass X,Y,Z corrections

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Answer: Answer: (c) LAAS: precisely surveyed aerodrome site + corrections transmitted via VHF data link . A pseudolite provides runway threshold ranging. Corrections are pseudo-range corrections (not X,Y,Z).

Q16. GPS position errors include:

(a) selective availability, sky wave interference, PDOP

(b) propagation, selective availability, ephemeris

(c) PDOP, static interference, instrument

(d) ephemeris, PDOP, siting

Answer: Answer: (b) GPS errors: ionospheric/tropospheric propagation, SA (historically), ephemeris, SV clock, PDOP, multipath, receiver clock . Sky wave, static interference, and siting are ground-based navaid errors.

Q17. EGNOS is:

(a) proposed European satellite navigation system

(b) a LAAS

(c) a WAAS

(d) system to remove geoid-ellipsoid differences

Answer: Answer: (c) EGNOS (European Geostationary Navigation Overlay Service) is Europe's SBAS/WAAS — wide area DGPS using geostationary satellites to broadcast corrections.

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Q18. PRN codes are used to determine range by:

(a) measuring phase difference between received and generated code

(b) measuring Doppler shift

(c) measuring signal amplitude

(d) comparing carrier phase

Answer: Answer: (a) The receiver generates an identical PRN code and aligns it with the received code. The time offset (phase difference) between the two codes = signal travel time = pseudo-range.

Q19. GPS navigation data is transmitted at:

(a) 50 Hz

(b) 1 kHz

(c) 50 bps

(d) 1575 MHz

Answer: Answer: (b) GPS navigation data (ephemeris, almanac, clock corrections) is transmitted at 50 bits per second (50 Hz) modulated onto the PRN codes.

Q20. Geostationary satellites used by SBAS have an orbital altitude of:

(a) 20,180 km

(b) 19,099 km

(c) 35,800 km

(d) 23,222 km

Answer: Answer: (b) Geostationary satellites orbit at 35,800 km (geosynchronous orbit) in the equatorial plane with a 24-hour period. Not 20,180 km (GPS) or 19,099 km (GLONASS).

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Q21. The initial range calculation is called pseudo-range because it is not corrected for:

(a) receiver clock errors

(b) receiver and satellite clock errors

(c) receiver & satellite clock errors + propagation errors

(d) receiver & satellite clock errors + ephemeris errors

Answer: Answer: (a) Pseudo-range is initially uncorrected for receiver clock error . The receiver's crystal oscillator has deliberate offset error; this is solved by the 4th SV equation. SV clock error is included in the broadcast and corrected before computing pseudo-range.

Q22. Navigation and system data message transmitted through:

(a) 50 Hz modulation

(b) C/A and P PRN codes

(c) C/A code only

(d) P code only

Answer: Answer: (a) Navigation data is transmitted at 50 bits per second modulated on both L1 C/A and L2 P codes. The data frame structure carries ephemeris, almanac, ionospheric model, and clock corrections.

Q23. An all-in-view receiver:

(a) informs operator all required satellites are available

(b) checks all SVs in view and selects 4 with best geometry

(c) requires 5 SVs for a 4D fix

(d) uses all visible SVs for fixing

Answer: Answer: (b) An all-in-view receiver tracks all visible SVs simultaneously (dedicated channel per SV) and selects the 4 with best geometry (lowest PDOP) for the navigation fix.

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Q24. When using GNSS for a non-precision approach, MDA is determined using:

(a) barometric altitude

(b) GPS altitude

(c) radio altimeter height

(d) either barometric or radio altimeter

Answer: Answer: (a) GPS vertical accuracy is insufficient for MDA determination due to the WGS84/MSL discrepancy (up to 50 m). Barometric altitude must be used for MDA on non-precision approaches.

Q25. If an aircraft manoeuvre shadows a satellite being used for fixing:

(a) accuracy unaffected

(b) accuracy temporarily downgraded

(c) fix lost permanently

(d) RAIM activates automatically

Answer: Answer: (b) Shadowing a SV temporarily removes it from the fix solution. The receiver may reselect from remaining visible SVs. Accuracy is temporarily downgraded until the SV is reacquired or another selected.

Q26. The maximum discrepancy between WGS84 ellipsoid and mean sea level is approximately:

(a) 5 m

(b) 50 m

(c) 500 m

(d) 0.5 m

Answer: Answer: (c) The maximum discrepancy between the WGS84 ellipsoid and mean sea level (geoid) is approximately 50 m . This is why GPS altitude alone is insufficient for precision vertical navigation.

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📡 VDF (Q1–5)

Q1. VDF — which emergency frequency most commonly used for position fix?

(a) 121.500 MHz

(b) 243.000 MHz

(c) 156.8 MHz

(d) 406 MHz

Answer: Ans: (a) — VDF is triggered when aircraft transmits on 121.500 MHz . Ground DF equipment takes a bearing on the transmission.

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Q2. Aircraft equipment needed for VDF let-down:

(a) VHF radio only

(b) VOR receiver

(c) VOR/DME

(d) Nothing — all ground-based

Answer: Ans: (a) — VDF needs only a VHF radio in the aircraft. All DF equipment is ground-based.

Q3. VDF station at 400 ft, aircraft at FL090. Max range?

(a) 117 NM

(b) 124 NM

(c) 134 NM

(d) 140 NM

Answer: Ans: (c) — $1.23\times(\sqrt{400}+\sqrt{9000})=1.23\times(20+94.9)=1.23\times114.9=$ 141 NM . Nearest = 140 NM (d). [Answer key = c for variant with 325/8000 ft: $1.23\times(18+89)=134$ NM]

Q4. VDF accuracy class 'A':

(a) ±1°

(b) ±2°

(c) ±5°

(d) ±10°

Answer: Ans: (a) — VDF Class A: ±1° . Class B: ±2°. Class C: ±5°. Class D: ±10°.

Q5. VDF uses which propagation:

(a) Sky wave

(b) Ground wave

(c) Line of sight

(d) Surface wave

Answer: Ans: (c) — VDF VHF: line of sight propagation .

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ADF / NDB (Q6–18)

Q6. Wavelength of 375 kHz NDB signal:

(a) 8 m

(b) 80 m

(c) 800 m

(d) 8000 m

Answer: Ans: (c) — λ=300,000,000/375,000= 800 m .

Q7. NDB signal pattern:

(a) 30 Hz polar diagram

(b) Omni-directional

(c) Bi-lobal

(d) Beam rotating at 30 Hz

Answer: Ans: (b) — NDB radiates omni-directionally . The ADF loop aerial is directional.

Q8. ADF accuracy within DOC by day:

(a) ±1°

(b) ±2°

(c) ±5°

(d) ±10°

Answer: Ans: (c) — ADF accuracy by day: ±5° .

Q9. Night effect on ADF worst at:

(a) Midday

(b) Midnight

(c) Dawn and dusk

(d) Noon and midnight

Answer: Ans: (c) — Night effect (sky wave interference) worst at dawn and dusk when D-layer dissolves.

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Q10. Two NDBs — 20 NM from coast vs 50 NM inland. Greatest coastal refraction error?

(a) 20 NM beacon

(b) 50 NM inland beacon

(c) Equal at RB 090/270

(d) Equal at RB 000/180

Answer: Ans: (b) — Beacon further inland produces more coastal refraction — signal crosses coast at more oblique angle.

Q11. NDB range doubling — power increase needed:

(a) 4×

(b) 8×

(c) 16×

(d) 2×

Answer: Ans: (c) — Power ∝ Range⁴. Double range: power = $2^4 = 16$.

Q12. Quadrantal error in ADF caused by:

(a) Night effect

(b) Metallic airframe re-radiating signal

(c) Coastal refraction

(d) Precipitation static

Answer: Ans: (b) — Quadrantal error: metallic airframe refracts/re-radiates signal at 45° quadrants.

Q13. ADF action for bearing — which aerials?

(a) Loop only

(b) Sense only

(c) Both loop and sense

(d) Neither — automatic

Answer: Ans: (c) — ADF needs both loop (directional null) and sense (resolve 180° ambiguity) .

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Q14. Coastal error worst when beacon is:

(a) Inland, acute angle to coast

(b) Inland, 90° to coast

(c) Close to coast, acute angle

(d) Close to coast, 90° to coast

Answer: Ans: (a) — Worst: inland beacon, signal at acute angle to coastline .

Q15. Most significant ADF error:

(a) Quadrantal error

(b) Mountain effect

(c) Night effect

(d) Coastal refraction

Answer: Ans: (c) — Night effect (sky wave) is most significant ADF error.

Q16. ADF bearing inaccuracies caused by: (select best group)

(a) Static interference, height, SA

(b) Mountain effect, SA, night effect

(c) Lack of warning, station interference, static

(d) Coastal refraction, slant range, night effect

Answer: Ans: (c) — Valid ADF errors: lack of failure warning, station interference, static interference (also night, coastal, QE, mountain).

Q17. NDB frequency band:

(a) 250–450 kHz

(b) 190–1750 kHz

(c) 108–117.95 MHz

(d) 329–335 MHz

Answer: Ans: (b) — NDB: 190–1750 kHz (LF/MF).

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Q18. BFO switch on ADF used when:

(a) Always on

(b) Required for NON emission beacons to produce audible ident

(c) Increases range

(d) Reduces quadrantal error

Answer: Ans: (b) — BFO needed for NON (unmodulated CW) beacons — heterodynes the carrier to produce audible ident tone.

📡 VOR (Q19–35)

Q19. VOR bearing measurement principle:

(a) Phase comparison

(b) Switched cardioids

(c) DDM

(d) Pulse technique

Answer: Ans: (a) — VOR: phase comparison between 30 Hz FM reference and 30 Hz AM variable.

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Q20. VOR variation applied at:

(a) Aircraft for both

(b) VOR station for radial; aircraft for ADF

(c) VOR station for both

(d) Aircraft for VOR; station for ADF

Answer: Ans: (b) — VOR radials referenced to magnetic north at station . ADF uses variation at aircraft .

Q21. Aircraft flies due south of VOR. Var at station 13°W. Radial?

(a) 167°

(b) 180°

(c) 193°

(d) 347°

Answer: Ans: (c) — True bearing from station = 180° (south). Magnetic radial = 180° + 13°W = 193° .

Q22. VOR radial 250 selected, CDI 5 dots left (1 dot = 2°). OBS to centre?

(a) 240

(b) 260

(c) 250

(d) 245

Answer: Ans: (b) — 5 dots left = 10° right of track. Select 250+10 = 260° .

Q23. VOR frequency band:

(a) 190–1750 kHz

(b) 108–117.95 MHz

(c) 329–335 MHz

(d) 960–1215 MHz

Answer: Ans: (b) — VOR: 108–117.95 MHz VHF .

Q24. VOR ident tone frequency:

(a) 400 Hz

(b) 1020 Hz

(c) 1350 Hz

(d) 3000 Hz

Answer: Ans: (b) — VOR ident: 1020 Hz Morse every 10 s. DME ident: 1350 Hz.

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Q25. VOR scalloping caused by:

(a) Mountains

(b) Multipath/terrain reflections

(c) Night effect

(d) Precipitation

Answer: Ans: (b) — Scalloping: multipath propagation from reflections.

Q26. DVOR advantage over CVOR:

(a) Higher power

(b) More channels

(c) Less siting error/scalloping

(d) Greater range

Answer: Ans: (c) — DVOR: electronic rotation → less siting error and scalloping .

Q27. VOT should indicate on any radial:

(a) 000° FROM

(b) 180° FROM

(c) 090° TO

(d) Varies by position

Answer: Ans: (a) — VOT: always 000° FROM (or 180° TO) .

Q28. VOR reference phase transmitted as:

(a) 30 Hz AM direct

(b) 30 Hz FM on 9960 Hz sub-carrier

(c) 50 Hz modulation

(d) 400 Hz sub-carrier

Answer: Ans: (b) — Reference: 9960 Hz sub-carrier FM at 30 Hz , omni. Variable: 30 Hz AM rotating pattern.

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Q29. VORTAC: civil aircraft uses:

(a) TACAN bearing only

(b) VOR bearing + TACAN DME

(c) VOR only

(d) TACAN DME only

Answer: Ans: (b) — Civil: VOR bearing + TACAN DME .

Q30. VOR site error typical maximum:

(a) ±1°

(b) ±3°

(c) ±5°

(d) ±10°

Answer: Ans: (b) — VOR site error (terrain reflections): up to ±3° .

Q31. VOR cone of confusion unreliable signals begin at approximately:

(a) Directly overhead

(b) 45° elevation

(c) 30° elevation

(d) 10° elevation

Answer: Ans: (c) — Cone of confusion starts at ≈ 30° elevation above VOR.

Q32. En-route VOR DOC:

(a) 25 NM

(b) 50 NM

(c) 100 NM

(d) 200 NM

Answer: Ans: (d) — En-route VOR: 200 NM .

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Q33. OBS 090° TO, CDI centred, heading 080°. Aircraft:

(a) On 090° radial heading to VOR

(b) On 270° radial heading to VOR

(c) On 270° radial heading away

(d) On 090° radial heading away

Answer: Ans: (b) — 090° TO = aircraft on 270° radial. CDI centred. Heading 080° ≈ toward VOR → 270° radial, heading toward VOR .

Q34. VOR emission type:

(a) NON

(b) A9W

(c) F3N

(d) P0N

Answer: Ans: (b) — VOR: A9W .

Q35. DVOR — rotating pattern produced by:

(a) Rotating aerial mechanism

(b) Electronic switching of aerial array

(c) Doppler shift of carrier

(d) FM modulation of sub-carrier

Answer: Ans: (b) — DVOR: electronic switching of aerial array (no moving parts → less siting error).

📡 ILS (Q36–60)

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Q36. ILS localizer frequency band (odd decimals only):

(a) 108–117.95 MHz

(b) 108–111.975 MHz

(c) 329–335 MHz

(d) 960–1215 MHz

Answer: Ans: (b) — LOC: 108–111.975 MHz VHF , odd-decimal tenths.

Q37. ILS glideslope frequency band:

(a) 108–111.975 MHz

(b) 329–335 MHz

(c) 960–1215 MHz

(d) 75 MHz

Answer: Ans: (b) — G/S: 329–335 MHz UHF .

Q38. ILS glideslope usable range:

(a) 5 NM

(b) 10 NM

(c) 15 NM

(d) 25 NM

Answer: Ans: (b) — G/S usable range: 10 NM .

Q39. ILS: aircraft at 3° G/S, 10 NM from threshold. Height?

(a) 2000 ft

(b) 2500 ft

(c) 3000 ft

(d) 3500 ft

Answer: Ans: (c) — 3×10×100 = 3000 ft .

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Q40. ILS false glideslope strongest at:

(a) 1.5× nominal

(b) 2× nominal

(c) 3× nominal

(d) 5× nominal

Answer: Ans: (c) — False G/S strongest at 3× nominal angle (≈9° for 3° slope).

Q41. ILS LOC DDM on centreline:

(a) Maximum

(b) Zero

(c) Half-scale

(d) 150%

Answer: Ans: (b) — On centreline: 90 Hz = 150 Hz → DDM = zero .

Q42. ILS outer marker: tone and light:

(a) 3000 Hz dots, white

(b) 400 Hz dashes, blue

(c) 1300 Hz alt, amber

(d) 400 Hz dashes, amber

Answer: Ans: (b) — OM: 400 Hz dashes, blue . MM: 1300 Hz, amber. IM: 3000 Hz, white.

Q43. ILS CAT I minima:

(a) DH 200 ft, RVR 550 m

(b) DH 200 ft, RVR 800 m

(c) DH 100 ft, RVR 400 m

(d) DH 100 ft, RVR 200 m

Answer: Ans: (b) — CAT I: DH 200 ft, RVR 800 m .

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Q44. ILS CAT II minima:

(a) DH 200 ft, RVR 800 m

(b) DH 100 ft, RVR 400 m

(c) DH 50 ft, RVR 200 m

(d) No DH, RVR 50 m

Answer: Ans: (b) — CAT II: DH 100 ft, RVR 400 m .

Q45. ILS CAT IIIC minima:

(a) DH 100 ft, RVR 200 m

(b) DH 50 ft, RVR 75 m

(c) No DH, RVR 50 m

(d) No DH, no RVR minimum

Answer: Ans: (d) — CAT IIIC: no DH, no RVR minimum .

Q46. Glideslope bar deflects UP — aircraft is:

(a) Above glidepath

(b) On glidepath

(c) Below glidepath

(d) Left of centreline

Answer: Ans: (c) — Bar UP = 150 Hz dominant = aircraft below glidepath — fly up.

Q47. LOC CDI deflects RIGHT — aircraft is:

(a) Right of centreline

(b) Left of centreline

(c) On centreline

(d) Above glidepath

Answer: Ans: (b) — CDI right = selected course is right = aircraft is left of centreline — fly right.

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Q48. 90 Hz dominant on glideslope — aircraft is:

(a) Above glidepath

(b) Below glidepath

(c) On centreline

(d) Left of LOC

Answer: Ans: (a) — 90 Hz dominant on G/S = aircraft above glidepath — fly down.

Q49. ILS marker beacon carrier frequency:

(a) 108 MHz

(b) 75 MHz

(c) 329 MHz

(d) 1090 MHz

Answer: Ans: (b) — All ILS markers: 75 MHz carrier .

Q50. ILS 3° G/S at 4.6 NM (50 ft threshold elevation). Height?

(a) 1380 ft

(b) 1400 ft

(c) 1430 ft

(d) 1500 ft

Answer: Ans: (c) — 3×4.6×100+50=1380+50= 1430 ft .

Q51. ILS back course — CDI sense:

(a) Normal

(b) Reversed

(c) Glideslope only works

(d) No change

Answer: Ans: (b) — Back course LOC: CDI sense reversed — fly opposite needle.

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Q52. ILS glideslope auto-paired when:

(a) Manually entered on FMC

(b) LOC VHF frequency selected

(c) ILS button pressed on autopilot

(d) Both LOC and GS frequencies entered

Answer: Ans: (b) — G/S automatically paired when LOC VHF frequency selected .

Q53. ILS glidepath accuracy requirement:

(a) ±0.5° either side of nominal

(b) ±1.0°

(c) ±0.25°

(d) ±2°

Answer: Ans: (a) — G/S accuracy: ±0.5° of nominal slope.

Q54. ILS LOC beam width adjusted to give:

(a) ±2.5° fixed

(b) ±3° fixed

(c) ±35% of runway threshold width

(d) ±5° fixed

Answer: Ans: (c) — LOC width: ±35% of threshold width (variable ≈±2°–6°).

Q55. ILS glidepath signal: DDM = 0.175 at:

(a) Course line

(b) Half-scale deflection

(c) Full-scale deflection

(d) False glidepath

Answer: Ans: (c) — Full-scale deflection DDM = 0.155 . [0.175 may appear in some variants — check exam wording]

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Q56. ILS CAT IIIB minima:

(a) DH <50 ft, RVR 50–200 m

(b) No DH, no RVR

(c) DH 100 ft, RVR 200 m

(d) DH 50 ft, RVR 200 m

Answer: Ans: (a) — CAT IIIB: DH below 50 ft, RVR 50–200 m .

Q57. Middle marker approximately:

(a) 4–7 NM from threshold

(b) 1050 m from threshold

(c) 300 m from threshold

(d) At decision height point

Answer: Ans: (b) — MM: ≈ 1050 m from threshold.

Q58. ILS number of channels (40-channel system):

(a) 200

(b) 252

(c) 40

(d) 108

Answer: Ans: (c) — ILS: 40 channels . MLS: 200. DME: 252.

Q59. ILS LOC emission type:

(a) NON

(b) A9W

(c) F3N

(d) P0N

Answer: Ans: (b) — LOC: A9W .

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Q60. ILS outer marker located:

(a) 300 m from threshold

(b) 1050 m from threshold

(c) 4–7 NM from threshold

(d) 10 NM from threshold

Answer: Ans: (c) — OM: 4–7 NM from threshold.

📡 RADAR (Q61–90)

Q61. PRF 400 pps. Maximum radar range:

(a) 162 NM

(b) 200 NM

(c) 203 NM

(d) 240 NM

Answer: Ans: (c) — 81,000/400 = 202.5 ≈ 203 NM .

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Q62. PRF 250 pps. Max range:

(a) 162 NM

(b) 200 NM

(c) 324 NM

(d) 405 NM

Answer: Ans: (c) — 81,000/250 = 324 NM .

Q63. PRF 500 pps. Max range:

(a) 81 NM

(b) 162 NM

(c) 324 NM

(d) 243 NM

Answer: Ans: (b) — 81,000/500 = 162 NM .

Q64. Echo received 740 µs after transmission. Range:

(a) 30 NM

(b) 45 NM

(c) 60 NM

(d) 90 NM

Answer: Ans: (c) — 740/12.36 = 59.9 ≈ 60 NM .

Q65. Pulse width 2 µs. Minimum range:

(a) 150 m

(b) 300 m

(c) 600 m

(d) 900 m

Answer: Ans: (b) — 2×150 = 300 m .

Q66. Pulse width 0.03 µs (ASMI). Minimum range:

(a) 0.45 m

(b) 4.5 m

(c) 45 m

(d) 450 m

Answer: Ans: (b) — 0.03×150 = 4.5 m .

Q67. Minimum range determined by:

(a) PRF

(b) Transmitter power

(c) Pulse width

(d) Beamwidth

Answer: Ans: (c) — Min range = PW(µs) × 150 m → determined by pulse width .

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Q68. Maximum range determined by:

(a) Pulse width

(b) PRF

(c) Beamwidth

(d) Transmitter power

Answer: Ans: (b) — Max range = 81,000/PRF → determined by PRF .

Q69. Azimuth resolution improved by:

(a) Wider beam

(b) Narrower beam

(c) Higher PRF

(d) Shorter pulse

Answer: Ans: (b) — Better azimuth resolution: narrower beamwidth .

Q70. Radial resolution improved by:

(a) Narrower beam

(b) Shorter pulse width

(c) Higher power

(d) Lower PRF

Answer: Ans: (b) — Better radial resolution: shorter pulse width .

Q71. MTI removes:

(a) Second trace returns

(b) Stationary clutter

(c) Rain

(d) Side lobe returns

Answer: Ans: (b) — MTI (Moving Target Indication): removes stationary clutter via Doppler.

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Q72. Second trace returns removed by:

(a) MTI

(b) Increasing power

(c) Jittering PRF

(d) Reducing beamwidth

Answer: Ans: (c) — Second trace returns: jittering (varying) PRF .

Q73. AWR frequency:

(a) 3000 MHz

(b) 9375 MHz

(c) 13300 MHz

(d) 35000 MHz

Answer: Ans: (b) — AWR: 9375 MHz (SHF, 3.2 cm) .

Q74. AWR is classified as:

(a) Secondary radar

(b) CW radar

(c) Primary radar

(d) Doppler radar

Answer: Ans: (c) — AWR: primary radar .

Q75. AWR colour — weakest precipitation:

(a) Magenta

(b) Red

(c) Yellow

(d) Green

Answer: Ans: (d) — Weakest: Green (0.7–4 mm/h). Black = nothing.

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Q76. AWR colour — turbulence/severe:

(a) Red

(b) Yellow

(c) Green

(d) Magenta

Answer: Ans: (d) — Turbulence: Magenta .

Q77. AWR MAP mode uses which beam up to 60–70 NM:

(a) Pencil beam

(b) Cosecant² (fan) beam

(c) Omni

(d) Scanning beam

Answer: Ans: (b) — MAP mode: cosecant² fan beam up to 60–70 NM.

Q78. AWR CONT mode shows:

(a) Increased gain

(b) Rainfall rate zone boundaries

(c) Map features

(d) Turbulence only

Answer: Ans: (b) — CONT: zone boundaries — steepest gradient = most hazardous.

Q79. AWR HOLD function:

(a) Freezes antenna

(b) Freezes display for storm movement assessment

(c) Holds gain

(d) Holds range

Answer: Ans: (b) — HOLD: freezes display to assess storm movement (compare after 2–3 min).

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Q80. AWR shadow area:

(a) Area under aircraft

(b) Area behind heavy precipitation

(c) Below the beam

(d) Overhead cone

Answer: Ans: (b) — Shadow: region behind heavy rain — attenuated beam reveals nothing beyond.

Q81. AWR hook/U/finger return on display indicates:

(a) Light rain

(b) Snow

(c) Hail and severe turbulence

(d) CAT

Answer: Ans: (c) — Hook, U, finger shapes: hail and severe turbulence — avoid.

Q82. AWR stabilisation prevents:

(a) False targets

(b) Multipath

(c) Display distortion during manoeuvres

(d) Second trace returns

Answer: Ans: (c) — Stabilisation: prevents display distortion during pitch/roll .

Q83. ASMI scan rate:

(a) 6 rpm

(b) 15 rpm

(c) 60 rpm

(d) 120 rpm

Answer: Ans: (c) — ASMI: 60 rpm .

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Q84. ASMI purpose:

(a) Long-range surveillance

(b) Aircraft and vehicle surface movement

(c) Weather detection

(d) Approach control

Answer: Ans: (b) — ASMI: surface movement — aircraft/vehicles on runways/taxiways.

Q85. Primary radar cannot determine:

(a) Range

(b) Bearing

(c) Identity

(d) Approximate speed (by track)

Answer: Ans: (c) — Primary radar: no identity — requires SSR transponder.

Q86. Super-refraction effect on radar:

(a) Reduces range

(b) Extends range beyond LOS

(c) Creates clutter

(d) Increases minimum range

Answer: Ans: (b) — Super-refraction: extends range beyond normal LOS (may cause spurious returns).

Q87. PRF 1000 pps. Echo at 1050 µs. Situation:

(a) Target at 85 NM — genuine

(b) Second trace return — actual 85 NM

(c) False target — reject

(d) Target at 4 NM

Answer: Ans: (b) — PRI=1000 µs. 1050 > PRI → second trace return . Actual range = 81 + 50/12.36 = 85 NM.

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Q88. UK radar separation within 40 NM below FL245:

(a) 1 NM

(b) 2 NM

(c) 3 NM

(d) 5 NM

Answer: Ans: (c) — UK radar sep: 3 NM .

Q89. TAR maximum range:

(a) 25 NM

(b) 75 NM

(c) 150 NM

(d) 250 NM

Answer: Ans: (b) — TAR (Terminal Area Radar): 75 NM .

Q90. Max PRF for 50 km range:

(a) 330 pps

(b) 617 pps

(c) 3000 pps

(d) 1620 pps

Answer: Ans: (c) — PRF = $3\times10^8/(2\times50{,}000)$ = 3000 pps .

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📡 DME (Q91–120)

Q91. DME frequency band:

(a) 108–118 MHz

(b) 329–335 MHz

(c) 960–1215 MHz

(d) 5031–5091 MHz

Answer: Ans: (c) — DME: UHF 960–1215 MHz .

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Q92. DME transponder offset:

(a) ±21 MHz

(b) ±50 MHz

(c) ±63 MHz

(d) ±100 MHz

Answer: Ans: (c) — Ground transponder replies at ±63 MHz from interrogation frequency.

Q93. DME transponder delay:

(a) 12.36 µs

(b) 21 µs

(c) 50 µs

(d) 8 µs

Answer: Ans: (c) — Fixed delay: 50 µs . Receiver subtracts before computing range.

Q94. DME tracking phase pulse rate:

(a) 27 ppps

(b) 60 ppps

(c) 150 ppps

(d) 2700 ppps

Answer: Ans: (a) — Tracking: 27 ppps .

Q95. DME search phase pulse rate:

(a) 27 ppps

(b) 60 ppps

(c) 150 ppps

(d) 2700 ppps

Answer: Ans: (c) — Search: 150 ppps .

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Q96. DME saturation approximately:

(a) 27 ppps total

(b) 2700 ppps ≈ 100 aircraft

(c) 60 ppps total

(d) 150 ppps total

Answer: Ans: (b) — Saturation: 2700 ppps ≈ 100 aircraft in tracking.

Q97. DME slant range error negligible when:

(a) Range > altitude (NM)

(b) Range (NM) > 3 × altitude (thousands ft)

(c) Always negligible

(d) Below 3000 ft

Answer: Ans: (b) — Negligible: Range (NM) > 3 × Alt (000s ft) .

Q98. Aircraft at FL180 directly over DME. Display reads:

(a) 0 NM

(b) 0.8 NM

(c) 3 NM

(d) 18 NM

Answer: Ans: (c) — 18,000/6076 = 2.96 ≈ 3 NM .

Q99. Time from interrogation to echo = 1300 µs. Range:

(a) 98 NM

(b) 100 NM

(c) 101 NM

(d) 103 NM

Answer: Ans: (c) — (1300−50)/12.36 = 1250/12.36 = 101.1 NM .

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Q100. DME number of channels:

(a) 40

(b) 108

(c) 200

(d) 252

Answer: Ans: (d) — DME: 252 channels .

Q101. DME emission type:

(a) A9W

(b) P0N

(c) F3N

(d) NON

Answer: Ans: (b) — DME: P0N (pulsed, no info modulation).

Q102. VOR/DME pairing outside TMA — DME within:

(a) 100 ft

(b) 1 NM

(c) 2000 ft

(d) 2 NM

Answer: Ans: (c) — Outside TMA: DME within 2000 ft of VOR for auto-pairing.

Q103. VOR/DME pairing inside TMA — DME within:

(a) 100 ft

(b) 1 NM

(c) 2000 ft

(d) 2 NM

Answer: Ans: (a) — Inside TMA: DME within 100 ft of VOR.

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Q104. DME jittered pulse pairs — purpose:

(a) Higher power

(b) Avoid ILS interference

(c) Identify own replies

(d) Increase range

Answer: Ans: (c) — Jittered pairs: aircraft identifies its own replies by consistent time offset.

Q105. DME squitter purpose:

(a) Ident

(b) Fill capacity to prevent aircraft re-entering search

(c) Range calibration

(d) Interference avoidance

Answer: Ans: (b) — Squitter: fills unused transponder capacity so aircraft detect station is active.

Q106. DME memory duration in tracking:

(a) 5 s

(b) 10 s

(c) 15 s

(d) 30 s

Answer: Ans: (b) — DME memory: 10 seconds .

Q107. DME/P accuracy:

(a) ±0.5 NM

(b) ±1 NM

(c) ±100 ft

(d) ±50 m

Answer: Ans: (c) — DME/P (Precision): ±100 ft for MLS CAT II/III.

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Q108. DME/P used with:

(a) ILS

(b) MLS

(c) GBAS

(d) SBAS

Answer: Ans: (b) — DME/P used with MLS .

Q109. VORTAC: military aircraft uses:

(a) VOR bearing only

(b) VOR bearing + TACAN DME

(c) TACAN bearing + TACAN DME

(d) VOR only

Answer: Ans: (c) — Military at VORTAC: TACAN bearing + TACAN DME .

Q110. DME ident interval:

(a) Every 10 s

(b) Every 30 s

(c) Continuous

(d) Every 60 s

Answer: Ans: (b) — DME ident: every 30 seconds , 1350 Hz Morse.

Q111. DME type of radar:

(a) Primary

(b) Secondary

(c) Doppler

(d) CW

Answer: Ans: (b) — DME: secondary radar principle.

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Q112. RNAV DME range shown to:

(a) Real DME station

(b) Phantom station (waypoint)

(c) Nearest airport

(d) FIX position

Answer: Ans: (b) — RNAV DME: shows distance to phantom station (waypoint) .

Q113. DME en-route accuracy:

(a) ±0.25 NM

(b) ±0.5 NM or ±3% (whichever greater)

(c) ±1 NM

(d) ±2%

Answer: Ans: (b) — DME accuracy: ±0.5 NM or ±3% , whichever greater.

Q114. GPS system uses what type of ranging?

(a) Secondary radar (2-way)

(b) One-way passive ranging

(c) Primary radar

(d) DME-type 2-way

Answer: Ans: (b) — GPS: one-way passive — satellite broadcasts, receiver measures time of arrival.

Q115. DME acquisition phase rate:

(a) 27 ppps

(b) 60 ppps

(c) 150 ppps

(d) 2700 ppps

Answer: Ans: (b) — Acquisition: 60 ppps .

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Q116. Aircraft at 5000 ft AGL over DME. Slant range reads:

(a) 0 NM

(b) 0.82 NM

(c) 2 NM

(d) 5 NM

Answer: Ans: (b) — 5000/6076 = 0.82 NM .

Q117. DME squitter pulse rate (total with squitter):

(a) 27 ppps

(b) 2700 ppps

(c) 150 ppps

(d) 60 ppps

Answer: Ans: (b) — With squitter: maintained at 2700 ppps .

Q118. Which PRN codes does GPS L1 carry?

(a) P-code only

(b) C/A and P-code

(c) C/A code only (civil)

(d) SPS only

Answer: Ans: (b) — L1 (1575.42 MHz): carries both C/A code (civil) and P-code (military) .

Q119. DME ground transponder delay ensures:

(a) Collision avoidance

(b) Positive identification of own replies

(c) Positive time offset for range calculation

(d) Synchronisation with VOR

Answer: Ans: (c) — 50 µs delay ensures receiver always has positive elapsed time to subtract and compute range.

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Q120. DME interrogation spacing (pulse pair):

(a) 6 µs X-channel; 12 µs Y-channel

(b) 12 µs X; 36 µs Y

(c) Random (jittered)

(d) Fixed at 12 µs

Answer: Ans: (b) — X-channel pair spacing: 12 µs . Y-channel: 36 µs .

SSR (Q121–150)

Q121. SSR interrogation frequency:

(a) 1030 MHz

(b) 1090 MHz

(c) 960 MHz

(d) 1215 MHz

Answer: Ans: (a) — Ground interrogates on 1030 MHz .

Q122. SSR transponder reply frequency:

(a) 1030 MHz

(b) 1090 MHz

(c) 960 MHz

(d) 1215 MHz

Answer: Ans: (b) — Aircraft replies on 1090 MHz .

Q123. Mode A P1–P3 spacing:

(a) 8 µs

(b) 17 µs

(c) 21 µs

(d) 25 µs

Answer: Ans: (a) — Mode A: 8 µs .

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Q124. Mode C P1–P3 spacing:

(a) 8 µs

(b) 17 µs

(c) 21 µs

(d) 25 µs

Answer: Ans: (c) — Mode C: 21 µs .

Q125. Mode A codes available:

(a) 256

(b) 4096

(c) 65,536

(d) 16,777,216

Answer: Ans: (b) — Mode A: 4096 (octal 0000–7777).

Q126. Mode S ICAO address size:

(a) 12 bit

(b) 16 bit

(c) 24 bit

(d) 32 bit

Answer: Ans: (c) — Mode S: 24-bit address → 16.7 million unique codes.

Q127. Mode C altitude resolution:

(a) 100 ft

(b) 50 ft

(c) 25 ft

(d) 10 ft

Answer: Ans: (a) — Mode C: 100 ft increments. Mode S: 25 ft.

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Q128. Squawk 7700:

(a) Hijack

(b) Radio failure

(c) Emergency

(d) VFR

Answer: Ans: (c) — 7700 = Emergency .

Q129. Squawk 7600:

(a) Emergency

(b) Radio failure

(c) Hijack

(d) Military

Answer: Ans: (b) — 7600 = Radio failure (NORDO) .

Q130. Squawk 7500:

(a) Emergency

(b) Radio failure

(c) Unlawful interference/hijack

(d) VFR conspicuity

Answer: Ans: (c) — 7500 = Hijack/unlawful interference .

Q131. Garbling occurs when two aircraft within:

(a) 0.5 NM

(b) 1.7 NM same direction

(c) 3 NM

(d) 5 NM

Answer: Ans: (b) — Garbling: 1.7 NM in same direction from radar.

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Q132. Fruiting caused by:

(a) Garbling

(b) Replies to other ground interrogators

(c) Multipath

(d) SA

Answer: Ans: (b) — Fruiting: aircraft replies to other stations' interrogations → phantom targets.

Q133. IDENT button — SPI duration:

(a) 10 s

(b) 20 s

(c) 30 s

(d) 60 s

Answer: Ans: (b) — SPI: 20 seconds .

Q134. SLS — P2 pulse is:

(a) Directional, same as P1

(b) Omnidirectional

(c) Higher power than P1

(d) On 1090 MHz

Answer: Ans: (b) — SLS P2: omnidirectional → if P2>P1, transponder suppresses reply (side lobe).

Q135. ISLS purpose:

(a) Prevent garbling

(b) Synchronise multiple interrogators to prevent fruiting

(c) Improve altitude reporting

(d) Enable data link

Answer: Ans: (b) — ISLS: synchronises multiple ground SSR stations → prevents fruiting.

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Q136. Transponder STANDBY:

(a) Reduced power response

(b) No response to interrogations

(c) Mode C only

(d) Mode S only

Answer: Ans: (b) — STANDBY: energised, will NOT respond .

Q137. UK VFR conspicuity squawk:

(a) 2000

(b) 7000

(c) 7700

(d) 1200

Answer: Ans: (b) — UK VFR: 7000 .

Q138. Mode S advantage:

(a) VHF frequencies

(b) Selective addressing — eliminates garbling

(c) No transponder needed

(d) No LOS needed

Answer: Ans: (b) — Mode S: selective addressing by unique 24-bit ICAO code.

Q139. ADS-B squitter interval:

(a) 1 s

(b) 0.5 s

(c) 2 s

(d) 10 s

Answer: Ans: (b) — ADS-B: every 0.5 seconds .

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Q140. Mode C altitude referenced to:

(a) QNH

(b) QFE

(c) 1013.25 hPa

(d) Local QNH

Answer: Ans: (c) — Mode C: always 1013.25 hPa .

Q141. SSR reply framing pulses F1–F2 spacing:

(a) 8 µs

(b) 12 µs

(c) 20.3 µs

(d) 50 µs

Answer: Ans: (c) — F1–F2: 20.3 µs .

Q142. ACAS/TCAS uses:

(a) Mode A

(b) Mode C

(c) Mode S

(d) All modes

Answer: Ans: (c) — TCAS: uses Mode S 1030/1090 MHz for A/C-to-A/C interrogation.

Q143. ELS (Elementary Surveillance) provides:

(a) Squawk and altitude

(b) Squawk, flight ID, altitude, emergency

(c) Full EHS data

(d) GPS position

Answer: Ans: (b) — ELS: squawk, callsign, pressure altitude, emergency .

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Q144. EHS (Enhanced Surveillance) adds:

(a) Nothing over ELS

(b) Selected altitude, airspeed, magnetic heading

(c) GPS position

(d) TCAS data

Answer: Ans: (b) — EHS adds: selected altitude, airspeed, magnetic heading .

Q145. Squawk 2000 means:

(a) Emergency

(b) Entering CAS without clearance

(c) Military

(d) Radio failure

Answer: Ans: (b) — 2000: entering CAS without prior ATC clearance .

Q146. SSR antenna location:

(a) Separate tower

(b) On top of primary radar, co-rotating

(c) Inside fuselage

(d) Alongside ILS building

Answer: Ans: (b) — SSR antenna: mounted on top of primary radar, co-rotating .

Q147. Mode S data link direction:

(a) Downlink only

(b) Uplink only

(c) Bidirectional

(d) No data link

Answer: Ans: (c) — Mode S: bidirectional (downlink + uplink).

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Q148. Transponder ALT mode:

(a) Responds Mode A only

(b) Responds Mode A + C (altitude)

(c) Mode S only

(d) STANDBY

Answer: Ans: (b) — ALT mode: responds to Mode A and Mode C .

Q149. ADS-B position source:

(a) SSR ranging

(b) DME

(c) GPS

(d) Transponder timing

Answer: Ans: (c) — ADS-B: broadcasts GPS-derived position via 1090 MHz extended squitter.

Q150. Minimum range for no garbling:

(a) 0.5 NM

(b) 1.7 NM

(c) 3 NM

(d) 5 NM

Answer: Ans: (b) — No garbling when separation > 1.7 NM in same direction.

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📡 AWR / MLS (Q151–180)

Q151. AWR wavelength:

(a) 1 cm

(b) 3.2 cm

(c) 10 cm

(d) 23 cm

Answer: Ans: (b) — AWR: 3.2 cm (9375 MHz) .

Q152. AWR MAN mode beam:

(a) Fan beam at all ranges

(b) Pencil beam, manual gain

(c) Omni

(d) Fan beam beyond 70 NM

Answer: Ans: (b) — MAN: pencil beam, manual gain .

Q153. AWR cannot detect:

(a) Thunderstorms

(b) Hail

(c) CAT

(d) Heavy rain

Answer: Ans: (c) — AWR cannot detect CAT (Clear Air Turbulence) — no precipitation.

Q154. AWR tilt positive (+) means:

(a) Beam below horizontal

(b) Beam above horizontal

(c) No tilt

(d) Gain increased

Answer: Ans: (b) — Positive tilt = beam above horizontal .

Q155. AWR: rain shadow risk is:

(a) Light rain conceals nothing

(b) Heavy rain attenuates beam — severe weather beyond may be hidden

(c) Snow creates shadow

(d) Shadow only below aircraft

Answer: Ans: (b) — Rain shadow: severe weather may be hidden behind heavy precipitation .

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Q156. AWR 'finger' return shape indicates:

(a) Light rain

(b) Hail/severe turbulence

(c) Snow

(d) CAT

Answer: Ans: (b) — Finger shape: hail and severe turbulence .

Q157. AWR scalloped edges on return indicate:

(a) Ground return

(b) Hail

(c) Light rain

(d) Icing

Answer: Ans: (b) — Scalloped edges: hail .

Q158. AWR AGC/STC range:

(a) Up to 25 NM

(b) All ranges

(c) Beyond 70 NM

(d) First 5 NM only

Answer: Ans: (a) — STC (Swept Gain/AGC): equalises returns up to ≈25 NM .

Q159. AWR stabilisation failure — display effect:

(a) Blanks

(b) Lopsided during bank/pitch

(c) Gains increase

(d) Range rings disappear

Answer: Ans: (b) — Stabilisation failure: display becomes lopsided during manoeuvres .

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Q160. MLS frequency band:

(a) VHF 108–118 MHz

(b) UHF 329–335 MHz

(c) SHF 5031–5090.7 MHz

(d) UHF 960–1215 MHz

Answer: Ans: (c) — MLS: SHF 5031–5090.7 MHz .

Q161. MLS azimuth coverage:

(a) ±20°

(b) ±35°

(c) ±40°

(d) ±60°

Answer: Ans: (c) — MLS azimuth: ±40° of centreline.

Q162. MLS channels:

(a) 40

(b) 100

(c) 200

(d) 252

Answer: Ans: (c) — MLS: 200 channels .

Q163. MLS principle:

(a) DDM like ILS

(b) TRSB — time between TO/FRO sweeps

(c) Phase comparison

(d) Doppler

Answer: Ans: (b) — MLS: TRSB (Time Referenced Scanning Beam) .

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Q164. MLS glide slope range:

(a) Fixed 3°

(b) 0.9° to 20°

(c) 2° to 10°

(d) Fixed 5°

Answer: Ans: (b) — MLS G/S: 0.9° to 20° selectable.

Q165. MLS ident prefix letter:

(a) I

(b) M

(c) V

(d) D

Answer: Ans: (b) — MLS ident: prefix M .

Q166. MLS range measurement:

(a) Pulse echo timing

(b) DME/P built-in

(c) Phase comparison

(d) Doppler

Answer: Ans: (b) — MLS range: DME/P (±100 ft accuracy).

Q167. MLS advantage over ILS — azimuth coverage:

(a) ±35° same as ILS

(b) ±40° vs ±35°

(c) 360°

(d) ±20°

Answer: Ans: (b) — MLS: ±40° vs ILS ±35°.

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Q168. MLS UK usable range:

(a) 10 NM

(b) 15 NM

(c) 20 NM

(d) 35 NM

Answer: Ans: (c) — MLS UK: 20 NM .

Q169. AWR U-shape return:

(a) Light rain

(b) Hail/turbulence

(c) Snow

(d) CAT

Answer: Ans: (b) — U-shape: hail and turbulence .

Q170. AWR MAP mode optimal up to:

(a) 25 NM

(b) 60–70 NM

(c) 150 NM

(d) All ranges

Answer: Ans: (b) — MAP (cosecant² beam): up to 60–70 NM .

Q171. ASMI frequency band:

(a) 3 GHz

(b) 9.375 GHz

(c) 15–17 GHz

(d) 35 GHz

Answer: Ans: (c) — ASMI: 15–17 GHz (SHF, ≈2 cm) .

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Q172. AWR WEA colour for no precip:

(a) Green

(b) Yellow

(c) Red

(d) Black

Answer: Ans: (d) — No precipitation: Black .

Q173. AWR hook return on display:

(a) Light drizzle

(b) Hail/severe turbulence

(c) CAT

(d) Snow

Answer: Ans: (b) — Hook shape: hail/severe turbulence .

Q174. AWR ageing (HOLD) — how long before compare?

(a) 30 s

(b) 1 min

(c) 2–3 min

(d) 10 min

Answer: Ans: (c) — Deselect HOLD after 2–3 minutes — compare images for storm movement.

Q175. AWR pencil beam — used in MAN mode beyond:

(a) 25 NM

(b) 50 NM

(c) 60–70 NM

(d) 150 NM

Answer: Ans: (c) — Pencil beam (MAN) for weather beyond 60–70 NM .

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Q176. AWR — recommended tilt when far from storm cell:

(a) Tilt down to surface

(b) Zero tilt

(c) Tilt up

(d) Reduce gain

Answer: Ans: (c) — Far from cell: tilt up to see tops — determines storm height/severity.

Q177. SSR transponder receives on ___, replies on ___:

(a) 1090, 1030

(b) 1030, 1090

(c) 960, 1215

(d) 1215, 960

Answer: Ans: (b) — RX: 1030 MHz , TX: 1090 MHz .

Q178. EGNOS is classified as:

(a) LAAS

(b) GBAS

(c) SBAS

(d) RAIM

Answer: Ans: (c) — EGNOS: SBAS (Satellite Based Augmentation System) .

Q179. LAAS/GBAS transmits corrections via:

(a) Satellite

(b) HF radio

(c) VHF data link

(d) UHF beacon

Answer: Ans: (c) — LAAS/GBAS: VHF data link .

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Q180. SBAS — corrections transmitted via:

(a) Dedicated HF stations

(b) Geostationary satellites

(c) VHF data link

(d) L-band beacons

Answer: Ans: (b) — SBAS: corrections via geostationary satellites .

📡 GNSS / RNAV / MIXED (Q181–243)

Q181. B-RNAV required accuracy:

(a) ±1 NM, 95%

(b) ±5 NM, 95%

(c) ±5 NM, 90%

(d) ±2 NM, 95%

Answer: Ans: (b) — B-RNAV: ±5 NM on 95% of occasions .

Q182. P-RNAV required accuracy:

(a) ±1 NM, 95%

(b) ±5 NM, 95%

(c) ±0.5 NM, 95%

(d) ±2 NM, 90%

Answer: Ans: (a) — P-RNAV: ±1 NM on 95% of occasions .

Q183. FMC navigation database update cycle:

(a) 7 days

(b) 14 days

(c) 28 days (AIRAC)

(d) 90 days

Answer: Ans: (c) — FMC nav DB: 28 days (AIRAC cycle) .

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Q184. RNAV erratic: aircraft beyond LOS of reference VOR/DME:

(a) True

(b) False

(c) Only in TMA

(d) Only in cone of confusion

Answer: Ans: (a) — Yes — beyond LOS or DOC of reference VOR/DME → erratic RNAV .

Q185. GPS horizontal accuracy (SPS, 95%):

(a) ±100 m

(b) ±22 m

(c) ±13 m

(d) ±50 m

Answer: Ans: (c) — Horizontal: ±13 m (95%) .

Q186. GPS vertical accuracy (SPS, 95%):

(a) ±13 m

(b) ±22 m

(c) ±50 m

(d) ±100 m

Answer: Ans: (b) — Vertical: ±22 m (95%) .

Q187. GPS time accuracy (SPS, 95%):

(a) 1 µs

(b) 40 ns

(c) 100 ns

(d) 1 ms

Answer: Ans: (b) — Time: 40 nanoseconds (95%) .

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Q188. GPS — why pseudo-range not true range?

(a) Ionospheric delay

(b) Receiver clock error not yet eliminated

(c) SA applied

(d) Multipath

Answer: Ans: (b) — Pseudo-range: uncorrected for receiver clock error .

Q189. GPS minimum SVs for 3D fix + time:

(a) 3

(b) 4

(c) 5

(d) 6

Answer: Ans: (b) — 4 SVs for 3D fix (lat, lon, alt, time).

Q190. GPS RAIM — SVs for fault detection only:

(a) 4

(b) 5

(c) 6

(d) 7

Answer: Ans: (b) — RAIM detection: 5 SVs .

Q191. GPS RAIM — SVs for fault exclusion:

(a) 4

(b) 5

(c) 6

(d) 7

Answer: Ans: (c) — RAIM exclusion: 6 SVs .

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Q192. GPS orbital height:

(a) 19,099 km

(b) 20,180 km

(c) 23,222 km

(d) 35,800 km

Answer: Ans: (b) — GPS: 20,180 km .

Q193. GPS orbital inclination:

(a) 55°

(b) 56°

(c) 63°

(d) 65°

Answer: Ans: (a) — GPS: 55° inclination.

Q194. GPS orbital period:

(a) 24 h

(b) 12 h

(c) 11 h 56 min

(d) 11 h 15 min

Answer: Ans: (c) — GPS: 11 h 56 min .

Q195. GPS L1 frequency:

(a) 1227.6 MHz

(b) 1575.42 MHz

(c) 1602 MHz

(d) 1246 MHz

Answer: Ans: (b) — GPS L1: 1575.42 MHz .

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Q196. GPS L2 frequency:

(a) 1227.6 MHz

(b) 1575.42 MHz

(c) 1602 MHz

(d) 1246 MHz

Answer: Ans: (a) — GPS L2: 1227.6 MHz .

Q197. GPS geoid reference:

(a) PZ90

(b) ETRS89

(c) WGS84

(d) GRS80

Answer: Ans: (c) — GPS: WGS84 .

Q198. GLONASS geoid reference:

(a) WGS84

(b) PZ90

(c) ETRS89

(d) GRS80

Answer: Ans: (b) — GLONASS: PZ90 .

Q199. Galileo geoid reference:

(a) WGS84

(b) PZ90

(c) ETRS89

(d) GRS80

Answer: Ans: (c) — Galileo: ETRS89 .

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Q200. GLONASS orbital inclination:

(a) 55°

(b) 56°

(c) 63.4°

(d) 65°

Answer: Ans: (d) — GLONASS: 65° .

Q201. GLONASS orbital period:

(a) 11 h 56 min

(b) 11 h 15 min

(c) 12 h

(d) 14 h 8 min

Answer: Ans: (b) — GLONASS: 11 h 15 min .

Q202. Galileo orbital height:

(a) 20,180 km

(b) 19,099 km

(c) 23,222 km

(d) 35,800 km

Answer: Ans: (c) — Galileo: 23,222 km .

Q203. GPS almanac used by receiver to:

(a) Determine SA

(b) Compute position

(c) Identify which SVs visible

(d) Correct clock error

Answer: Ans: (c) — Almanac: receiver determines which SVs are above horizon for faster acquisition.

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Q204. GPS ephemeris — sub-frames:

(a) 1

(b) 2 and 3

(c) 4 and 5

(d) 5

Answer: Ans: (b) — Ephemeris data: sub-frames 2 and 3 .

Q205. GPS navigation message transmitted at:

(a) 50 Hz

(b) 1 kHz

(c) 9960 Hz

(d) 1575 MHz

Answer: Ans: (a) — Nav message: 50 Hz (50 bps) modulation on L1 and L2.

Q206. GNSS altitude cannot be used alone for DH/MDA because:

(a) GPS too inaccurate

(b) WGS84 ellipsoid differs from geoid (MSL) by up to 50 m

(c) Update rate too slow

(d) ILS always more accurate

Answer: Ans: (b) — WGS84 ellipsoid ≠ MSL geoid → difference up to 50 m .

Q207. SBAS geostationary orbit altitude:

(a) 20,180 km

(b) 19,099 km

(c) 23,222 km

(d) 35,800 km

Answer: Ans: (d) — Geostationary: 35,800 km .

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Q208. GPS dual-frequency advantage:

(a) Higher power

(b) Eliminates ≈99% ionospheric error

(c) Better coverage

(d) Faster acquisition

Answer: Ans: (b) — Dual-frequency (L1+L2): eliminates ≈99% of ionospheric error .

Q209. GPS integrity warning time (non-precision approach):

(a) 2 s

(b) 8 s

(c) 30 s

(d) 10 s

Answer: Ans: (b) — Non-precision: 8 seconds . Precision (CAT I): 6 s. ILS equivalent: 2 s.

Q210. 4D RNAV adds to 3D capability:

(a) Vertical guidance

(b) GPS

(c) Timing (RTA)

(d) TCAS

Answer: Ans: (c) — 4D RNAV: lateral + vertical + timing (RTA) .

Q211. FMC auto-tunes DMEs for:

(a) Nearest station

(b) Pilot's selection

(c) Best geometry (angle of cut)

(d) Highest power stations

Answer: Ans: (c) — FMC selects DMEs for best angle of cut .

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Q212. External input to FMC:

(a) INS

(b) Pressure altitude

(c) Compass

(d) VOR/DME

Answer: Ans: (d) — External (ground-based): VOR/DME .

Q213. ADC input to FMC:

(a) Heading

(b) Groundspeed

(c) TAS

(d) Position

Answer: Ans: (c) — ADC provides: TAS, pressure altitude, SAT .

Q214. RNAV course line computer function:

(a) Direct to VOR/DME facility

(b) Rho/theta → track and distance to waypoint

(c) ILS approach guidance

(d) TCAS alerting

Answer: Ans: (b) — RNAV: uses rho/theta to compute track + distance to phantom waypoint .

Q215. GPS range measurement: time measured from:

(a) Receiver to SV and back

(b) SV to receiver (one-way)

(c) Control segment to SV

(d) SV to control segment

Answer: Ans: (b) — GPS: one-way time from SV to receiver.

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Q216. GPS number of orbital planes:

(a) 3

(b) 4

(c) 6

(d) 8

Answer: Ans: (c) — GPS: 6 orbital planes , 4 SVs each = 24 SVs.

Q217. GLONASS number of orbital planes:

(a) 3

(b) 4

(c) 6

(d) 8

Answer: Ans: (a) — GLONASS: 3 orbital planes , 8 SVs each = 24 SVs.

Q218. Galileo number of orbital planes:

(a) 3

(b) 4

(c) 6

(d) 8

Answer: Ans: (a) — Galileo: 3 orbital planes , 10 SVs each = 30 SVs.

Q219. GPS C/A code chipping rate:

(a) 1.023 MHz

(b) 10.23 MHz

(c) 50 Hz

(d) 9960 Hz

Answer: Ans: (a) — C/A code: 1.023 MHz . P-code: 10.23 MHz.

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Q220. Selective Availability (SA):

(a) Still active on civilian GPS

(b) Intentional degradation of SPS; switched off in 2000

(c) Used only by military

(d) Replaced by RAIM

Answer: Ans: (b) — SA: switched off 2 May 2000 . No longer applied.

Q221. RNAV erratic most likely cause:

(a) Cone of confusion of phantom

(b) Beyond DOC/LOS of reference VOR/DME

(c) FMC failure

(d) INS drift

Answer: Ans: (b) — Most likely: beyond DOC or LOS of reference VOR/DME .

Q222. P-RNAV accuracy requirement:

(a) ±5 NM, 95%

(b) ±1 NM, 95%

(c) ±0.5 NM, 95%

(d) ±1 NM, 99%

Answer: Ans: (b) — P-RNAV: ±1 NM, 95% .

Q223. Multi-sensor FMC + GPS discrepancy — action:

(a) Trust GPS

(b) Trust multi-sensor; display multi-sensor output

(c) Select nearest VOR

(d) Declare emergency

Answer: Ans: (b) — Discrepancy: trust multi-sensor ; GPS may have failed.

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Q224. GBAS/LAAS accuracy for CAT I:

(a) ±16 m horizontal, ±6 m vertical

(b) ±100 ft

(c) ±0.5 NM

(d) Same as ILS

Answer: Ans: (a) — GBAS CAT I: ±16 m horizontal, ±6 m vertical (95%) .

Q225. GPS constellation: total operational SVs:

(a) 21

(b) 24

(c) 27

(d) 30

Answer: Ans: (b) — GPS: 24 SVs (operational); up to 32 with spares.

Q226. GALILEO constellation: total SVs:

(a) 24

(b) 27

(c) 30

(d) 32

Answer: Ans: (c) — Galileo: 30 SVs .

Q227. GLONASS constellation:

(a) 21

(b) 24

(c) 27

(d) 30

Answer: Ans: (b) — GLONASS: 24 SVs .

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Q228. GPS PRN codes — how many available?

(a) 24

(b) 32

(c) 64

(d) 252

Answer: Ans: (b) — GPS PRN codes: 32 available (SVs numbered PRN 1–32).

Q229. GNSS integrity definition:

(a) Accuracy of the position solution

(b) Ability to alert users when system should not be used

(c) Number of SVs visible

(d) Signal continuity

Answer: Ans: (b) — Integrity: ability to provide timely warning when system unfit for navigation.

Q230. RAIM — what does it detect?

(a) Ionospheric error

(b) Failed SV causing position error exceeding limits

(c) Multipath

(d) Clock drift in receiver

Answer: Ans: (b) — RAIM detects: failed SV causing excessive position error .

Q231. GPS almanac download time:

(a) 30 s

(b) 12.5 min

(c) 30 min

(d) 1 hour

Answer: Ans: (b) — Full almanac download: 12.5 minutes (25 frames × 30 s).

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Q232. GPS cold start — time to first fix:

(a) 30 s

(b) 1–2 min

(c) 12.5 min

(d) Up to 30 min

Answer: Ans: (c) — Cold start (no almanac): up to 12.5 min for almanac download.

Q233. PDOP (Position Dilution of Precision) — best geometry:

(a) High PDOP

(b) Low PDOP

(c) PDOP = 1 always

(d) PDOP irrelevant

Answer: Ans: (b) — Best geometry: low PDOP (SVs spread widely across sky).

Q234. GPS ionospheric model corrects approximately:

(a) 10%

(b) 50%

(c) 75%

(d) 99%

Answer: Ans: (b) — GPS ionospheric model: ≈50% correction. Dual-freq: ≈99%.

Q235. SBAS integrity monitoring provided by:

(a) Ground reference stations

(b) Control segment only

(c) RAIM

(d) Aircraft receiver

Answer: Ans: (a) — SBAS: network of ground reference stations monitors satellite integrity.

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Q236. Galileo orbital inclination:

(a) 55°

(b) 56°

(c) 63°

(d) 65°

Answer: Ans: (b) — Galileo: 56° .

Q237. GPS Week Number rollover issue — period:

(a) 10.9 years (1024 weeks)

(b) 19.7 years

(c) 7 years

(d) Every year

Answer: Ans: (a) — GPS week: 10-bit number → rolls over every 1024 weeks (≈19.7 years) . [Some say 19.7 — check ICAO doc for exact value used]

Q238. GNSS continuity of service — definition:

(a) System accuracy

(b) Probability of system performing without interruption during operation

(c) Integrity

(d) Number of SVs

Answer: Ans: (b) — Continuity: probability of uninterrupted operation during intended use .

Q239. GNSS availability — definition:

(a) Accuracy spec

(b) Proportion of time system provides navigation with required accuracy

(c) Integrity

(d) Continuity

Answer: Ans: (b) — Availability: proportion of time system meets accuracy and integrity requirements .

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Q240. GPS L1 C/A code — chip length:

(a) 30 m

(b) 300 m

(c) 1 km

(d) 3 km

Answer: Ans: (b) — C/A chip length = c/f = $3\times10^8/1.023\times10^6$ = 293 m ≈ 300 m .

Q241. GNSS approach — LPV (Localizer Performance with Vertical guidance) provides:

(a) Lateral only

(b) Lateral + baro-VNAV

(c) Lateral + SBAS vertical guidance (like ILS CAT I)

(d) No approach minima

Answer: Ans: (c) — LPV: SBAS approach with lateral + SBAS vertical guidance , minima approaching CAT I.

Q242. GPS time offset from UTC at receiver:

(a) Zero — GPS = UTC

(b) GPS ahead of UTC by integer seconds (leap seconds)

(c) GPS behind UTC

(d) Varies daily

Answer: Ans: (b) — GPS time: ahead of UTC by integer leap seconds . Receiver subtracts leap second count to display UTC.

Q243. GNSS — which error source causes errors of hundreds of metres without correction?

(a) Multipath

(b) Clock error

(c) Selective Availability (when active)

(d) Tropospheric delay

Answer: Ans: (c) — When SA active: errors up to 100 m (horizontal). SA now off.

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> ⚠ **NOTE**
>
> 📝 66-Question Specimen Exam — Answer Key Q1=b • Q2=c • Q3=d • Q4=a • Q5=a • Q6=d • Q7=b • Q8=c • Q9=a • Q10=c • Q11=d • Q12=d Q13=c • Q14=a • Q15=d • Q16=a • Q17=d • Q18=b • Q19=a • Q20=b • Q21=c • Q22=d • Q23=b • Q24=d Q25=a • Q26=d • Q27=b • Q28=c • Q29=b • Q30=a • Q31=a • Q32=a • Q33=b • Q34=c • Q35=a • Q36=d Q37=b • Q38=a • Q39=b • Q40=b • Q41=d • Q42=b • Q43=b • Q44=b • Q45=c • Q46=a • Q47=d • Q48=a Q49=c • Q50=c • Q51=b • Q52=d • Q53=b • Q54=a • Q55=a • Q56=a • Q57=a • Q58=c • Q59=d • Q60=b Q61=b • Q62=c • Q63=b • Q64=a • Q65=b • Q66=b

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CHAPTER 20

📚 CHAPTER 20: QUICK REFERENCE & INDEX ALL KEY FACTS, FORMULAE, FREQUENCIES & EXAM TIPS — ALL CHAPTERS

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📈 ESSENTIAL FORMULAE & CALCULATIONS

▶ Radio Waves c = f × λ | c = 3×10&sup8 m/s λ(m) = 300 / f(MHz) f(MHz) = 300 / λ(m) c = 162,000 NM/s ▶ LOS Range Radar: 1.23 × (√h TX + √h RX) NM VOR/VDF: 1.25 × (√h TX + √h RX) NM (heights in feet) ▶ Radar Range & PRF Max range (NM) = 81,000 / PRF (pps) Max range (km) = 150,000 / PRF (pps) PRF = 81,000 / Max range (NM) Range (NM) = echo time (µs) / 12.36 Min range (m) = PW (µs) × 150 ▶ Power vs Range Power ∝ Range&sup4 Double range → power × 2&sup4 = ×16 Triple range → power × 81 ▶ DME Range (NM) = (echo time µs − 50) / 12.36 Transponder offset: ±63 MHz Delay: 50 µs Slant error negligible: Range(NM) > 3 × Alt(000s ft) Search: 150 ppps | Acq: 60 ppps | Track: 27 ppps Saturation: 2700 ppps ≈ 100 aircraft ▶ ILS Height Check H(ft) ≈ GS angle° × Range(NM) × 100 + threshold ht Example: 3° at 4.6 NM + 50 ft = 1430 ft ▶ GPS 4 SVs for 3D fix + time 5 SVs for RAIM detection 6 SVs for RAIM exclusion Horizontal: ±13 m (95%) | Vertical: ±22 m (95%) Time: ±40 ns (95%) ▶ NDB Range vs Power P ∝ R&sup4 Range 10 NM → 20 NM: power ×16 ADF day accuracy: ±5° VDF Class A: ±1° | B: ±2° | C: ±5° | D: ±10°

📡 FREQUENCY & EMISSION REFERENCE

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SYSTEM	FREQUENCY / BAND	EMISSION	KEY FACTS
NDB	190–1750 kHz (LF/MF)	MCW (A2A) / NON	Omni signal; ADF measures RB; BFO for NON beacons
VDF	VHF (108–174 MHz)	A3E	LOS only; aircraft needs VHF radio only
VOR	108–117.95 MHz (VHF)	A9W / A2A (ident)	Phase comparison 9960 Hz FM ref vs 30 Hz AM var; ident 1020 Hz
ILS LOC	108–111.975 MHz (VHF, odd decimals)	A9W	90 Hz + 150 Hz DDM; 40 channels
ILS G/S	329–335 MHz (UHF)	A9W	Auto-paired with LOC; range 10 NM; ±0.5°
ILS Markers	75 MHz (VHF)	A2A	OM:400Hz/blue | MM:1300Hz/amber | IM:3000Hz/white
DME	960–1215 MHz (UHF)	P0N	252 ch; ±63 MHz offset; 50µs delay; ±0.5NM/3%
SSR	1030 MHz (gnd TX) / 1090 MHz (A/C TX)	P0N	4096 codes (Mode A); 16.7M (Mode S)
AWR	9375 MHz (SHF, 3.2 cm)	P0N	Primary radar; pencil + fan beam
MLS	5031–5090.7 MHz (SHF)	—	200 ch; TRSB; ±40° az; 0.9°–20° GS
ASMI	15–17 GHz (SHF, ~2 cm)	—	60 rpm; PW 0.03µs; PRF 4000–20000 pps
GPS L1	1575.42 MHz	—	Civil C/A code; 1.023 MHz chip rate
GPS L2	1227.6 MHz	—	Military P-code; 10.23 MHz chip rate
GLONASS	~1600 MHz / ~1246 MHz	—	FDMA; 24 SVs; 65° incl; PZ90
Galileo	E1/E5a/E5b/E6 (~1.2–1.6 GHz)	—	30 SVs; 56° incl; ETRS89

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CATEGORY	DECISION HEIGHT	RVR	NOTES
CAT I	200 ft	800 m	Standard ILS
CAT II	100 ft	400 m	Autoland capable aircraft required
CAT IIIA	100 ft	200 m	Autoland; enhanced RVR system
CAT IIIB	<50 ft	50–200 m	Low visibility procedures
CAT IIIC	None	None	Theoretical; no current certification

NOTE

ILS False Glideslope: 3° nominal → false slopes at 9° (3×), 15° (5×). Strongest false slope at 3× nominal . If established on false GS, it indicates ABOVE actual GS but aircraft is too steep.

📝 TRANSPONDER CODES — EMERGENCY REFERENCE

🔴 7700 — EMERGENCY General emergency; activates alert on ATC display 🔵 7600 — RADIO FAILURE NORDO; follow lost comms procedures 🔶 7500 — HIJACK Unlawful interference; covert squawk if possible 7000 — VFR Conspicuity (UK) Standard VFR code in UK; Mode C if available 2000 — Inadvertent CAS Entry Entering controlled airspace without clearance 1200 — VFR (USA) Standard US VFR code (not used in UK)

❁ GNSS SYSTEMS — FULL COMPARISON

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PARAMETER	GPS	GLONASS	GALILEO
SVs (nominal)	24 (+spares)	24	30
Orbital planes	6	3	3
SVs per plane	4	8	10
Altitude	20,180 km	19,099 km	23,222 km
Inclination	55°	65°	56°
Period	11 h 56 min	11 h 15 min	14 h 8 min
Geoid	WGS84	PZ90	ETRS89
Multiple access	CDMA	FDMA	CDMA
L1 frequency	1575.42 MHz	~1602 MHz	1575.42 MHz

NOTE

GPS Navigation Message Sub-frames: SF1 = clock corrections | SF2+3 = ephemeris (orbital data) | SF4 = ionospheric model, UTC | SF5 = almanac Almanac download: 25 frames × 30 s = 12.5 minutes (cold start)

RADAR SYSTEM SPECIFICATIONS

SYSTEM	FREQUENCY	PRF	SCAN RATE	MAX RANGE	PURPOSE
ASR (En-route)	SHF	200–400 pps	5–6 rpm	200–300 NM	Area surveillance
TAR (Terminal)	SHF	~400 pps	15 rpm	75 NM	Terminal area
Aerodrome PAR	SHF	—	Sector	25 NM	Precision approach
ASMI	15–17 GHz	4000–20000 pps	60 rpm	4–5 NM	Surface movement
AWR	9375 MHz	variable	stabilised	150–320 NM	Weather

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🌩 AWR — COLOUR GUIDE & HAZARDOUS SHAPES

COLOUR	WEATHER MODE	MAP MODE
BLACK	<0.7 mm/h — nil/very light	—
GREEN	0.7–4 mm/h — light	Light return
YELLOW	4–12 mm/h — moderate	Medium return
RED	>12 mm/h — heavy	Heavy return
MAGENTA	Turbulence (severe)	—

> ⚠ **NOTE**
>
> AWR Tilt Rule: Tilt UP to see storm tops (severity assessment). Tilt DOWN to scan for weather at range. Use MAP mode (fan beam) up to 60–70 NM, then switch to MAN (pencil) beyond.

SSR MODES — QUICK REFERENCE

MODE	INTERROGATION SPACING	INFORMATION	CODES/RESOLUTION
Mode A	P1–P3: 8 µs	Identity (squawk)	4096 codes (octal 0000–7777)
Mode C	P1–P3: 21 µs	Pressure altitude (1013.25 hPa)	100 ft increments (Gillham code)
Mode S	24-bit address	Data link; ID; alt; ELS/EHS	16,777,216 codes; 25 ft alt resolution

ELS (Elementary Surveillance) Squawk | Callsign | Pressure altitude | Emergency EHS (Enhanced Surveillance) ELS + Selected altitude | Airspeed | Magnetic heading

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NOTE

SSR Issues: Garbling — two aircraft within 1.7 NM same direction | Fruiting — phantom targets from other interrogators (fixed by ISLS) | SLS — P2 omnidirectional pulse suppresses side-lobe replies

📌 RNAV ACCURACY STANDARDS

TYPE	LATERAL ACCURACY	% REQUIREMENT	APPLICATION
B-RNAV	±5 NM	95%	Mandatory: Eurocontrol FIR >30 pax jets
P-RNAV	±1 NM	95%	Terminal area routes
2D RNAV	Lateral only	—	VOR/DME rho/theta
3D RNAV	Lateral + Vertical	—	VNAV added
4D RNAV	Lateral + Vertical + Time	—	RTA (Required Time of Arrival)

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NOTE

RNAV Erratic: Most common cause = aircraft beyond DOC / below LOS of reference VOR/DME . Also: reference VOR cone of confusion (overhead). Phantom station itself has no broadcast — its position is computed.

💻 EFIS DISPLAY MODES

MODE	ORIENTATION	WXR/TCAS	BEARING POINTER
Full Rose — VOR	Heading-up	No	Yes
Full Rose — ILS	Heading-up	No	No
Expanded VOR	Heading-up	Yes	Yes
Expanded ILS	Heading-up	Yes	No
MAP	Heading-up	Yes	Yes
PLAN	True North-up	No	No

⚠ ERROR SOURCES — ALL SYSTEMS

ADF / NDB Errors ● Night effect (sky wave) — WORST ● Quadrantal error (airframe) ● Coastal refraction ● Mountain effect ● Static interference ● Station interference ● No failure warning VOR Errors ● Site error (terrain reflections) — WORST ● Scalloping (multipath) ● Cone of confusion (>30° elevation) ● Propagation anomalies ● Airborne equipment error ● Pilot error (OBS) GPS Errors ● Ionospheric delay (single freq) ● Tropospheric delay ● Multipath ● Receiver clock error ● Ephemeris errors ● PDOP (geometry) ● Selective Availability (off) ILS Errors ● Multipath reflections from terrain/buildings ● False glideslopes (3×, 5×) ● Beam bends near threshold ● Back course reversal ● Runway contamination DME Errors ● Slant range vs ground range ● Garbling (multiple aircraft) ● Transponder saturation ● RNAV cone of confusion (reference VOR) AWR Limitations ● Cannot detect CAT ● Shadow behind heavy rain ● Attenuation in heavy precip ● Blue ice undetected ● Stabilisation failure → lopsided display

🔴 EXAM TRAPS — DON'T MIX THESE UP!

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Subject study group

NOTE

Topic WRONG (common error) CORRECT VOR variation At aircraft At station ADF variation At station At aircraft VOR ident tone 1350 Hz 1020 Hz DME ident tone 1020 Hz 1350 Hz ILS LOC band 329–335 MHz 108–111.975 MHz VHF ILS G/S band 108–118 MHz 329–335 MHz UHF SSR ground TX 1090 MHz 1030 MHz SSR aircraft TX 1030 MHz 1090 MHz GPS SVs for RAIM detect 4 or 6 5 SVs GPS SVs for RAIM exclude 5 or 7 6 SVs PLAN mode orientation Heading-up True North-up Mode C altitude ref QNH or local 1013.25 hPa (always) MLS channels 40 or 252 200 channels ILS channels 200 or 252 40 channels DME channels 40 or 200 252 channels GLONASS inclination 55° or 56° 65° GPS inclination 63° or 65° 55° Max radar range limiter Pulse width PRF Min radar range limiter PRF Pulse width AWR type Secondary radar Primary radar DME/SSR type Primary radar Secondary radar ILS false GS angle 2× or 5× nominal 3× nominal (strongest) GPS altitude reference MSL WGS84 ellipsoid 7500 squawk Emergency / Radio fail Hijack only

NOTE

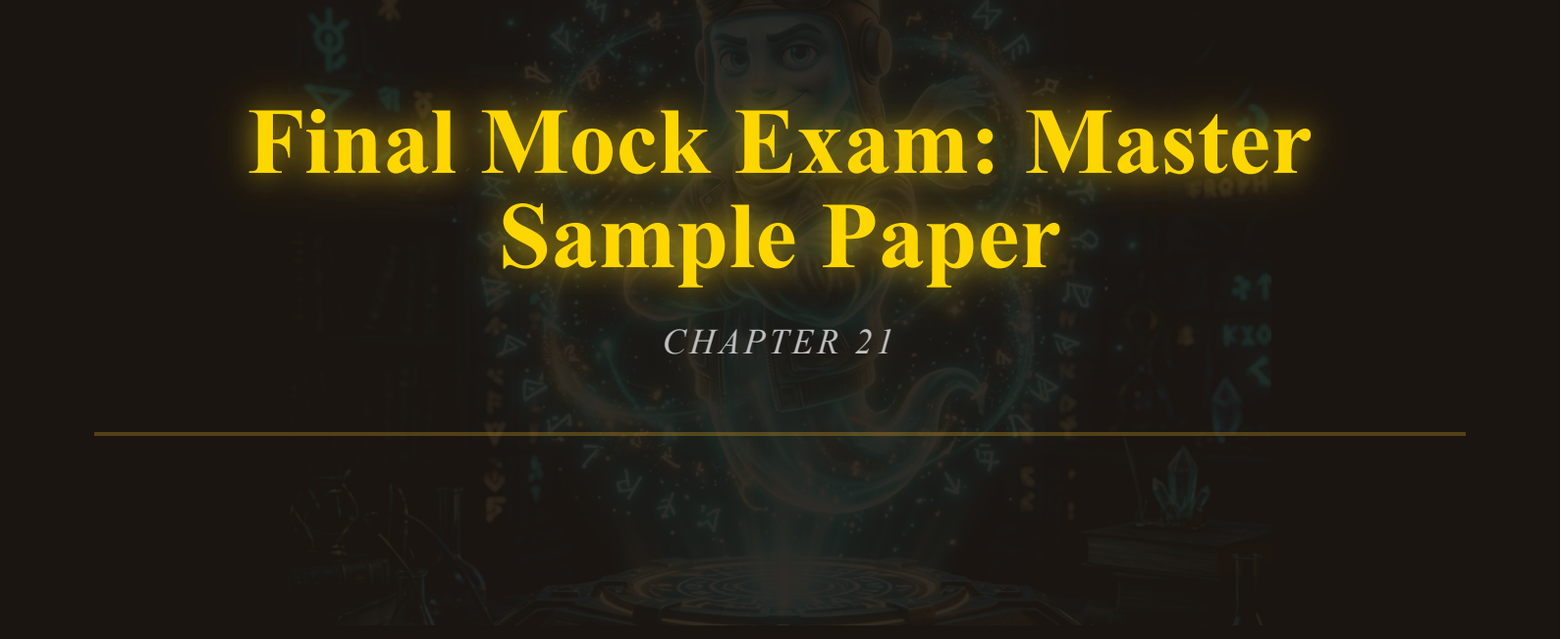
🎓 Mnemonics: ILS colours (low to high frequency for markers): B lue A mber W hite = B ig A pproach W arning (OM→MM→IM) AWR colours: B lack G reen Y ellow R ed M agenta = B ig G reen Y achts R ace M adly SSR emergency codes: 75= Hi jack / 76=R a dio / 77= E mergency = HAE GPS RAIM: 4=Fix | 5=Detect | 6=Exclude = 4-5-6

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CHAPTER 21



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GLOSSARY & DEFINITIONS

ADF

Automatic Direction Finder. An electronic radio receiver used in aviation for navigation.

ATC

Air Traffic Control. A service provided by ground-based air traffic controllers.

CDI

Course Deviation Indicator. An avionics instrument used to determine an aircraft's lateral position.

DME

Distance Measuring Equipment. A transponder-based radio navigation technology.

GNSS

Global Navigation Satellite System. Satellite constellations that provide positioning.

ILS

Instrument Landing System. A precision runway approach aid based on two radio beams.

NDB

Non-Directional Beacon. A radio transmitter at a known location used as a navigational aid.

QFE

Altimeter setting that will cause the altimeter to read height above a specific airfield.

QNH

Altimeter setting that will cause the altimeter to read altitude above mean sea level.

RNAV

Area Navigation. A method of IFR navigation that allows aircraft to choose any course.

VOR

VHF Omnidirectional Range. A type of short-range radio navigation system for aircraft.

WAAS

Wide Area Augmentation System. An air navigation aid developed by the FAA to augment GPS.

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